VITA 46.11 Tier-1 IPMC Software Specification

IPMIRustFRU: Rust IPMC firmware for the NUCLEO-F401RE, x86 emulator and bench tools

Revision 10-25-2026

Quick Start

1. Host tests and the emulator, from the repository root (Rust stable, GNU host): `cargo test --workspace`, then `cargo run -p ipmc-emu -- replay vectors/sbc_ab.vec --b-fitted 1 --b-enabled 1 --policy alternate --quiet` (the capture file is local only; on a fresh clone the capture tests skip with a message).
2. Firmware build and flash (NUCLEO-F401RE on USB): `powershell -ExecutionPolicy Bypass -File crates/ipmc-nucleo-f401re/gen.ps1 flash`. `flash` programs over SWD when OpenOCD finds the ST-LINK probe and copies the `.bin` to the drive otherwise (section 12.1; the two paths treat the FRU store differently, section 9.1). The console is COM3 at 115200 8N1; `gen.ps1 console "sys status"` sends one command and prints the reply.
3. Bench check with an independent IPMI implementation (Aardvark on IPMB-A, PB8/PB9): `tools\python-ipmi\.venv\Scripts\ipmitool.exe -I aardvark -t 0x82 raw 0x06 0x01` answers `00 01 80 00 01 02 09 00 00 00 4b 49 00 00 00 00`; `python tools/board_test_report.py` runs every bench check and writes `Documentation/board_test_report.md`.
4. Capture log and application port: `DEBUG DUMP` on the console prints the IPMB capture log (section 10.11); `python -I tools\capture\capture2pcap.py dump.txt -o dump` turns a saved dump into a file Wireshark opens. A USB-serial adapter on PB6/PB7 (COM5 on the bench) reaches the application port (section 7.6): `tools\python-ipmi\.venv\Scripts\python.exe tools\app-port\app_cli.py --port COM5 HELLO STATUS`.

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The section 10 diagrams are Mermaid state and sequence diagrams and one WaveDrom timing diagram; the block drawings are the `ipmc-software` drawing set (`diagrams/src/ipmc-software.js`, rendered to `diagrams/out/png/`).

1. Overview

IPMIRustFRU is an Intelligent Platform Management Controller (IPMC) for a VPX plug-in module, written in Rust to ANSI/VITA 46.11-2022 Tier 1. It runs on an STM32F401RE (NUCLEO-F401RE board), answers a Chassis Manager over the redundant Intelligent Platform Management Bus (IPMB-A and IPMB-B, I2C at 100 kHz; 400 kHz FAST-MODE at Tier-3), serves the FRU 0 inventory and the Device SDR repository, reports the seven mandatory VITA sensors and keeps Write FRU Data in on-chip flash across a reset. The same IPMI code builds for the PC as an emulator that replays captured Chassis Manager traffic or serves a live Chassis Manager over TCP. This document explains how the code works, crate by crate and module by module, for a software engineer who maintains or extends it.

The primary function is a Tier-1 IPMC: the 13 Tier-1 mandatory commands plus Broadcast Get Device ID, the 7 mandatory sensors, FRU 0 with Common Header, Board Info and Product Info areas, the FRU 0 Operational State machine (M0, M1, M4, M6; M5 at Tier-3), dual-bus link rules (alternation, resend on the other bus, Set IPMB State), a startup self test and NVMRO write protection.

The firmware runs on the STM32F401RE with the bit-bang IPMB master (default) or the hardware I2C master. The x86 emulator offers capture replay (`replay`), real-time TCP serving (`serve`) and an SDR dump (`sdr`).

The firmware has two serial ports, both interrupt driven. The console (USART2, the ST-LINK virtual COM port) carries the command line, including the DEBUG group: the IPMB capture log records every whole frame received and sent and every trace line, and the console dumps it as CSV or as trace rows for a spreadsheet or Wireshark (section 10.11). The application port (APP_UART, USART1 on PB6 and PB7) lets a payload CPU on the same board ask for status, run IPMI requests locally and write the per-unit FRU fields (serial numbers, part numbers, asset tag, manufacturing date); its protocol is the `ipmc-app` crate (section 7.6).

The public build is the Tier-1 IPMC described here. A commercial release build adds a private Tier-2/3 extension through the hook interface of section 4 (Build modes and the Tier-2/3 extension); the commands and behaviour it adds are listed in section 7.2, each marked "commercial release build". The public build answers those commands C1h. The commercial release build is Tier-3 by default (`IPMC_TIER` 3).

The public build leaves out Tier-2 and Tier-3 functions (events, payload power control by the Chassis Manager, FRU activation policy; they belong to the commercial release build), optional Tier-1 commands (Get Mandatory Sensor Numbers, FRU Control Capabilities, Get IPMB Link Info, Get FRU Hash, Get Device GUID, Set SEL Time: all answer C1h), subsidiary FRUs, and IPMI bridging. The built FRU image has no Internal Use or MultiRecord area; `verify` still checks those areas when a written image contains them.

The code is the authority for every statement here. Where the VITA 46.11-2022 text and the captured live Chassis Manager disagree, the code follows the specification.

2. System Context

- **Target device:** STM32F401RET6 (Cortex-M4F, 512 KB flash, 96 KB RAM) on the NUCLEO-F401RE, 84 MHz system clock from the 8 MHz ST-LINK MCO clock in HSE bypass.
- **Operating environment:** bare metal, `no_std`, one cooperative main loop plus interrupt handlers that move bytes, stamp times and drive the LED (the I2C slaves, the USART1 and USART2 rings, SysTick). No RTOS, no heap.
- **Host build:** `x86_64-pc-windows-gnu` (the same sources build for `x86_64-unknown-linux-gnu`); the emulator is a `std` binary.
- **Dependencies:** `cortex-m 0.7.10`, `cortex-m-rt 0.7.7`, `stm32f4 0.16.0` (PAC, feature `stm32f401`), `heapless 0.8.0` (SPSC queues), `panic-halt 1.0.0`, `embedded-hal 1.0.0` (bus traits of the example external sensor drivers). Every version is an exact `=` requirement.
- **Peers on the IPMB:** a Chassis Manager at 8-bit address 20h (`IPMB_CM_ADDR8`; the captured live Chassis Manager uses 10h). The IPMC answers at (40h + GA[4:0]) x 2: slot 1 is 82h.
- **Bench:** an Aardvark I2C host adapter on IPMB-A with its pull-ups on, driven by python-ipmi as the Chassis Manager; the ST-LINK USB port carries the console (COM3) and the drag-and-drop flash drive (`NOD_F401RE`); a USB-serial adapter (COM5) on PB6 and PB7 carries the application port (`tools/app-port/app_cli.py`).
- **Normative documents:** ANSI/VITA 46.11-2022 (local, licensed, not in the repository), IPMI v2.0 rev 1.1, IPMI Platform Management FRU Information Storage Definition v1.0 rev 1.3, RM0368 (STM32F401 reference manual), DS10086 (datasheet), UM10204 (I2C-bus specification).

Term	Meaning
IPMB	Intelligent Platform Management Bus: I2C with IPMI framing and two checksums
rsSA / rqSA	responder / requester slave address, 8-bit form (7-bit address shifted left, R/W bit 0)
NetFn	network function; even for a request, odd (request + 1) for its response
VSO	VITA Standards Organization; its Group Extension defining-body byte is 03h
SDR	Sensor Data Record; the Device SDR repository lives in the IPMC
FRU	Field Replaceable Unit; FRU 0 is the module that holds the IPMC
GA, GAP	geographic address straps GA[4:0] and their odd-parity strap
NVMRO	non-volatile memory read-only strap; asserted, the IPMC writes no non-volatile memory
T1..T9, C1	VITA 46.11 Table 4.1.1-1 timing and retry parameters
APP_UART	the application UART (USART1) between the IPMC and the payload CPU

3. Requirements

Functional requirements

ID	Requirement	Rule	Where
FR-001	Answer the Tier-1 mandatory commands of Tables 10.1.1-1 and 10.1.2-1; every other command answers C1h	Rules 4.2-2, 4.1.1-16	`ipmc-core/src/commands.rs`
FR-002	Answer Broadcast Get Device ID (general-call 00h before the frame) at the own address only	IPMI 20.9	`link.rs` `on_frame`
FR-003	FRU Device ID other than 0 answers C9h (Get FRU Address Info ignores the ID when a physical address key is given)	Rules 4.2-3, 4.1.1-17	`cmd/mod.rs` `check_fru_device_id`
FR-004	Power-up bus state: IPMB-A enabled, IPMB-B isolated until Set IPMB State	Rule 4.1.1-4	`link.rs` `BusPair::power_up`
FR-005	Accept requests on both buses; alternate transmissions between enabled buses	Rules 4.1.1-5, 4.1.1-6	`link.rs` `choose_tx_bus`
FR-006	Resend a failed transmission on the other bus, at most `IPMB_TX_RETRIES` times	Rule 4.1.1-7	`link.rs` `on_tx_result`
FR-007	Refuse a Set IPMB State that leaves no bus enabled (CCh)	Rule 4.1.8-3	`link.rs` `plan_ipmb_state`
FR-008	Bus state changes only by Set IPMB State or a persistent local fault	Rule 9.3-10	`link.rs`, `ipmb_link.rs`
FR-009	A retried request (same requester, sequence, command and data) is answered from a cache and runs no handler	Obs 4.1.1-6 (a retry keeps its sequence number)	`dupcache.rs`
FR-010	Report sensors 0..6 with the layouts of Tables 10.2.1-1 to 10.2.7-1, and sensor 7 (Payload Mode, Table 10.2.8-1) at Tier-3	Table 10.2-1	`sensor.rs`, `dispatch.rs` `read_sensor`
FR-011	FRU 0 holds Common Header, Board Info and Product Info areas; Chassis Info offset 00h	Rules 8.1-2, 8.1-3	`ipmc-fru/src/records.rs`
FR-012	FRU data readable whatever the payload power	Rule 8-1	`cmd/storage.rs`

ID	Requirement	Rule	Where
FR-013	Write FRU Data while NVMRO is asserted answers 80h and changes nothing; an application port `SET` answers `ERR NVMRO` and changes nothing; no flash write while NVMRO is asserted	Rules 8-3, 8-5, Rec 4.1.10-1	`cmd/storage.rs`, `app_port.rs`, `persist.rs`, `bsp/flash.rs`
FR-014	FRU 0 state machine M0 to M1 on the initial checks, M1 to M4 on valid payload power, M4 to M6 on power loss, M6 to M1 on a complete shutdown; at Tier-3 also M4 to M5 on a deactivation request, M5 to M4 when it is cleared and M5 to M6 on power loss	Section 4.1.5, Rules 4.1.5.4.1-2, 4.1.5.4.2-3, 4.1.5.4.3-2, 4.1.5.4.4-2, 4.1.5.4.4-5, 4.1.5.4.5-1	`fru_state.rs`
FR-015	A payload power-on request in M6 is refused; in the commercial release build it is also refused while the FRU is activation-locked; one check serves the boot power-on, the handler hand-off and the console	Rules 4.1.5.4.5-2, 4.1.5.4.2-1	`fru_state.rs` `check_power_on`, `ext.rs` `check_power_on`
FR-016	An invalid hardware address (even parity or GA = 0) isolates both buses and keeps the FRU in M0	Rules 4.1.5.4.1-4, 4.1.5.4.1-5	`addr.rs`, `link.rs`
FR-017	Startup self test of the FRU image and the SDR chain; 55h 00h or 57h with the failure bits; a failure keeps the FRU in M0; rerun after each accepted Write FRU Data	IPMI Table 20-5, Rule 4.1.5.4.1-9, Rec 8-2	`selftest.rs`
FR-018	Write FRU Data and an application port `SET` survive a reset (`IPMC_FRU_PERSIST 1`); the FRU store is flash sector 1 alone	owner decisions 10-23-2026 and 10-25-2026	`persist.rs`, `fru_log.rs`, `memory.x`, `link.x`
FR-019	No IPMC pull-ups on the IPMB lines	Rule 9.2.3-2	`bsp/ipmb_pins.rs`, `i2c_hw.rs`
FR-020	A (netFn, cmd) the Tier-1 table does not own goes to the extension hook; the Tier-1 build has none and answers C1h	Rules 4.2-2, 4.1.1-16	`ext.rs`, `dispatch.rs`
FR-021	A Tier-2/3 build sends requests one at a time: a response wait of `IPMB_REQ_TIMEOUT_MS` (T3), up to `IPMB_REQ_RETRIES` retries with the same sequence number, a failed transmission retried on the other bus after T6 (60 ms minimum), T4 between new requests, T9 before a sequence number is reused; a Tier-1 build sends none	Rules 4.1.1-9, 4.1.1-10, 7-4, Table 4.1.1-1	`requester.rs`, `link.rs`
FR-022	A Cold Reset request restarts the processor once its response has left the egress queue and an accepted FRU write is saved, or after `IPMC_RESET_WAIT_MS`	IPMI v2.0 20.2	`reset_gate.rs`, `ipmb_link.rs`

ID	Requirement	Rule	Where
FR-023	The optional speed byte of Set IPMB State sets each bus to 100 kHz (00b) or, at Tier-3 on a fitted bus, 400 kHz (01b); every other value, a reserved bit and 400 kHz below Tier-3 answer CCh with no change; every bus starts and restarts at 100 kHz	Rules 4.1.1-4.3, 4.1.1-4.4, 4.1.1-14	`cmd/vita.rs`, `link.rs` `plan_ipmb_speed`
FR-024	A Warm Reset request (commercial release build) restarts the link state and the requester without a processor reset, once its response has left and an accepted FRU write is saved, or after `IPMC_RESET_WAIT_MS`	IPMI v2.0 20.3	`reset_gate.rs`, `ipmb_link.rs`, `link.rs` `restarted`
FR-025	A Cold Reset of the payload (commercial release build) holds PAYLOAD_RST_OUT_N (PA9, open drain) low for `IPMC_PAYLOAD_RESET_PULSE_MS` without blocking the main loop; the FRU state does not change	Rule 4.1.2-9	`reset_pulse.rs`, `sensing.rs`
FR-026	Every whole IPMB frame the IPMC receives (before the parser) or sends (with its outcome) and every trace line goes into a RAM ring of `IPMC_LOG_SIZE` bytes, oldest overwritten first; the console dumps it as CSV or as trace rows	owner, Phase 11 (PLAN, 10-25-2026)	`capture.rs`, `debug_cmds.rs`, `ipmb_link.rs`
FR-027	A payload CPU on APP_UART gets exactly one answer line (`OK ...` or `ERR <CODE> ...`) per request line, with the optional tag echoed; a `SET` of a unit field rewrites that field in place, moves no area and goes to the flash through the Write FRU Data save path	owner, Phase 11 (PLAN, 10-25-2026)	`ipmc-app`, `app_port.rs`, `ipmc-fru/src/unit.rs`

Performance requirements

Item	Target	Notes
Response time	at most 227 ms (T5)	request STOP to response STOP; `RESPONSE_DEADLINE_MS`, late ones counted
One master write	at most T2 + T1 = 80 ms	retries included; `WRITE_BUDGET_NS`; one write per pass
Clock stretch	at most 3 ms (T8)	`IPMB_STRETCH_TIMEOUT_US 3000`
Bus-free wait	at least 60 ms (T2)	`IPMB_BUS_FREE_TIMEOUT_MS 60`; ends at the write budget
One IPMB frame	at most 20 ms (T1)	`ipmb_common.rs` `T1_FRAME_NS`

Item	Target	Notes
IPMB clock	100 kHz maximum; 400 kHz maximum in FAST-MODE (Tier-3)	half bit 5000 ns; measured SCL period 12.0 us (83 kHz). FAST-MODE: bit-bang 6795 ns (about 147 kHz), hardware master 2981 ns (about 335 kHz)
Flash sector erase	typical 250 ms, at most 500 ms (16 KB sector, x32 programming), core stalled; 198 ms measured on the bench	DS10086 Table 45; IPMB slaves off the bus meanwhile; one erase per 60 saves, and only when the bus is quiet

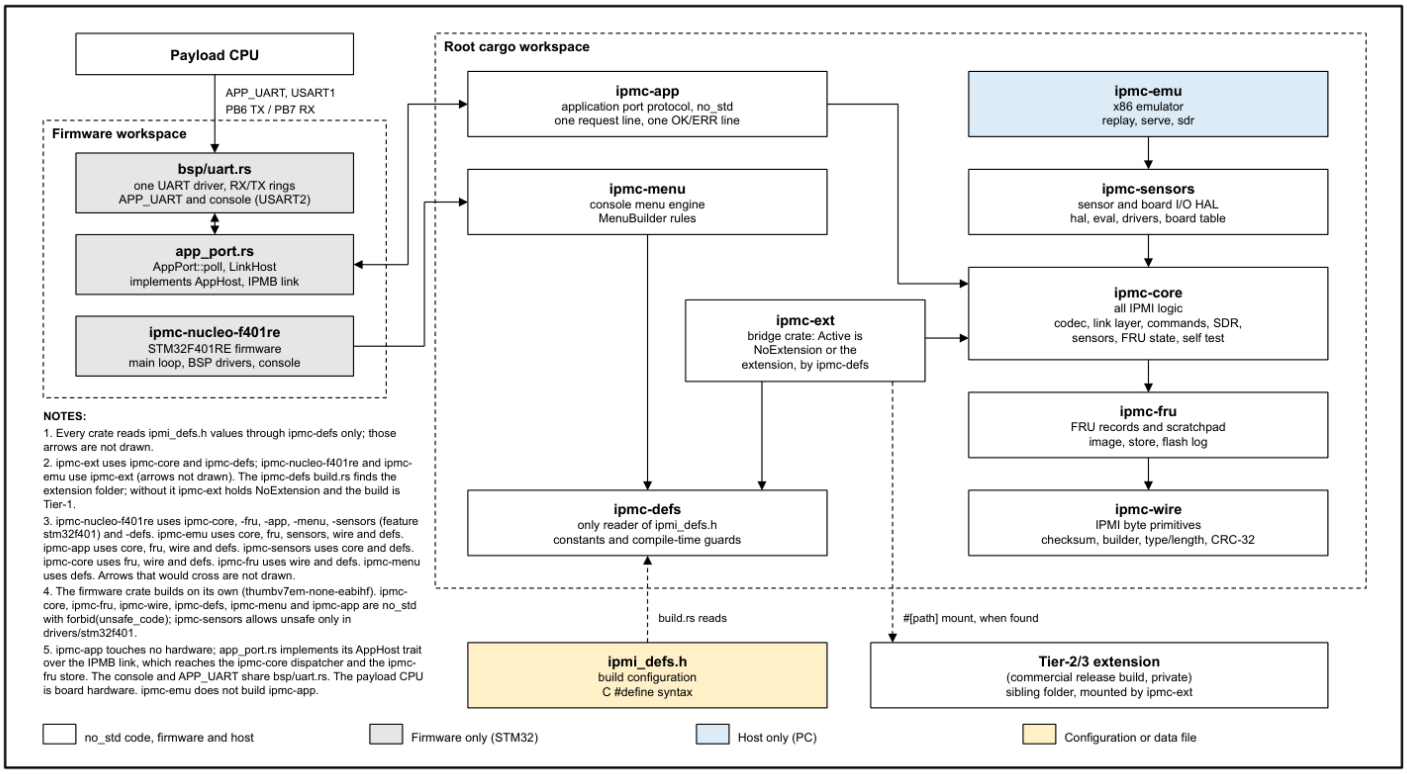
Compliance requirements

ANSI/VITA 46.11-2022 Tier 1, Layer 1 IPMC (VSO revision 22h); IPMI v2.0 rev 1.1 message formats, completion codes (Table 5-2) and the Group Extension defining-body rule (Table 5-1, section 5.3.1). The spec-only suite in `tools/conformance/` (section 13) checks the IPMC against those texts.

4. Architecture

Crate structure

Figure 4-1. Crate structure and dependencies (ipmc-software, sheet 1)



Source: `diagrams/src/ipmc-software.js`, sheet 1.

The code is a Cargo workspace of nine crates plus the firmware crate, which is its own workspace (it builds only for `thumbv7em-none-eabihf`). Dependencies point one way: the two binaries at the top, the IPMI logic in the middle, the configuration reader at the bottom. The bridge crate `ipmc-ext` sits between the IPMI logic and the two binaries; it is the only place the private Tier-2/3 extension enters the build.

Crate	Kind	Holds
`ipmc-defs`	`no_std` library	the only Rust reader of `ipmi_defs.h` (and of the extension's header when the extension is built in): one `pub const` per define, compile-time guards on every value, the IPMB timing arithmetic (`i2c_timing.rs`) and the one detection of the Tier-2/3 extension (`build.rs`, `tier23_detect.rs`, `dep_cfg.rs`)
`ipmc-wire`	`no_std` library	IPMI byte primitives: zero checksum, const record builder, type/length ASCII field, CRC-32, hex-byte parser

Crate	Kind	Holds
`ipmc-fru`	`no_std` library	every FRU 0 record (image build, decoder, `verify`), the per-unit field editor (`unit.rs`), the RAM scratchpad (`FruStore`) and the flash store (`persist`)
`ipmc-core`	`no_std` library, `forbid(unsafe_code)`, no allocator	every IPMI decision: frame codec, address decode, link layer with the IPMB requester, duplicate cache, queues, byte ring, capture log, command table, dispatcher, extension hook interface, handlers, SDR repository, Tier rules, event enables, sensor encoders, FRU state machine, self test, reset gate, trace hook
`ipmc-ext`	`no_std` library, `forbid(unsafe_code)`	the bridge: `ipmc_ext::Active` is `NoExtension` in the Tier-1 build and the extension's state when `ipmc-defs` found it; the firmware and the emulator hold one `Active`
`ipmc-sensors`	`no_std` library	sensor and board-I/O HAL: four traits, hardware-free evaluation, drivers (simulation, STM32F401 ADC/GPIO behind feature `stm32f401`, example I2C/SPI parts), board table, sampling monitor, payload reset pulse timing, GPIO pin ownership
`ipmc-menu`	`no_std` library, no dependencies besides `ipmc-defs`	the console menu engine (MenuBuilder rules, tiered groups)
`ipmc-app`	`no_std` library, `forbid(unsafe_code)`, no allocator	the application port protocol: line assembler, command table `COMMANDS`, handlers, and the `AppHost` trait through which the firmware gives the protocol its view of the board (section 6.8)
`ipmc-emu`	host `std` binary and library	the x86 emulator: vector reader, `Emulator`, capture replay (also the Event Receiver), real-time `serve`, SDR dump
`ipmc-nucleo-f401re`	firmware binary	start-up, board support (one module per hardware access path), IPMB drivers, UART driver, flash driver, console, the application port transport, composition of the crates above

Layering rules

- `ipmc-core` decides; the firmware and the emulator move values. A handler never touches hardware; a driver never decides an IPMI response.
- No IPMI code runs in an interrupt. The I2C interrupt posts whole frames to a per-bus single-producer single-consumer queue; the main loop runs the link layer and the handlers.
- One owner per hardware resource: `ports` owns GPIOA..GPIOC and hands out pin claims; `bsp/i2c_hw.rs` owns I2C1 and I2C3; the bit-bang master owns the IPMB pins only while it transmits; `bsp/flash.rs` owns the FLASH registers after clock start-up; `uart.rs` owns USART1, USART2 and their pins; `led.rs` owns PA5; `ga.rs` owns the strap pins (GA, GAP, NVMRO and the BOOTLOADER pin); `ipmb_link.rs` owns the static Link, the dispatch Context and the capture log; `app_port.rs` moves bytes between the APP_UART rings and `ipmc-app` and owns nothing else.

- The application protocol never touches hardware: `ipmc-app` asks the board through the `AppHost` trait, and the per-unit FRU fields reach the flash through the one path a Chassis Manager's Write FRU Data uses (the RAM image, the dirty mark, the save step).
- The dispatcher asks the extension hook only after the Tier-1 table has no row for a command, so a Tier-1 command cannot be replaced. Nothing specific to a Tier-2/3 command, event or hook is written in this repository: board support, processor functions (reset, flash, timers, the I2C speed) and the link layer, the IPMB requester included, live here; the Tier-2/3 commands, hooks and sensor events live in the extension.
- One implementation of each function, judged by behaviour (`tools/dupcheck.py` is a gate): one checksum, one frame parser and encoder, one GA decode, one sensor-read function, one record builder, one retry loop shared by both master methods.
- One file for every tunable value: `ipmi_defs.h` (section 5). Values fixed by a specification or datasheet stay in the module that owns them, with the rule cited beside each.

Key design choices

- The link layer (`ipmc-core/src/link.rs`) is a pure state machine: frames go in with the STOP time, responses come out with the bus to use, results come back. The same object runs in the firmware and the emulator, so the dual-bus rules are tested on the PC against 309 captured exchanges.
- The bit-bang master (`bsp/i2c_bitbang.rs`) drives the slave's own pins. The I2C peripheral stays a slave; for a transmission it is switched off, the same pins are driven open-drain as GPIO, then the peripheral is switched back on. This follows the earlier PIC32 controller and avoids the STM32 I2C master state machine; the hardware master stays available behind `IPMB_MASTER_MODE`.
- The response bus is chosen at first transmission (`Link::next_tx`), not at queue time, so alternation follows the frames really handed to a driver, resends included.
- The duplicate cache (`dupcache.rs`) is keyed by the exact request, so a retry runs no handler and a retried Write FRU Data or Set IPMB State takes effect once.
- Set IPMB State uses a hand-off (`PendingIpmbState`). The handler cannot reach the Link; it validates against a read-only view and leaves the change for `Link::poll`, which applies it before the response is queued.
- The FRU flash store (`ipmc-fru/src/persist.rs`) is an append-only log in one sector (flash sector 1, 16 KB, 60 records of 272 bytes) with a commit word programmed last and a CRC-32 per record; the highest sequence number wins at boot, and a power cut at any word of an append leaves the previous record loadable. The save that finds the sector full erases it and writes the newest image first (owner decision 10-25-2026, no second copy): a power cut between that erase and that write loses the stored image, and the next start-up loads the built image.
- The capture log (`ipmc-core/src/capture.rs`) is one byte ring of whole records. Every trace sink the firmware hands to the core is a `LoggedTrace`, which records the line and prints it on the console unless the IPMI console trace is off (section 10.11).
- Both UART ports run one driver (`bsp/uart.rs`) with a receive and a transmit ring each; a write never waits for the wire, and a line that does not fit the transmit ring is dropped and counted (section 6.6).
- Configuration is checked at compile time (`ipmc-defs`). A malformed or out-of-range define fails the build with its line number or the guard that names it.
- The requester (`requester.rs`) sits in the link layer. The request sequence number, response matching and the resend timing sit beside the response path of `Link`, so one `next_tx` hands out both and responses keep priority over requests. The extension decides what to send and when; a Tier-1 build never sends a request (Rule 7-4).
- A processor reset uses a hand-off (`reset_gate.rs`). A handler asks for a Cold Reset through `Context::pending_reset`; the owner of the processor performs it when the gate says nothing is lost (section 8.2).

- Processor and payload services use a hand-off (`dispatch.rs`). A handler never switches payload power, pulses the payload reset output or restarts the link; it leaves a request in the `Context` (`pending_power`, `pending_payload_reset`, `pending_reset`) and the owner of that resource performs it (section 6.3, section 8.2).
- The tier rules live in one place (`tier.rs`). Get Device ID, the Device SDR records, the event enables, the FRU state machine, the sensor count and the IPMB speed derive their Tier-dependent behaviour from six rules (section 9.6).

Build modes and the Tier-2/3 extension

One source tree builds two kinds of image. The public build is the Tier-1 IPMC of this document. The commercial release build adds the Tier-2/3 extension, which lives in a private sibling folder named `IPMIRustFRU_Tier23`; this repository holds only the hook interface, the bridge and the detection. The public Tier-1 build answers every Tier-2/3 command C1h (section 7.2).

Build	Selected by	`IPMC_TIER`	`ipmc_ext::Active`
Public Tier-1	sibling folder absent, or `IPMC_TIER23=0`	1	`NoExtension`
Commercial release, Tier-2	folder present, its header sets `IPMC_TIER` 2	2	the extension
Commercial release, Tier-3 (the default)	its header sets `IPMC_TIER` 3	3	the extension
Evaluation (section 8.4)	`IPMC_EVAL=1` with the extension, release profile	as the commercial build	the extension

`crates/ipmc-defs/build.rs` is the one detection point; its rules are pure functions in `tier23_detect.rs`, tested by `tests/tier23.rs`.

`IPMC_TIER23`	Result
unset	on when both marker files exist (the extension's header `ipmi_defs_tier23.h` and its library source), off when neither exists; exactly one present fails the build ("half a Tier23 checkout")
`0`	off, whatever exists: the public Tier-1 build
`1`	on; a missing marker file fails the build and names it
any other value	fails the build

- Defines: off uses `ipmi_defs.h` plus `IPMC_TIER` = 1 in the generated file; on, the extension's header is parsed by the same parser and merged; a name in both headers fails the build, and so does `IPMC_TIER` in `ipmi_defs.h` or missing from the extension's header. The guards of the extension's defines are mounted from the extension folder.
- `ipmc-defs` publishes the decision. It declares `links = "ipmc_defs"` and prints `cargo:tier23=1` or `0` (and `cargo:eval=1` or `0`); cargo hands these to each direct dependent as `DEP_IPMC_DEFS_TIER23` and `DEP_IPMC_DEFS_EVAL`. `build.rs` also sets `cfg(tier23)` for `ipmc-defs` itself and exports `ipmc_defs::TIER23_BUILT`.
- The dependents share one build-script step. `ipmc-ext` and the firmware crate include `crates/ipmc-defs/dep_cfg.rs` and call `cfg_from_defs(key, cfg, path)`, which turns the published value into `cfg(tier23)` or `cfg(ipmc_eval)`. They read no file and no other variable, so they cannot disagree with `ipmc-defs`.

- With `cfg(tier23)`, `ipmc-ext` mounts the extension's sources through a relative `#[path]` and `Active` is the extension's state type; otherwise `Active` is `NoExtension`. A compile-time assertion requires a build without the extension to report `IPMC_TIER` 1.
- The build script cannot watch a folder that is missing, so after adding it to a tree already built without it, run `cargo clean -p ipmc-defs`. `gen.ps1 build` sets `IPMC_TIER23` from the folder.
- `IPMC_EVAL` is unset or 0 (off) or 1 (on); `IPMC_EVAL=1` without the extension, or outside the release profile, fails the build. On, it prints a build warning with the length and sets `cfg(ipmc_eval)`.

The hook interface is `ipmc_core::ext::Extension`, five methods with a default for the last four.

Method	Called	Purpose
`command(ctx, req, out)`	by the dispatcher, after the Tier-1 table has no row for (netFn, cmd) and the LUN and VSO checks passed	answers a command the extension owns with its name and a `Handler`-style result; `None` gives C1h. For NetFn 2Ch the request data and `out` exclude the VSO byte, as for a `Handler`
`poll(ctx, requester, now_ms, trace)`	once per `Link::poll`, after the queued requests are answered and `ctx.link` is refreshed	the extension sends its requests through the `Requester`
`on_sample(ctx, transition, now_ms, trace)`	once per sensor sample, after `Context::update_sensors` stored the snapshot and stepped the FRU state machine	`transition` is the FRU state change that sample made, if any
`warm_reset()`	once, after the owner applied a Warm Reset (`Context::warm_reset` ran)	the extension re-initialises its volatile state (a request in flight, its event timers) and keeps what a Warm Reset keeps (section 8.2)
`power_on_allowed(ctx)`	by `ext::check_power_on`, after the FRU state machine's own check passed	whether payload power may come on now; the default allows it, so the Tier-1 build is unchanged, and a Tier-3 extension answers false while the FRU is activation-locked (Rule 4.1.5.4.2-1)

`NoExtension` answers nothing and takes the defaults of the hooks. `ext::feed_sensors(ctx, ext, now_ms, snapshot, trace)` is the one sensor-sample entry of the firmware and the emulator: it calls `update_sensors`, then `on_sample` with the transition, and returns the transition. `ext::check_power_on(ctx, ext)` is the one combined power-on check: `FruMachine::check_power_on` first (`PowerOnRefused::DeactivationInProgress` in M6), then `power_on_allowed` (`PowerOnRefused::ActivationLocked`); the first refusal is returned. The boot power-on, the `pending_power` On hand-off and the console `POWER ON` all call it.

5. Build-Time Configuration

`ipmi_defs.h` at the repository root is the single build configuration. It keeps C `#define` syntax: plain `#define NAME <integer>` lines (a signed value carries the tag `[i32]` as the first token of its trailing comment) and `#define NAME "text"` lines (one printable-ASCII literal, no escapes). `crates/ipmc-defs/build.rs` parses it with `defs_parser.rs` into `$OUT_DIR/ipmi_defs.rs`; every crate reads a value as `ipmc_defs::NAME`. Any other line form fails the build with `path:line: reason`, and `ipmc-defs/src/lib.rs` holds more than 100 compile-time guards (written out, and from the FRU-string, SDR-name and rail guard macros) that tie values to their rules (for example `IPMB_STRETCH_TIMEOUT_US` at most 3000, the T8 limit; `IPMC_FRU_SIZE` a multiple of 8; every FRU string at most 63 bytes). `types.rs` fixes each define to the type its consumers expect, so a wrong `[i32]` tag fails the build.

Prefixes: `IPMB_` for bus values, `IPMC_` for controller and application values. The tables below group the defines; the header itself carries the rule beside each line.

Bench and test switches

The block at the top of `ipmi_defs.h` holds the switches to check before a system build (PLAN 3.7 decision 24).

Define	Default	Meaning
`IPMC_EVENTS_AT_BOOT`	0	Tier-2/3 event generator only, no effect in a Tier-1 build. 0 (bench): at start-up the IPMC sends itself Set Event Receiver FFh, so no Platform Event is retried toward a Chassis Manager that never acknowledges, and the console prints `Events: off from start-up`. 1 (system build): event messages go to the Event Receiver 20h from power-up (Rule 7-5). `DEBUG EVENTS ON` or `OFF` changes it at run time. The firmware start-up reads it; the emulator and the conformance runs are unchanged
`IPMC_LOG_AT_BOOT`	1	the capture log records from start-up; `DEBUG LOG ON` or `OFF` changes it at run time
`IPMC_I2C_TRACE_AT_BOOT`	0	one console line per IPMB frame in and out from start-up; `DEBUG TRACE I2C ON` or `OFF` changes it at run time; a row that does not fit the console TX ring whole is dropped and counted (`tx_lines_dropped`), and the frame stays in the capture log

IPMB link and drivers

Define	Default	Meaning
`IPMB_MASTER_MODE`	`IPMB_MASTER_BITBANG` (1)	master method of both buses; `IPMB_MASTER_HW` (0) selects the hardware I2C master
`IPMB_BUS_A_ENABLE`	1	IPMB-A built in; must be 1 (Rule 4.1.1-1)

Define	Default	Meaning
`IPMB_BUS_B_ENABLE`	0	IPMB-B built in; 0 on the one-Aardvark bench (Get VSO Capabilities then reports one interface)
`IPMB_I2C_HZ`	100000	IPMB clock, STANDARD-MODE
`IPMB_I2C_FAST_HZ`	400000	IPMB FAST-MODE clock (Rule 4.1.1-4.4; Tier-3, entered with the Set IPMB State speed byte); fixed at 400000, the hardware master derives its CCR and TRISE from it (RM0368 18.6.8, 18.6.9)
`IPMB_MSG_MAX`	32	whole frame, rsSA to the second checksum
`IPMB_TX_POLICY`	`IPMB_TX_ALTERNATE` (0)	response bus: alternate (Rule 4.1.1-6) or `IPMB_TX_ARRIVAL` (1), the request's bus
`IPMB_CM_ADDR8`	20h	Chassis Manager address; only guarded, no code reads it (responses go to each request's rqSA)
`IPMB_DUP_CACHE`	64	duplicate-cache entries (8..=64)
`IPMB_DUP_WINDOW_MS`	3000	retry window: covers 11 attempts of 250 ms, below the 3.846 s sequence reuse of the captured Chassis Manager
`IPMB_TX_RETRIES`	3	resends of one response on the other bus (3..=10, C1)
`IPMB_REQ_TIMEOUT_MS`	250	wait for the response to a request this IPMC sent, before a retry (60..=250; T3max, Rule 4.1.1-9; T6 spaces the retries)
`IPMB_REQ_RETRIES`	10	retries of a request after its first transmission (3..=10; C1max, Rule 4.1.1-10): 11 transmissions x 250 ms = 2.75 s, inside the T9 expiry of 5 s
`IPMB_REQ_GAP_MS`	5	least time from one new request to the next (at least 5; T4, Table 4.1.1-1)
`IPMC_RESET_WAIT_MS`	3000	longest wait before a pending Cold Reset runs anyway (1..=5000): the response leaving the egress queue, one flash sector erase (at most 500 ms) and the FRU write fit inside it
`IPMC_PAYLOAD_RESET_PULSE_MS`	100	time PAYLOAD_RST_OUT_N (PA9) is held low by a Cold Reset of the payload (1..=1000; Rule 4.1.2-9)
`IPMB_INGRESS_DEPTH`	8	received requests held for `Link::poll` (1..=32)
`IPMB_TXQ_DEPTH`	4	response frames in the egress queue
`IPMB_RX_QUEUE_DEPTH`	4	frames one bus interrupt can post before the main loop drains them (2..=16)
`IPMB_BITBANG_HALF_NS`	5000	bit-bang half bit

Define	Default	Meaning
`IPMB_BITBANG_FAST_LOW_NS`	1400	bit-bang SCL low time in FAST-MODE (1300..=100000; the I2C fast-mode tLOW minimum is 1300 ns)
`IPMB_BITBANG_FAST_HIGH_NS`	1000	bit-bang SCL high time in FAST-MODE (600..=100000; the I2C fast-mode tHIGH minimum is 600 ns)
`IPMB_STRETCH_TIMEOUT_US`	3000	longest stretched SCL (T8)
`IPMB_BUS_FREE_TIMEOUT_MS`	60	T2 share of the master write budget (T2 + T1); the free-bus wait ends at that budget
`IPMB_ARB_RETRIES`	3	same-bus retries after arbitration loss (0..=10)
`IPMB_FAULT_STREAK`	8	consecutive bus-fault errors before a bus is marked faulted (2..=64)
`IPMB_FAULT_CLEAR_MS`	1000	both lines idle-high this long before a fault clears
`IPMC_IPMI_TRACE`	1	one trace line per dispatched request, and the link and event lines (failed transmissions, resends, requests sent). 0 removes the link and dispatcher lines at compile time, so they are absent from the capture log as well as from the console; the status lines stay (FRU log saves, bus faults, bus speed changes, resets, and the FRU state change line)
`IPMC_LOG_SIZE`	8192	capture log ring, bytes (1024..=65536, a power of two); the usable size is one byte less, because the ring keeps a slot empty; each `DEBUG DUMP` row walks the log from its start, so a larger log makes the dump slower per row
`IPMC_LOG_TEXT_MAX`	160	longest trace line the log keeps, bytes (32..=200); a longer line is cut and ends in `...`

Straps, identity and FRU

Define	Default	Meaning
`IPMB_GA_SOURCE`	`IPMB_STRAP_FORCED`	GA from this file (bare Nucleo, no straps wired) or `IPMB_STRAP_FROM_PINS`
`IPMB_GA_FORCED`	01h	slot 1, hardware address 41h, IPMB 82h; GAP generated for odd parity
`IPMB_NVMRO_SOURCE`, `IPMB_NVMRO_FORCED`	forced, 0	NVMRO deasserted (writes allowed)
`IPMC_DEVICE_ID`, `IPMC_DEVICE_REVISION`, `IPMC_DEVICE_PROVIDES_SDRS`	01h, 0, 1	Get Device ID byte 2, byte 3 bits 3:0, byte 3 bit 7

Define	Default	Meaning
`IPMC_IPMI_VERSION`	02h	IPMI 2.0 (least significant digit in bits 7:4)
`IPMC_SUPPORT_*`	sensor 1, FRU inventory 1, others 0	Get Device ID byte 7
`IPMC_MANUFACTURER_ID`	000000h	unspecified until an IANA enterprise number is registered
`IPMC_PRODUCT_ID`	494Bh	product ID
`IPMC_FRU_BOARD_*`, `IPMC_FRU_PRODUCT_*`, `IPMC_FRU_FILE_ID`, `IPMC_FRU_ASSET_TAG`	strings	FRU board and product area fields
`IPMC_FRU_MFG_MINUTES`	16204320	board manufacturing date, minutes since 1996-01-01 (10-23-2026)
`IPMC_FRU_SIZE`	256	FRU inventory area size
`IPMC_FRU_PERSIST`	1	Write FRU Data and the application port survive reset (the flash store, sector 1 alone, 16 KB); 0 keeps the FRU in RAM and a reset restores the built image
`IPMC_FRU_BOARD_SERIAL_MAX`, `IPMC_FRU_BOARD_PART_MAX`, `IPMC_FRU_PRODUCT_PART_MAX`, `IPMC_FRU_PRODUCT_SERIAL_MAX`, `IPMC_FRU_ASSET_TAG_MAX`	24 each	room the built areas reserve for each per-unit text field, bytes, whatever the default string's length (the serial numbers 17..=63, so that a MAC address text fits; the others 2..=63). An update up to this length rewrites its field in place and no area moves
`IPMC_FRU_PERSIST_DELAY_MS`	1000	quiet time after the last write before a save
`IPMC_FRU_SCAN_CHUNK`	64	bytes per read when the log checks a range for erased flash (a multiple of 4, 16..=272; the buffer lives on the stack)
`IPMC_FRU_ERASE_QUIET_MS`	500	run-time erase waits for an empty egress queue and this long without a received frame
`IPMC_FRU_PAUSE_WAIT_MS`	40	longest wait for an IPMB slave to go idle before an erase

Define	Default	Meaning
`IPMC_PAYLOAD_MODES`	3	payload operational modes this payload supports, bit n = mode n (modes 0..=14, bit 0 set, bit 15 clear): 3 is modes 0 and 1. The Payload Mode sensor SDR record (sensor 7, Tier-3) carries it as its reading and assertion masks, and the commercial release build answers it in Get Payload Mode Capabilities (Table 10.2.8-1)
`IPMC_SDR_*_NAME`	strings	SDR ID strings, 2..=16 bytes (locator at most 6); `IPMC_SDR_PAYLOAD_MODE_NAME` names sensor 7 (Tier-3 only)

Console, clock, LED, sensors and emulator

Define	Default	Meaning
`IPMC_CONSOLE_BAUD`	115200	console UART, 8N1 (`gen.ps1` reads the same line)
`IPMC_CONSOLE_BYTES_PER_POLL`	16	console bytes per main-loop pass; at most one command line per pass
`IPMC_UART_RX_BUF`	256	console receive ring, bytes; a power of two, one slot stays empty, and the usable bytes (size - 1) cover `IPMC_MENU_LINE_MAX`
`IPMC_UART_TX_BUF`	4096	console transmit ring, bytes, a power of two; 4096 bytes are 356 ms at 115200 baud
`IPMC_APP_UART_ENABLE`	1	the application port APP_UART (USART1) runs; 0 leaves USART1 off and parks its pins
`IPMC_APP_UART_BAUD`	115200	application port baud rate, 8N1
`IPMC_APP_UART_FLOW`	0	1 = RTS/CTS hardware handshake on PA12/PA11; 0 = TX and RX only, and PA11/PA12 stay unclaimed
`IPMC_APP_UART_RX_BUF` , `IPMC_APP_UART_TX_BUF`	256, 1024	application port rings, bytes, powers of two; the receive ring holds at least one line (`IPMC_APP_LINE_MAX` + 2)
`IPMC_APP_LINE_MAX`	128	longest application port request line, bytes (16..= `IPMC_APP_UART_RX_BUF` - 3)
`IPMC_APP_BYTES_PER_POLL`	32	most application port bytes one main-loop pass takes

Define	Default	Meaning
`IPMC_APP_FRU_WRITE`	1	the application port may `SET` the unit fields and run the IPMI Write FRU Data request; 0 refuses both with `ERR WRITE_DISABLED` (the table `WRITE_REFUSED` in `commands.rs` lists the refused NetFn and command pairs), while `GET` and the other `IPMI` requests (FRU Control, Set IPMB State, Set Event Receiver) still run, as from the Chassis Manager
`IPMC_MENU_LINE_MAX`, `IPMC_MENU_TOKENS_MAX`	128, 40	console line and token limits
`IPMC_FRU_DUMP_BYTES_PER_LINE`	16	bytes on one line of `FRU DUMP` (1..=16)
`IPMC_FW_NAME`, `IPMC_CONSOLE_PROMPT`	`ipmc-nucleo-f401re`, `ipmc`	firmware name of the banner, `SYS VERSION` and the application `HELLO` answer; root prompt text
`IPMC_HSE_HZ`, `IPMC_SYSCLK_HZ`	8000000, 84000000	clock input and system clock (PLL values checked at compile time)
`IPMC_LED_*`, `IPMC_FAULT_BLINK_CHZ`	1 Hz heartbeat, 10 Hz activity, 5 Hz fault	LD2 patterns (centihertz)
`IPMC_ADC_INTERNAL_SOURCE`	`IPMC_SRC_HW`	VDDA and die temperature from ADC1
`IPMC_ADC_EXTERNAL_SOURCE`	`IPMC_SRC_SIM`	rails A0..A3 simulated (unwired on the bench)
`IPMC_FAULT_SOURCE`, `IPMC_POWER_SOURCE`	`IPMC_SRC_HW`	fault inputs and PWR_EN on the pins
`IPMC_POWER_AT_BOOT`	1	request payload power at start-up
`IPMC_PGOOD_TIMEOUT_MS`	1000	longest wait for PGOOD after PWR_EN
`IPMC_SENSOR_PERIOD_MS`	1000	sensor sample period
`IPMC_ADC_AVERAGE`	8	conversions averaged per ADC read
`IPMC_SENSOR_TRACE`	`IPMC_TRACE_EVENTS`	sensor console lines at start-up
`IPMC_RAIL0..4_*`, `IPMC_RAIL_NONCRIT_PCT`, `IPMC_RAIL_CRIT_PCT`	5 rails, 5 %, 10 %	rail names, nominal mV, dividers, windows; rail 4 is `+3.3V_AUX` (3300 mV, divider 2:1, A3 on PB0)
`IPMC_TEMP_*_C`	LNC 0, LC -10, UNC 70, UC 85, UNR 100	temperature thresholds
`IPMC_SIM_*`	25 C, every input pin high	simulator start values: PGOOD asserted, FAULT_N, OVERTEMP_N and PAYLOAD_RESET_N deasserted

Define	Default	Meaning
`IPMC_EMU_PORT`, `IPMC_EMU_SERVE_POLL_MS`, `IPMC_EMU_TX_DONE_DELAY_MS`	5470, 1, 1	emulator TCP port, step period, transmit time
`IPMC_EMU_VECTOR_LINE_MAX`, `IPMC_EMU_REPORT_LINES_MAX`, `IPMC_EMU_ERROR_ECHO_MAX`	512, 8, 80	emulator line and report limits

The four request and reset defines in the IPMB table are read by the link layer, the reset gate and the requester in every build; a Tier-1 build compiles them and sends no request. The FAST-MODE and payload defines are read in every build as well; the timing in `ipmc-defs` `i2c_timing.rs` is used only after a Set IPMB State with a 400 kHz speed byte (Tier-3), and the reset pulse and payload mode values only by the Tier-3 handlers and the Payload Mode sensor record.

Tier-2/3 header

A build with the extension reads a second header, `ipmi_defs_tier23.h`, from the extension folder, with the same syntax and parser as `ipmi_defs.h` (section 4, Build modes and the Tier-2/3 extension). The public code reads two names from it:

Define	Meaning
`IPMC_TIER`	the Tier the IPMC reports, 1..=3 (Get VSO Capabilities, Get Device ID, the SDR records, `LinkConfig::requests_allowed`); 1 when the extension is not built. `ipmi_defs.h` itself never defines it
`IPMC_EVAL_MINUTES`	evaluation length in minutes, read by the evaluation image only (section 8.4); the firmware build checks 1..=7158

The header also carries the SEL size (`IPMC_SEL_ENTRIES`, the number of 16-byte SEL records the Tier-3 SEL holds in RAM, section 7.2), which only the commercial release build reads. The header's other defines belong to the extension. `IPMC_TIER` is 3 in the commercial release build. A name defined in both headers fails the build, and the public `ipmi_defs.h` names no Tier-2/3 feature.

Pin assignments are not in `ipmi_defs.h`: they live in the owning board-support module (`bsp/ipmb_pins.rs`, `bsp/ga.rs`, `bsp/board_pins.rs`, `bsp/uart.rs`) and in `Documentation/nucleo_f401re_pinout.md`. Python bench-tool settings live in `tools/chassis-manager/cm_settings.py`; only values the IPMC and a tool must agree on (such as `IPMC_EMU_PORT`) are read by the tools from `ipmi_defs.h`.

6. Module Descriptions

Each module file opens with a comment that states its purpose, its owner rules and the specification tables it implements; the paragraphs below summarise them and say how the modules connect.

6.1 `ipmc-defs`, `ipmc-wire`

`crates/ipmc-defs/` (`build.rs`, `defs_parser.rs`, `src/lib.rs`, `src/types.rs`, `src/i2c_timing.rs`)

`build.rs` reads `ipmi_defs.h` (or the file named by the `IPMI_DEFS_H` environment variable, to build against another header) and writes one constant per define. It is also the one detection point of the Tier-2/3 extension (section 4, Build modes and the Tier-2/3 extension): it reads `IPMC_TIER23` and `IPMC_EVAL`, checks the extension's marker files, merges the extension's header when present and publishes the decision. `tier23_detect.rs` holds the pure rules (`parse_mode`, `decide`, `merge`, `tier_of`, `parse_eval`, `check_eval`); `dep_cfg.rs` is the build-script step `ipmc-ext` and the firmware crate share. The parser accepts integers in decimal or `0x` hex, references to earlier defines, `[i32]`-tagged signed values and string literals, and rejects everything else. `lib.rs` re-exports the constants and holds the cross-checks; `types.rs` fixes types and boolean ranges.

`i2c_timing.rs` is the IPMB timing arithmetic, hardware-free and host-tested, shared by the bit-bang master, the hardware master and the bus-free waits. `cycles_for(ns, core_hz)` converts a time to core cycles, rounding up. `BitPhases::standard` and `BitPhases::fast` give the bit-bang SCL low and high phases in cycles (100 kHz: `IPMB_BITBANG_HALF_NS` each; 400 kHz: `IPMB_BITBANG_FAST_LOW_NS` and `IPMB_BITBANG_FAST_HIGH_NS`) with the slack above the minimum high time (4.0 us standard mode, 0.6 us fast mode, I2C-bus specification Table 5). `hw_ccr(pclk1, fast)` is the hardware master's CCR divisor (standard mode ceil(PCLK1 / (2 x 100000)); fast mode with F/S = 1 and DUTY = 0 ceil(PCLK1 / (3 x 400000)), RM0368 18.6.8) and `hw_trise(freq_mhz, fast)` its TRISE value (FREQ + 1 for 1000 ns; FREQ x 300 ns + 1 for 300 ns, RM0368 18.6.9). At PCLK1 = 42 MHz the divisors are 210 and 35 and TRISE is 43 and 13.

`crates/ipmc-wire/src/lib.rs`

The byte primitives shared by frames, FRU and SDR records and the tools: `zero_checksum` (the two's-complement checksum of IPMB frames and FRU areas, IPMI 13.8 and FRU spec sections 8, 11, 12), `Builder<N>` (a `const` byte-record builder: the FRU image is built at compile time; the SDR records are built by `SdrRepository::new` with the resolved address when `Context::new` runs), `Builder::push_type_length_ascii` (the IPMI type/length field, 8-bit ASCII), `crc32` (IEEE 802.3, used by the FRU flash record) and `hex_byte` (the one hex-token parser of the emulator and tools).

6.2 `ipmc-fru`: FRU records, unit fields, scratchpad and flash store

`crates/ipmc-fru/src/records.rs`

Builds the 256-byte FRU 0 image at compile time from the `IPMC_FRU_*` strings: the 8-byte Common Header (version 1, Internal Use 00h, Chassis Info 00h, Board Info at offset 01h = byte 8, Product Info at offset 0Eh = byte 112 (70h), MultiRecord 00h, checksum F0h), the Board Info Area (104 bytes: language, manufacturing date, manufacturer, product name, serial, part number, FRU file ID, C1h end marker, room for the unit fields, padding, checksum) and the Product Info Area (120 bytes: language, manufacturer, name, part or model number, version, serial, asset tag, an empty file ID, C1h, room, padding, checksum), 232 bytes in all (`IMAGE_LEN`), then 00h padding to `IPMC_FRU_SIZE`. The room is a run of 00h bytes after the C1h

marker that FRU Information Storage Definition section 10 allows up to the area length: each unit field reserves its `IPMC_FRU_*_MAX` bytes whatever the default string's length. A string the area cannot hold or an image larger than the area fails the build. The decoder (`common_header`, `area`, `Area::fields`) and `verify` check a possibly rewritten image: header version and checksum, area versions, lengths, zero checksums and end-of-fields markers.

`crates/ipmc-fru/src/unit.rs`

The per-unit field editor (section 7.6 gives the protocol that calls it). `UnitField` names the six fields: `BOARD_SERIAL`, `BOARD_PART`, `MFG_DATE`, `PRODUCT_PART`, `PRODUCT_SERIAL`, `ASSET_TAG`; `max_len` is the room `IPMC_FRU_*_MAX` reserves (16 characters for the date text). `FruStore::set_unit_field` decodes the area, writes its fields back in the same order with the one field replaced, then C1h, 00h up to the area's old length and the zero checksum, so the area length never changes and no other area moves. The rewritten image is built beside the old one, passes `verify`, and only then replaces it. The date is the text `YYYY-MM-DD HH:MM` (UTC); the image holds minutes since 1996-01-01 00:00 in 3 bytes, least significant byte first. A refusal leaves the image as it was: `FieldError` has `TooLong`, `BadLength` (a text of one byte is reserved by the type/length rule), `NotAscii`, `NoRoom` (an image a Chassis Manager wrote may have no spare room), `Nvmro` (the firmware checks the pin, the store cannot see it), `BadDate`, `AreaMissing` and `Corrupt`. `write_unit_field` prints a field as stored.

`crates/ipmc-fru/src/store.rs`

`FruStore` is the RAM scratchpad behind Read FRU Data and Write FRU Data: `read(offset, count)`, `contains`, `write(offset, data)` (sets a dirty mark), `take_dirty`, `bytes` and `image`. It holds the built image (`new`) or an image loaded from flash (`from_image`). No other crate holds FRU bytes. It also holds the Tier-3 non-volatile settings byte (bit 0 Default-Activation-Locked, bit 1 Commanded-Deactivation-Ignored, Rule 4.1.2-10.2): `nv_settings`, `load_nv_settings` (from the flash log, not dirty) and `set_nv_settings` (masked with `NV_SETTINGS_MASK`; a change sets the dirty mark, so the flash log saves it as it saves a Write FRU Data). The byte is 00h at power-up and stays 00h in a Tier-1 and Tier-2 build.

`crates/ipmc-fru/src/persist.rs` (tests in `persist/tests.rs`)

The FRU store is one flash sector (sector 1, 16 KB, `STORE_START` and `STORE_LEN`) behind the `FlashLog` trait (`sector_len`, `read`, `program`, `erase`, `millis`). `Persist::boot` scans the sector and loads the valid record with the highest sequence number; it erases nothing when it finds one, because the sector holds the only copy. A sector that is not blank and holds no valid record (damage only, nothing lost) is erased at boot unless NVMRO is asserted. `Persist::poll` is the save policy (dirty mark, quiet time, NVMRO, erase gate); `Persist::save` appends one record of the image and the settings byte, and when no slot is free it erases the sector and writes the record as the first one; `Persist::restore` writes the built image as a new record at once (console `FRU RESTORE`; the settings byte stays as it is). `store_region_ok` checks the linker's store region against the sector the log erases (section 9.1). Section 9.4 gives the record format and section 10.8 the save flow. The tests cut power at every word of a record and check the previous record still wins; the erase of a full sector is the one operation that can lose the image.

6.3 `ipmc-core`: IPMI logic

`crates/ipmc-core/src/ipmb.rs` - frame codec. `parse` checks length (7..=32 bytes), the header checksum, the address bytes (bit 0 clear), the response minimum and the data checksum, and returns a `Frame` (header, completion code for a response, data borrowed from the input). It rejects and never clamps; each failure is an `IpmbError` variant. `encode_request`, `encode_response` and `response_header_for` build frames with both checksums. A message is a response when its NetFn is odd.

`crates/ipmc-core/src/addr.rs` - geographic address. `resolve(raw_pins)` applies `IPMB_GA_SOURCE` and `IPMB_NVMRO_SOURCE` to the raw strap word (bit n = level of PC(n)): GA[4:0] are logic 1 when grounded, the hardware address is 40h + GA, the IPMB address twice that, parity must be odd over GA0..GA4 and GAP, and GA 0 is invalid. A forced GA gets a generated GAP. `AddrInfo::usable` is the one decision the link layer and the FRU initial checks read. `nvmro_asserted(raw)` is the one NVMRO rule, shared by start-up and run-time sampling.

`crates/ipmc-core/src/link.rs` - the dual-bus link layer (section 10.2). `Link` holds the bus states, per-bus counters, the ingress FIFO, a drop log, the duplicate cache, the egress queue and the bus of the latest transmission. Its main-loop API is `on_frame`, `poll`, `next_tx`, `on_tx_result` (section 11.1). A driver reports each transmission as a `TxOutcome`: `Sent`, or `Failed(TxFault)` with the cause the master saw (address NACK, data NACK, arbitration lost, bus busy, stretch timeout, line stuck, unspecified). A failure prints one line that names the destination, the sequence number and the cause, for example `IPMB-A response to 20h seq 01h not delivered, address 20h not acknowledged: no device answers at 20h; resend 1 of 3 on IPMB-B`, and a response given up on says `dropped after N resends`; a request the requester sends prints `sent`, `retry n of 10` or `not delivered` lines of the same form. `IPMC_IPMI_TRACE` 0 removes these lines at compile time. `plan_ipmb_state` is the one rule of Set IPMB State. `BusPair` power-up: IPMB-A Local Control, IPMB-B isolated, both isolated when the address is unusable. `Link` also owns the `Requester`: `on_frame` offers each valid response to it, `next_tx` hands out a response first and a due request only when the egress queue is empty, `on_tx_result` books a request transmission, and `poll` steps the requester timing, traces a request that ended and calls `Extension::poll` once. `LinkConfig::requests_allowed` is `IPMC_TIER >= 2`; a Tier-1 image does not reach the requester code. `with_first_request_seq`, `next_request_seq` and `recent_request_seqs` carry the request sequence number and the sequence numbers sent inside T9 across a processor reset. `Link::restarted(addr, now_ms)` is the one rebuild of a link for a Cold or Warm Reset: the same configuration and address, the power-up bus state, empty queues and caches, and the request sequence state of the old link carried over. Each `BusState` also holds a `BusSpeed` (`Standard` 100 kHz or `Fast` 400 kHz, power-up `Standard`, Rule 4.1.1-4.3); `plan_ipmb_speed` is the one rule of the speed byte (FAST-MODE on a bus that is not fitted is refused with `BusNotFitted`; STANDARD-MODE is always accepted) and `Link::set_ipmb_speed` applies it. `plan_ipmb_state` and `plan_ipmb_speed` share one loop, `plan_per_bus`. The driver owner reads `bus_state(bus).speed` and applies it between transfers (section 10.7).

`crates/ipmc-core/src/requester.rs` - the IPMB requester, the one request this IPMC sends at a time. A Tier-1 IPMC sends none (Rule 7-4); a Tier-2/3 IPMC is an Event Generator (Rule 7-3) and its extension calls `submit(&Request, now_ms)`, which builds the frame (rqSA own address, rqLUN 0, next 6-bit rqSeq) and returns the sequence number. The module holds the request state only; it never touches a bus. The `Link` hands the frame to a driver and reports the outcome and the responses it receives. `Phase` runs Idle, Due, Sending, Backoff, Waiting, Cancelling and Done.

Rule	Value	Where
response wait, T3	`IPMB_REQ_TIMEOUT_MS` (250 ms) from the end of a successful transmission	Rule 4.1.1-9
retries	`IPMB_REQ_RETRIES` (10): 11 transmissions, the same sequence number each time	Rule 4.1.1-10, Observation 4.1.1-6
failed transmission, T6	`T6_MIN_MS` (60 ms) after a failed transmission the retry goes out on the other bus, when it is enabled and a transmission is left; otherwise the wait for a response starts and the same bus is tried at the next timeout	Table 4.1.1-1, Rule 4.1.1-7
retry bus	the bus other than the one of the request's own previous transmission, the other enabled bus when that one is disabled	Rule 4.1.1-6

Rule	Value	Where
new requests, T4	`IPMB_REQ_GAP_MS` (5 ms) from the end of the previous request's first transmission	Table 4.1.1-1
sequence reuse, T9	`SEQ_EXPIRY_MS` (5 s) after a sequence number's latest transmission; a blocked next sequence number is skipped to the next free one, in order, wrapping	Table 4.1.1-1
priority	a queued response always goes out before a request	link layer

`submit` refuses with `NotAllowed` (Tier-1 build), `Busy` (a request or an untaken completion exists), `TooSoon` (T4), `SeqNotExpired` (all 64 sequence numbers were sent less than T9 ago) or `TooLong` (the frame does not fit or a header field is out of range), and a refused call changes nothing. A response matches when destination, source, NetFn + 1, command, sequence number and both LUNs equal the request's. The request ends with an `Answer` (completion code and data) or `NoResponse` after the last transmission; `take_completion` returns it once. `cancel` drops the request and any untaken completion, lets a frame already on the wire finish its one transmission and keeps the sequence number used for T9; the extension calls it when the request lost its purpose. `Requester::new(allowed, first_seq)` starts the sequence count at `first_seq` and `with_recent_seqs` books the sequence numbers the previous run sent inside T9, so a Cold Reset continues the count and no number is reused before its T9 (section 8.2).

`crates/ipmc-core/src/reset_gate.rs` - `ResetGate`, the one decision when a pending processor reset runs. `poll(pending, now_ms, egress_empty, fru_clean, trace)` returns true when the egress queue is empty (the response left or was dropped) and the FRU store holds no unsaved change, or when `IPMC_RESET_WAIT_MS` has passed since the request was first seen; the second case prints one trace line naming what was still open. With no reset pending it forgets the first-seen time.

`crates/ipmc-core/src/tier.rs` - the Tier rules, each a `const fn` of a tier argument so a test checks all three Tiers while the build runs at one: `event_generator` (Tier 2 and 3, Rule 7-3), `sel_device` (Tier 3, Rule 7-8), `per_state_event_enables` (Tier 3), `activation_control` (Tier 3: the IPMC controls payload power and uses M5, Rules 4.1.2-10.1 and 4.1.5.4.3-2), `payload_mode_sensor` (Tier 3: sensor 7, Rule 6.1-17) and `fast_mode` (Tier 3: 400 kHz, Rule 4.1.1-4.4). The constants `EVENT_GENERATOR`, `SEL_DEVICE`, `PER_STATE_EVENT_ENABLES`, `ACTIVATION_CONTROL`, `PAYLOAD_MODE_SENSOR` and `FAST_MODE` apply them to `IPMC_TIER`.

`crates/ipmc-core/src/event_enable.rs` - `EventEnables`, one `SensorEvents` (events on or off, assertion mask, deassertion mask) per sensor of the Tier, 0..=6 and sensor 7 at Tier-3 (`sdr::sensor_count`). The power-up values come from the SDR event masks (`sdr::event_masks`), so the records and the enables cannot disagree: all off with zero masks at Tier 1, all on with the record masks at Tier 2 and 3. Get Sensor Reading reads status bit 7 from it; the Tier-2/3 code writes it.

`crates/ipmc-core/src/ext.rs` - the `Extension` trait (the hooks `command`, `poll`, `on_sample`, `warm_reset` and `power_on_allowed`), `NoExtension`, `feed_sensors` and `check_power_on` (section 4, Build modes and the Tier-2/3 extension).

`crates/ipmc-core/src/dupcache.rs` - `DupCache` keeps the response frame of every answered request for `IPMB_DUP_WINDOW_MS` under a `Key` of requester address, sequence number, NetFn, command, requester LUN, responder LUN and the exact request data. When every entry is live the oldest is evicted and counted. Time is a wrapping millisecond counter.

`crates/ipmc-core/src/queue.rs` - `Fifo<T, N>`, a fixed-capacity FIFO that rejects and counts on full and never overwrites, and `EgressQueue`, the response frames with their arrival bus, arrival time, assigned bus and resend count. The ingress FIFO and

the drop log use `Fifo` too. `Bus` (A or B) with `other()` lives here.

`crates/ipmc-core/src/commands.rs` - the command table `COMMANDS`, one row per command: `cmd(netfn, cmd, name, spec, handler)`. Adding a command is adding one row (section 12.3); `find` is a linear scan over 13 rows. `tools/sync_ipmi_commands.py` keeps `Documentation/ipmi_commands.md` in step with this table, and a test fails while they disagree.

`crates/ipmc-core/src/dispatch.rs` - `Context` (section 11.2), the completion codes (`cc`), and the dispatcher. `run_request` answers C2h for a destination LUN other than 0 before any lookup; for NetFn 2Ch it echoes the request's defining-body byte as response data byte 0 on every path, answers C7h with no data for an empty request and C1h for a defining body other than 03h, and hands the handler the data after the VSO byte. `build_response_with` checks the output buffer before any handler runs; `dispatch_with` reserves an egress slot first, so a response that cannot be queued never causes a handler side effect. `run_command` looks the command up in the table; a (netFn, cmd) with no row goes to `Extension::command`, and only a miss there answers C1h. `Context` carries the Tier-2/3 hand-offs and state, each written by one owner (section 11.2): `pending_reset` (`Some(ResetRequest::Cold)` or `Some(ResetRequest::Warm)`, set by a handler and applied by the owner of the processor, section 8.2; `Context::warm_reset` clears the Context part of a Warm Reset), `pending_power` (a `PowerRequest`, On or Off, for the owner of the payload power), `pending_payload_reset` (a pulse request for the payload reset output), `deactivation_requested` (a Tier-3 standing request that `update_sensors` feeds to the FRU state machine, M4 to M5, only where `tier::activation_control`), `payload_mode` (the state of sensor 7, power-up mode 0 complete; `PayloadMode` in `sensor.rs`), `event_enables`, and the settings byte accessors `nv_settings` and `set_nv_settings` over `FruStore`. `trace_request` writes one line per request, `FN(06h) CMD(01h) Get Device ID -> CC 00h, 15 bytes`, when `IPMC_IPMI_TRACE` is 1 and returns at once when it is 0. `Context::read_sensor` is the one function that maps a sensor number to its encoder, used by Get Sensor Reading and the console.

`crates/ipmc-core/src/cmd/` - handlers, one file per NetFn: `app.rs` (Get Device ID, Get Self Test Results), `se.rs` (Get Device SDR Info, Reserve Device SDR Repository, Get Device SDR, Get Sensor Reading), `storage.rs` (Get FRU Inventory Area Info, Read FRU Data, Write FRU Data), `vita.rs` (Get VSO Capabilities, Set IPMB State, Get Device Locator Record ID, Get FRU Address Info). Get VSO Capabilities builds the IPMB capabilities byte from the Tier and the fitted buses (`ipmb_capabilities`: bits 5:4 the maximum frequency, 00b 100 kHz, 01b 400 kHz at Tier-3; bits 1:0 the interface count minus one). Set IPMB State parses the optional speed byte per Tier (`parse_speed`) and checks it with `plan_ipmb_speed`, as section 10.7 describes. `mod.rs` holds `require_no_data` and `check_fru_device_id`. A handler writes only its data bytes into the slice it is given and returns `Ok(len)` or `Err(completion code)`.

`crates/ipmc-core/src/identity.rs` - `DEVICE_IDENTITY`, assembled from `ipmc_defs` values; the firmware revision comes from the crate version (0.1), and a guard ties `IPMC_FRU_PRODUCT_VERSION` to the same major.minor. No literal identity value lives here. `additional_support(tier)` builds Get Device ID byte 7 from the `IPMC_SUPPORT_*` defines and the Tier rules (section 9.6).

`crates/ipmc-core/src/sdr.rs` - `SdrRepository`: eight records (nine at Tier-3) built once from the `IPMC_SDR_*_NAME` strings and the IPMB address, record IDs 0..=7 (0..=8 at Tier-3, section 9.5), the reservation counter (`reserve`, `reservation_valid`), `resolve` (0000h first, FFFFh last), `next_id`, `describe` and `locator_record_id` (7; 8 at Tier-3). `sensor_count(tier)` is 7, or 8 with the Payload Mode sensor (sensor 7) at Tier-3, and `record_count(tier)` one more; the tables are sized for the most (`SENSOR_COUNT_MAX`, `RECORD_COUNT_MAX`) and the records after the locator stay unused below Tier-3. At Tier-3 the FRU State record adds M5 and the IPMB Link record adds the 400 kHz states (section 9.6). The file header lists every place the records differ from the captured live IPMC and why. Sensor initialization, sensor capabilities, the event masks and the locator's global initialization and device capabilities follow the Tier (section 9.6).

`crates/ipmc-core/src/sensor.rs` - the sensor numbers and types of Table 10.2-1, the `SensorSnapshot` that `ipmc-sensors` fills, and one encoder per reading layout: `encode_summary` (FRU Health, FRU Voltage), `encode_temperature`, `encode_link`, `encode_fru_state`, `encode_payload_test_results` (unavailable: no payload test runs) and `encode_payload_test_status` (not in progress), and at Tier-3 `encode_payload_mode` (sensor 7, four bytes) over `PayloadMode` (current mode, previous mode, `TransitionStatus`; modes 0..=14). `encode_link` sets the 400 kHz bits 4 and 5 of the link state from `LinkBus::fast`. `apply_event_enable` is the one place that writes status bit 7. `LocalStatus` is the 3-bit IPMB local status of Table 10.2.2-1.

`crates/ipmc-core/src/fru_state.rs` - `FruMachine`, pure: `step(now_ms, inputs)` makes at most one transition and returns it; `check_power_on` refuses in M6. The states are M0, M1, M4, M5 and M6; M5 (FRU Deactivation Request) is reachable only at Tier-3 (`State::reachable_at`), through `Inputs::deactivation_requested`, which is false below Tier-3. `PowerOnRefused` names the two refusals (`DeactivationInProgress`, `ActivationLocked`). Section 10.4 has the diagram.

`crates/ipmc-core/src/selftest.rs` - `run(fru, sdr)`: `ipmc_fru::verify` on the image and a walk of the SDR chain (each record exists, carries the ID used to reach it, has a matching length byte, the walk ends at FFFFh after `record_count` records). Result 55h 00h, or 57h with bit 5 (FRU), bit 6 (SDR chain) or bit 3 (SDR empty).

`crates/ipmc-core/src/trace.rs` - the one `Trace` trait (`line(format_args!(..))`), implemented by the firmware console, the emulator's stdout sink, `NoTrace` and `RecordingTrace` (tests).

`crates/ipmc-core/src/ring.rs` - `ByteRing<N>`, the one byte FIFO of the project: N a power of two, N - 1 bytes usable (one slot stays empty), no allocator, every index masked and every slice access checked. The capture log stores its records in it and the firmware's UART driver keeps its receive and transmit queues in it.

`crates/ipmc-core/src/capture.rs` - the capture log (section 10.11). `CaptureLog<N>` is a `ByteRing` of whole records, `DefaultLog` the size of `IPMC_LOG_SIZE`; `frame` adds a whole IPMB frame with its bus, direction and `FrameStatus` (`Received`, `Injected`, `Sent` or `Failed(TxFault)`), `text` adds a trace line cut at `IPMC_LOG_TEXT_MAX`, and a record that does not fit drops the oldest whole records first. `Record` and `Kind` hand a held record out; the printers `CsvHeader`, `CsvRow` and `TraceRow` are `Display` types, the same ones the live I2C trace uses. `Names` and `CoreNames` give a command its name (the Tier-1 table, then the extension, then Platform Event, the one request the IPMC sends). `DumpCursor` walks the records held when a dump starts, a few rows per main-loop pass, and counts a record the ring overwrote meanwhile. `CaptureTrace` is the `Trace` that records every line and forwards it to the console when asked; `format_cut` and `Bounded` format a line into a stack buffer and cut it with `...`.

6.4 `ipmc-sensors`: sensors and board I/O

`crates/ipmc-sensors/src/board_sensors.rs` is the place to start: it lists every value the board measures and how each Tier-1 reading is built. FRU Voltage is good when every rail of the board table is inside its non-critical window; FRU Temperature carries the threshold flags of the die temperature; FRU Health is bad on FAULT_N, OVERTEMP_N, PWR_EN without PGOOD, a critical rail or a temperature at or above UC.

Layer	Files	Holds
Interfaces	`hal/analog.rs`, `hal/digital.rs`, `hal/power.rs`	`AnalogSource`, `DigitalInputs`, `PowerSwitch`, `ResetOutput` (the payload reset output), the shared `SourceError`
Evaluation	`eval/rail.rs`, `eval/temperature.rs`, `eval/faults.rs`, `eval/power_fsm.rs`, `eval/led.rs`	rail windows, calibration arithmetic (VREFIN_CAL, TS_CAL1, TS_CAL2), input assertion bitmap, PWR_EN/PGOOD state machine, LED level as a pure function of time

Layer	Files	Holds
Drivers	`drivers/sim/`, `drivers/stm32f401/` (feature), `drivers/i2c/` (INA226, PCA9555), `drivers/spi/` (MCP3208)	trait implementations; the external parts are host-tested examples with no bus assigned
Routing	`routing.rs`	per source switch, the on-chip driver or the simulator
Sampling	`monitor.rs`, `trace.rs`, `reset_pulse.rs`	one sample: inputs, power state machine, PWR_EN drive, snapshot, change lines; `Monitor::apply_power_request` carries out a handler's `PowerRequest`; `PayloadResetPulse` times one payload reset pulse
Pin ownership	`pin_claims.rs`, `drivers/stm32f401/ports.rs`	one owner per pin; a second claim fails naming both owners

`Monitor::apply_power_request(request, power_on_check, now_ms, io, trace)` is the one path from `Context::pending_power` to the PWR_EN drive, for the firmware main loop and the emulator. On asks the result of `ext::check_power_on` first (M6 or an activation lock): a refusal powers nothing and prints one line; Off is never refused.

`PayloadResetPulse` (`reset_pulse.rs`, hardware-free) times one pulse on the payload reset output (Rule 4.1.2-9). The owner calls `pulse_payload_reset(out, now_ms, ms)` when a handler set `Context::pending_payload_reset`, which asserts the output through `ResetOutput::set_reset`, and `service(out, now_ms)` on every main-loop pass, which releases it when the time is up (wrapping compare). Nothing waits, so no handler and no loop pass is held for the pulse; a second request restarts the time; a failed release is tried again on the next call. The STM32F401 driver (`drivers/stm32f401/gpio.rs`) claims PA9 with the other board I/O as an open-drain output, released (high impedance) before the pin becomes an output, so the payload sees no pulse at start-up.

`unsafe` is allowed only in `drivers/stm32f401/`; without the feature the crate builds on the host with the simulators and forbids `unsafe`.

6.5 `ipmc-menu`: console engine

`crates/ipmc-menu/src/` (`menu.rs`, `line.rs`, `args.rs`, `syntax.rs`; user documentation in `MenuDocumentation.md`). `LineEditor::push_byte` turns received bytes into echo, erase, line-complete and line-too-long events; `Menu::process_line` matches a line against the group tables (keywords case-insensitive and exact, `[REQUIRED]` and `{OPTIONAL}` parameters, optional last, and as the last token a `{NAME...}` that takes every remaining word, as in `DEBUG IPMI [NETFN] [CMD] {DATA...}`), prints help in fixed columns for `HELP` and `?`, and calls the handler with the caller's context and an `Args` view. A line holds at most `IPMC_MENU_LINE_MAX` (128) bytes and `IPMC_MENU_TOKENS_MAX` (40) tokens. `Menu::validate_tables` is a `const fn`; `console.rs` calls it in a compile-time assertion, so a bad syntax string in the firmware tables fails the build.

6.6 `ipmc-nucleo-f401re`: firmware

File	Role
`src/main.rs`	start-up order (section 8.3) and the main loop: `console.poll`, `console.service_app`, `console.service_sensors`, `console.service_link`, `wfi` (the evaluation image also checks the evaluation period)
`src/ipmb_link.rs`	owns the static `Link`, the dispatch `Context`, the `ipmc_ext::Active` extension, the `ResetGate`, the frame-queue consumers, the master, the fault trackers and the FRU log; `service` is the IPMB main-loop step (section 10.2) and ends with `service_reset` and `apply_bus_speeds`; `service_fru_log` saves at once while a Cold or Warm Reset is pending; `warm_reset` rebuilds the link (section 8.2); `power_on_check` and `take_hand_offs` give the sensing owner the payload power and payload reset requests of the handlers with the power-on check; `feed_sensors` calls `ext::feed_sensors`; owns the capture log: `LoggedTrace` (the trace sink that records every line and prints it unless the IPMI console trace is off), `record_frame` (every frame taken from a driver queue before the parser, and every frame handed to the master with its outcome), `status` (a line that prints whatever the console switch says), `local_request` and `inject` (the paths of `DEBUG IPMI` and `DEBUG INJECT`), `set_unit_field`; `IPMB STATS` and `IPMB STATE` commands; `stop_for_eval` (evaluation image only)
`src/fru_log.rs`	the boot load of the FRU log before the I2C drivers start; owns `Flash` and `Persist`
`src/sensing.rs`	builds the sensor backends from the source switches, owns the `Monitor` and the `PayloadResetPulse`; `serve_hand_offs` (power requests through `apply_power_request`, the payload reset pulse and its release, every main-loop pass); `start` requests payload power at boot when the power-on check allows; `SENSORS`, `POWER`, `SIM` commands
`src/console.rs`	command tables (eight groups, section 7.3) and the `SYS`, `LED`, `IPMB` handlers on the `ipmc-menu` engine; banner; `SYS STATUS` with the failure counters of both UARTs; `poll_dump`, which continues a running `DEBUG DUMP` one main-loop pass at a time; `service_app`
`src/debug_cmds.rs`	the `DEBUG` group (section 10.11): `STATUS`, `LOG`, `DUMP` (the `DumpJob` that prints rows while the console transmit ring has room for the longest one), `CLEAR`, `TRACE`, `EVENTS`, `IPMI`, `INJECT`, `CACHE`; `events_off_at_boot`; `local`, the function behind `DEBUG IPMI` and the application port `IPMI` command
`src/app_port.rs`	the transport of the application port (section 7.6): `AppPort::poll` takes bytes from the APP_UART receive ring, feeds the `ipmc-app` line assembler and writes the answer; `LinkHost`, the `AppHost` implementation over the link layer; the count of dropped answers
`src/data_cmds.rs`	`FRU SHOW`, `FRU DUMP`, `FRU STORE`, `FRU RESTORE`, `SENSORS SDR`
`src/bsp/clock.rs`	HSE bypass (HSI fallback) to 84 MHz, PLL M 8 / N 336 / P 4 / Q 7, compile-time check of the PLL against `IPMC_SYSCLK_HZ`
`src/bsp/i2c_hw.rs`	I2C1 and I2C3: slave interrupts (EV and ER), the per-bus frame queues, `suspend` / `resume` for the bit-bang master, `set_speed` (CCR, TRISE and F/S for 100 or 400 kHz while the slave is idle), `HwMaster`, `clear_stuck_busy`, driver counters

File	Role
`src/bsp/i2c_bitbang.rs`	`BitbangMaster`: START, bits, ACK, STOP on open-drain GPIO with DWT timing at the phases of the bus speed (`BitPhases`), clock-stretch read-back, arbitration check, bus recovery (9 clocks and a STOP, always at 100 kHz)
`src/bsp/ipmb_master.rs`	the `IpmbMaster` trait (with `speed` and `set_speed`), `MasterError`, counters, `SelectedMaster` chosen at compile time by `IPMB_MASTER_MODE`
`src/bsp/ipmb_common.rs`	DWT cycle counter, `Deadline`, `wait_for`, `wait_bus_free` (tBUF, slave idle, stuck-BUSY signature), `write_retrying` (arbitration retries inside one T2 + T1 budget)
`src/bsp/ipmb_pins.rs`	claims PB8, PB9, PA8, PB4 and parks them as floating inputs (PB4 resets with a pull-up that would load IPMB-B SDA)
`src/bsp/flash.rs`	word program, sector erase and reads of the store (sector 1) behind `FlashLog`; at start-up compares the linker symbols with the sector (`store_region_ok`) and refuses to start the store unless the region is exactly sector 1, the vector table ends before it and the program starts after it; refuses erase and program while NVMRO is asserted
`memory.x`, `link.x`	the flash regions VECTORS, STORE and FLASH and the section placement (section 9.1); `link.x` is cortex-m-rt's script with the three added asserts
`src/bsp/ga.rs`	strap pins PC0..PC6 with pull-ups, raw level read, live NVMRO sample; PC10 BOOTLOADER (input, pull-down, owner `boot_pin`, printed in the banner and `SYS STATUS`)
`src/bsp/systick.rs`	1 ms tick, `millis`, `seconds`, `catch_up` (adds the time the core stalled on a flash erase, measured with the DWT counter)
`src/bsp/uart.rs`	the one UART driver of both ports (`Port::Console` on USART2, PA2/PA3; `Port::App` on USART1, PB6/PB7 and optional PA11/PA12), interrupt driven in both directions; described below the table
`src/bsp/led.rs`	LD2 on PA5: heartbeat, 10 Hz activity window after each IPMB frame, manual modes, fault blink
`src/bsp/board_pins.rs`	ADC channels and GPIO pins of the sensor signals, the payload reset output (PA9) and the PAYLOAD_RESET_N input (PA6)
`src/bsp/reset_cause.rs`	reset-cause decode; `system_reset` (AIRCR SYSRESETREQ, so the next banner reads "software reset"); `keep_sequence` and `kept_sequence`: four `.uninit` words, untouched by start-up code, that carry the next request sequence number and the mask of the sequence numbers sent inside T9 across a software reset, with a check word, read back only after a software reset
`src/bsp/fault.rs`, `static_slot.rs`	fail-stop handlers that record and park; the one `Mutex<RefCell<Option<T>>>` access pattern
`src/bsp/eval_timer.rs`, `src/eval.rs`	evaluation image only (`cfg(ipmc_eval)`): the TIM2 run-time counter and the evaluation period check (section 8.4)

The UART driver (`bsp/uart.rs`) runs one code path for both ports; a `Port` value selects the peripheral, the pins, the ring sizes and the baud rate. Each port has a receive and a transmit `ByteRing` . One interrupt body (`service`) moves a received byte into the receive ring and sends the next byte of the transmit ring; the main loop reads with `getc` and writes with `write` , which copies as much as the ring takes and never waits. `print` (the `out!` macro) writes a whole console message and waits only while the ring is full; `print_line_or_drop` writes a trace line and its CR LF whole when the ring has room for all of it, and otherwise drops it and counts it (`tx_lines_dropped`), so a trace line never stalls the IPMB; `print_row_or_drop` does the same for the longer per-frame I2C trace row. `write` copies at most 64 bytes per critical section and loops. `flush` waits, bounded, until the last byte has left; it runs before a reset and before the fault park. Every wait polls the same `service` body, so it also ends where the interrupt cannot run (a HardFault). Receive overflow, overrun, transmit timeout and dropped lines are counted per port and printed by `SYS STATUS` . The interrupts run at NVIC priority 2 << 4, below the I2C slaves (1 << 4). The interrupt body runs inside the critical section of the port state, so an I2C interrupt that arrives meanwhile waits for it, a few microseconds. With `IPMC_APP_UART_FLOW` 1 the driver enables RTS/CTS in the peripheral (CTSE and RTSE, RM0368 19.3.14) and leaves a byte in the data register when the receive ring is full, so RTS stays deasserted until `getc` makes room. RTS deasserts only when the data register holds an unread byte, so a sender with a byte in flight can still overrun by one byte; the overrun counter counts it and the application port answers the line `ERR LINE_DAMAGED` ; PA11 reads "not clear to send" when it is open.

6.7 `ipmc-emu` : x86 emulator

File	Role
`src/emulator.rs`	`Emulator` : `addr::resolve` , `Link::new` , `Context::new` and a `Monitor` over the simulators, built as the firmware builds them; `rx` , `poll` , `next_tx` , `tx_done` mirror the driver calls
`src/vectors.rs`	the one reader of the vector format (`vectors/sbc_ab.vec` : time, bus tag, hex bytes)
`src/replay.rs`	feeds each captured request at its capture time, reports each response 1 ms later, compares with the captured response: Match, SameCC, CCDiff, NotImpl, Unans, NoCap, NoBus
`src/serve.rs`	`Board` , the emulator in real time behind the TCP line protocol (section 7.4); one client at a time; a new client or `#reset` starts a fresh board
`src/trace.rs` , `src/main.rs`	stdout trace sink and hex formatter; command line `replay` , `serve` , `sdr` , `help`

The emulator holds one `ipmc_ext::Active` extension (`Emulator::with_extension` replaces it for a test) and calls `ext::feed_sensors` for each sample. A pending Cold or Warm Reset runs through the same `ResetGate` as the firmware. Both rebuild the link with `restart_link` (`Link::restarted` plus IPMB-B enabled when configured), so the request sequence counter continues from the old link. A Cold Reset also rebuilds the `Context` , extension, sensor board and payload power as at a boot and restores the FRU store from the flash copy; a Warm Reset calls `Context::warm_reset` and `Extension::warm_reset` and keeps everything else, payload power included. The flash copy (`FlashCopy` : the FRU image and the settings byte) is a RAM image the emulator updates where the firmware saves (at once while a reset is pending), so an accepted Write FRU Data or a settings change survives the reset. `Emulator::flash` returns it (completing a pending save first, as a clean power cycle does) and `Emulator::new_with_flash` starts a board from it, so a power cycle of the emulator keeps the flash log; the boot power-on asks the power-on check with the stored settings, so a stored default activation lock keeps the payload off. `follow_payload_supply` makes PGOOD follow PWR_EN while `IPMC_SIM_PGOOD_HIGH` is 1 (with 0 PGOOD stays low, so the PGOOD timeout path stays reachable), so a deactivation ends in the complete shutdown that takes the FRU from M6 to M1 (the simulated inputs alone hold PGOOD at its configured level for ever); `service_hand_offs` applies

`pending_power` through `apply_power_request` and models the payload reset pulse by one trace line `payload reset pulse <ms> ms`. `replay` also plays the Event Receiver role of the captured Chassis Manager: a request the IPMC sends to 20h is answered one step later, 00h for a Platform Event and C1h for anything else, and counted in the summary line (`ipmc_requests`, `events`). Without the extension no request is sent and the counts are 0. The emulator otherwise keeps the FRU store in RAM only (no flash log driver) and simulates every sensor group.

6.8 `ipmc-app`: application port protocol

`crates/ipmc-app/src/` (`lib.rs`, `commands.rs`, `host.rs`, `line.rs`, `tests.rs`; the module comment of `lib.rs` is the protocol description and the recipe for a new command)

The crate is `no_std` and `forbid(unsafe_code)`; it never touches hardware. A request line travels `UART byte -> LineAssembler::push -> respond -> answer -> COMMANDS row -> handler -> AppHost`:

File	Holds
`line.rs`	`LineAssembler<N>` collects bytes into lines at CR or LF (an empty line is not a request; a CR LF pair yields one line) and reports a line longer than `N` or holding a byte other than printable ASCII once, at its end, and drops the bytes; `AnswerBuf<N>`, the fixed buffer an answer is formatted into
`commands.rs`	`COMMANDS`, one `AppCommand` row (keyword, argument summary, one sentence of help, handler) per command; `answer` (strips the optional tag, matches the keyword ignoring case, calls the handler); `respond` (answers what `push` returned, including the `LINE_TOO_LONG` and `ARGS` answers for a lost line); the handlers, each of which writes nothing until it knows whether the answer is `OK` or `ERR`
`host.rs`	the `AppHost` trait, the one door to the board: `version`, `status`, `fru_state`, `unit_field`, `set_unit_field`, `ipmi`, `save_state`; and the plain types it passes (`Status`, `SaveState`, `AppError`, `IpmiData`)
`tests.rs`	`REQUEST_TABLE`, one row per request line with the answer line the test host must produce; the test host (`FakeHost`) is an `AppHost` over a plain `FruStore`

The firmware implements `AppHost` as `LinkHost` in `app_port.rs`, over the IPMB link layer: the dispatch `Context` (addresses, FRU image, FRU state, latest sensor snapshot), the flash log status, the live NVMRO pin and the SysTick time. A handler returns without waiting for hardware and never runs in an interrupt. Section 7.6 gives the lines the port accepts and answers.

6.9 Tools

Path	Role
`tools/trace2vec.py`	merges the two logic-analyser I2C exports (IPMB-A and IPMB-B, one time base) into `vectors/sbc_ab.vec` with bus tags, redacted
`tools/sync_ipmi_commands.py`	rewrites the implemented column of `Documentation/ipmi_commands.md` from `COMMANDS`; `--check` for the gate
`tools/captured_sdr.py`	decodes the 40 captured SDRs next to ours into `Documentation/captured_sdr.md` (local only)

Path	Role
`tools/dupcheck.py`	scores every pair of Rust function bodies with identifiers normalised; fails on a production pair at or above 0.85 or a test pair at or above 0.92
`tools/capture/`	`capture2pcap.py` turns a saved `DEBUG DUMP` into a pcap file (LINKTYPE 260) and a CSV file (section 10.11); `test_capture2pcap.py`
`tools/app-port/`	`app_cli.py`, the bench client of the application port (section 7.6): tagged requests from the command line or an interactive prompt on COM5, exit status 0 when every answer is `OK`; `app_settings.py` (port, answer timeout, tag form); `test_app_cli.py` with a fake transport
`tools/python-ipmi/`	venv with python-ipmi and pyaardvark; `README.md` is the Tier-1 command-line reference
`tools/chassis-manager/`	VITA 46.11 Chassis Manager on python-ipmi: `cm_link.py` (requester rules: T3 250 ms, retries with the same sequence, alternation, per-target IPMB-B), `cm_commands.py`, `chassis_manager.py`, `crosscheck.py` (compares a run with the captured Chassis Manager, every difference tagged), `ipmc_emu_interface.py` (TCP transport), `cm_settings.py` (tool settings), `cm_events.py` (the Chassis Manager as Event Receiver: answers each Platform Event request and records its decoded fields), `test_cm_link.py`
`tools/chassis-manager/` options for events	`--cm-tier 1|2` (2: Event Receiver, answers a Platform Event with 00h; 1: not an Event Receiver, answers C1h), `--withhold-events N` (the first N Platform Event frames get no answer, retries count, so a run shows the IPMC retrying with the same sequence number), `--set-event-receiver` (sends Set Event Receiver to each IPMC found after discovery; a Tier-1 IPMC answers C1h, which is not an error), `--cold-reset-at-s T` (sends Cold Reset once to each IPMC found, T seconds into the polling)
`tools/conformance/`	spec-only Tier-1 suite: `TRANSPORT.md` (its only interface), `adapter.py` (maps it onto the Aardvark or the emulator), `run_conformance.py`, eight `tests_*.py` files, `REQUIREMENTS.md`
`tools/board_test_report.py`	runs the command line, Chassis Manager, conformance, link-health and FRU persistence checks against the board and writes `Documentation/board_test_report.md`

7. Interfaces

7.1 IPMB frame format

All addresses are 8-bit (the 7-bit I2C address shifted left, bit 0 clear); byte 0 is the destination as it appears in the I2C address phase. The I2C slave interrupt prepends the matched address byte (own address, or 00h for a general call) to the bytes it receives, so the link layer always sees a whole frame.

Byte	Request	Response
0	rsSA	rqSA
1	NetFn << 2 | rsLUN	(NetFn + 1) << 2 | rqLUN
2	chk1: bytes 0..2 sum to 0 mod 256	chk1
3	rqSA	rsSA
4	rqSeq << 2 | rqLUN	rqSeq << 2 | rsLUN
5	command	command
6..	request data	completion code, then response data
last	chk2: bytes 3..n-1 sum to 0 mod 256	chk2

A request is 7 to 32 bytes, a response 8 to 32 (`IPMB_MSG_MAX`). For NetFn 2Ch/2Dh the first data byte is the defining body, 03h for VSO; "byte 3" of a VITA response table is response data byte 0 after the VSO byte.

7.2 Command set

NetFn	Cmd	Command	Request data	Response data (after CC)	Notes
06h	01h	Get Device ID	none	15 bytes from `DEVICE_IDENTITY`	also the broadcast form
06h	04h	Get Self Test Results	none	55h 00h, or 57h and failure bits	
04h	20h	Get Device SDR Info	none or operation	count, flags 01h	07h sensors, or 08h records with bit 0 set; 08h sensors and 09h records at Tier-3

NetFn	Cmd	Command	Request data	Response data (after CC)	Notes
04h	21h	Get Device SDR	reservation, record ID, offset, count	next ID, bytes	C5h for a non-zero offset without the current reservation; CBh no record; C9h range; CAh too large
04h	22h	Reserve Device SDR Repository	none	reservation ID	each call returns a new non-zero ID
04h	2Dh	Get Sensor Reading	sensor number	reading, status, state	CBh for a number above 6 (above 7 at Tier-3, section 9.3)
0Ah	10h	Get FRU Inventory Area Info	FRU ID	00h 01h 00h	256 bytes, byte access
0Ah	11h	Read FRU Data	FRU ID, offset, count	count, bytes	C9h count 0 or range; CAh too large for one frame
0Ah	12h	Write FRU Data	FRU ID, offset, bytes	count written	80h while NVMRO is asserted; reruns the self test; saved to flash after the quiet time
2Ch	00h	Get VSO Capabilities	VSO	03h, 00h, IPMB capabilities, 00h, 22h, 00h, 00h	IPMB capabilities 00h (one interface) or 01h (two), 100 kHz; 10h or 11h at Tier-3, 400 kHz (section 9.6)
2Ch	09h	Set IPMB State	VSO, A state, B state, optional speed	03h	CCh for an unsupported link ID, speed, unfitted bus or both buses isolated (section 10.7)
2Ch	0Dh	Get Device Locator Record ID	VSO, FRU ID	03h 07h 00h	record 0008h (03h 08h 00h) at Tier-3
2Ch	40h	Get FRU Address Info	VSO, then none, FRU ID, or FRU ID, key type, site, site type	03h, HW address, IPMB address, FFh, 00h, site, 00h	CCh key type other than 03h; D3h another site

Common completion codes: C1h unknown command (and every Tier-1 not-applicable command), C2h a destination LUN other than 0, C7h a wrong request length, C9h a FRU ID other than 0 (except the physical-address form of Get FRU Address Info, which ignores the ID). Reservation IDs, record IDs and FRU offsets are two bytes, LS byte first; a Get Device SDR count of FFh reads the rest of the record; a Set IPMB State bus byte is 00h (isolate), 01h (Local Control) or FFh (no change). The

optional speed byte holds the IPMB-A speed in bits 1:0 and the IPMB-B speed in bits 3:2: 00b is 100 kHz and 01b is 400 kHz at Tier-3 on a fitted bus; every other value, a non-zero bits 7:4 and 400 kHz below Tier-3 answer CCh with no change (section 10.7).

Tier-2/3 commands (commercial release build)

The commercial release build adds the commands below; the public Tier-1 build answers each of them C1h. "From" is the Tier that makes the command mandatory for an IPMC (Tables 10.1.1-1 and 10.1.2-1); the codes match `Documentation/ipmi_commands.md`. Tier-3 is built and is the default of the commercial release build (`IPMC_TIER` 3). Deviation: a Tier-3 build runs with IPMB-A only on this bench (the owner's recorded deviation from Rule 4.1.1-1.1, and from Rule 4.1.1-4.4 on IPMB-B), and Get VSO Capabilities then reports one interface (Rule 4.1.1-3). A Tier-3 IPMC needs both buses, so the dual-bus Tier-3 build waits for the second Aardvark (section 13.5).

NetFn	Cmd	Command	From	Notes
06h	02h	Cold Reset	Tier-2	answers 00h, then the processor restarts (section 8.2)
04h	00h	Set Event Receiver	Tier-2	sets the Event Receiver address, the Chassis Manager; FFh disables event generation
04h	01h	Get Event Receiver	Tier-2	returns the address set
04h	02h	Platform Event	Tier-2	sent by this IPMC to the Event Receiver (Rule 7-3); a Platform Event request received here answers C1h, the IPMC is not an Event Receiver
2Ch	1Eh	FRU Control Capabilities	Tier-2	answers 00h at Tier-2 and 11h at Tier-3 (bit 4 Set FRU Activation, bit 0 cold reset)
06h	03h	Warm Reset	Tier-3	answers 00h; no processor reset, the link state restarts and payload power stays (section 8.2)
06h	2Fh	Get BMC Global Enables	Tier-3	answers 08h (System Event Logging on, no System Interface bits)
04h	28h	Set Sensor Event Enable	Tier-3	per-state event enables of sensors 0 to 7; the scanning bit is accepted and ignored (scanning is always on); enable bits the SDR record does not support stay clear; a sensor with no record answers CBh
04h	29h	Get Sensor Event Enable	Tier-3	returns the enables Set Sensor Event Enable writes
0Ah	40h	Get SEL Info	Tier-3	the SEL holds `IPMC_SEL_ENTRIES` (64) records of 16 bytes in RAM; the operation-support byte states that it does not survive a power cycle or a Cold Reset
0Ah	42h	Reserve SEL	Tier-3	

NetFn	Cmd	Command	From	Notes
0Ah	43h	Get SEL Entry	Tier-3	
0Ah	47h	Clear SEL	Tier-3	
0Ah	48h	Get SEL Time	Tier-3	time counts seconds from start-up plus the Set SEL Time offset
0Ah	49h	Set SEL Time	Tier-3	
2Ch	04h	FRU Control	Tier-3	Cold Reset option 00h only (resettable payload): a pulse on PAYLOAD_RST_OUT_N (PA9), FRU state unchanged, in M4 or M5 (section 8.2); the other options answer CCh
2Ch	0Ah	Set FRU State Policy Bits	Tier-3	Default-Activation-Locked and Commanded-Deactivation-Ignored; kept in the flash log (section 9.4)
2Ch	0Bh	Get FRU State Policy Bits	Tier-3	returns the policy bits
2Ch	0Ch	Set FRU Activation	Tier-3	activates or deactivates the FRU (payload power and M5); Activate in M0 answers D5h, Deactivate in M1 answers 00h and does nothing
2Ch	44h	Get Mandatory Sensor Numbers	Tier-3	
2Ch	45h	Get FRU Hash	Tier-3	CRC-32 over the FRU image, recomputed at start-up and after each accepted Write FRU Data
2Ch	46h	Get Payload Mode Capabilities	Tier-3	answers `IPMC_PAYLOAD_MODES`: modes 0 (normal) and 1
2Ch	47h	Set Payload Mode	Tier-3	modes 0 and 1, held by the IPMC (no payload signal); the change completes at once (sensor 7 reads transition complete)

Externally visible behaviour of the commercial release build:

- The Tier is reported in Get VSO Capabilities (identifier bits 1:0 = `IPMC_TIER` - 1), Get Device ID byte 7 (09h, 29h, 2Dh) and the SDR records; section 9.6 lists every byte that changes.
- A Tier-2/3 IPMC is an Event Generator. It sends Platform Event requests (NetFn 04h, command 02h) to the Event Receiver set by Set Event Receiver, for a state change of FRU State, System IPMB Link, FRU Health, FRU Voltage and FRU Temperature (sensors 0 to 4), and at Tier-3 for the 400 kHz states of System IPMB Link (offsets 4 and 5) and for Payload Mode (sensor 7). The Payload Test sensors never change on this module and send nothing. Set Event Receiver FFh stops event generation; the sensors keep their state and the active events (at Tier-3 the 400 kHz Link states and the current payload mode as well) are sent again at the next Set Event Receiver (Rule 4.1.6-1). A C1h answer from the receiver stops events until the next Set Event Receiver.

- Every event goes out through the requester (section 6.3): a response wait of 250 ms, up to 10 retries with the same sequence number (11 transmissions), a failed transmission retried on the other bus after at least 60 ms, 5 ms between new requests, and a sequence number held out of use for 5 s. An event that gets no answer after the last transmission is dropped, except a FRU State event, which is sent again at a period set in the extension's header until the receiver answers 00h or the state changes (Rule 4.1.5.2-4).
- The FRU State event carries cause 3h (programmatic action) for a change made by a Chassis Manager command or by the policy-bit automation (Rule 4.1.5.2-5), 0h (normal) for a change that follows the initial checks and payload power, and 6h for a power-failure change (Rules 4.1.5.2-6 and 4.1.5.2-7).
- At Tier-3 the FRU State event cause is 1h for a change commanded by Set FRU Activation, 3h for a change made by the policy-bit automation (Rule 4.1.5.2-5), 6h for a power failure (it wins over both) and 0h otherwise.
- At Tier-3 payload power follows the FRU activation policy. The Default-Activation-Locked policy bit, kept in the flash log, makes the FRU Activation-Locked at M0 to M1 (Rule 4.1.5.4.1-11); the default is clear, so the FRU activates by itself as at Tier-1 and Tier-2. While the FRU is Activation-Locked every payload power-on is refused (the boot power-on, a Set FRU Activation hand-off, the console `POWER ON`), as it is in M6 (section 10.4).
- At Tier-3 every event its sensor's enables allow is added once to the SEL, whether or not an Event Receiver is set; an event the enables turn off is neither sent nor logged, so a Chassis Manager that turns event messages off with Set Sensor Event Enable also leaves the SEL empty. The events sent again to a new Event Receiver are not logged again.
- Set FRU Activation answers before the owner applies the payload power request. If the power controller then refuses it (a latched fault, a failed PWR_EN write), the console prints one line and the FRU stays in M1 with no M4 event; the Chassis Manager sees the FRU State sensor unchanged.
- Event messages are enabled for sensors 0 to 6 at power-up (the SDR event masks, section 9.6), and for sensor 7 at Tier-3; Get Sensor Reading reports that in status bit 7.

7.3 Console

COM3, 115200 8N1 (`IPMC_CONSOLE_BAUD`), prompt `ipmc`. `HELP` or `?` lists the groups; `<GROUP> HELP` lists a group; a command is typed in full from the top level (`SENSORS SHOW`) or after entering its group, and `UP` or `..` leaves the group. A line holds at most 128 bytes and 40 tokens. A syntax token in `[ ]` is required, in `{ }` optional, and a last `{NAME...}` takes every remaining word; `crates/ipmc-menu/MenuDocumentation.md` has the command reference with the help text of every command.

Group	Commands
top level	`STATUS` (= `SYS STATUS`), `RESET` (= `SYS RESET`)
`SYS`	`STATUS`, `RESET`, `VERSION`
`LED`	`ON`, `OFF`, `BLINK [HZ]`, `AUTO`, `STATUS`
`IPMB`	`ADDR`, `STATS`, `STATE`
`SENSORS`	`SHOW`, `SDR`, `TRACE {LEVEL}`
`POWER`	`ON`, `OFF`, `CLEAR`, `STATUS`
`FRU`	`SHOW`, `DUMP {OFFSET} {COUNT}`, `STORE`, `RESTORE`
`SIM`	`RAIL {N} {MV}`, `TEMP {DC}`, `FAULT {NAME} {STATE}`, `SHOW`

Group	Commands
`DEBUG`	`STATUS`, `LOG {STATE}`, `DUMP {FORMAT}`, `CLEAR`, `TRACE {WHAT} {STATE}`, `EVENTS {STATE}`, `IPMI [NETFN] [CMD] {DATA...}`, `INJECT [BUS] {BYTES...}`, `CACHE {ACTION}`

The DEBUG group switches and reads the capture log, runs a request locally and injects a frame (section 10.11). `DEBUG DUMP` is a job: it prints a header line, then rows on the following main-loop passes while the console transmit ring has room for the longest row, then `# end of capture`; any received byte except a line feed stops it with `# dump stopped`, and a new command line waits until it ends.

The banner after reset prints the firmware and core versions, build profile, reset cause, clocks, geographic address with its source, IPMB address, the application port line (`App port: USART1 115200 8N1, RTS/CTS off`, or `App port:  off`), the `BOOTLOADER pin:` line and the self-test result (`Self test:  55h 00h (passed)`). With `IPMC_EVENTS_AT_BOOT` 0 in a build that sends events, the line `Events: off from start-up` follows. `SYS STATUS` adds the failure counters of both UARTs (receive overflow, transmit timeout, overrun, trace lines dropped, free transmit bytes) and the count of application port answers dropped.

7.4 Emulator TCP line protocol (`ipmc-emu serve`)

ASCII, one message per line, at most `IPMC_EMU_VECTOR_LINE_MAX` characters, on 127.0.0.1 port `IPMC_EMU_PORT` (5470).

Direction	Line	Meaning
client to emulator	`A <hex bytes>` / `B <hex bytes>`	a frame written on that bus; first byte the 8-bit destination (00h for a broadcast)
client to emulator	`#bus A down` / `#bus A up` (and B)	break or repair a bus: frames are lost and the IPMC's transmissions on it fail
client to emulator	`#reset`	power-cycle the board (fresh board, both buses up, clock restarts; the flash log keeps its content, so a saved FRU write and the policy bits survive)
client to emulator	`#quit`	end the session
emulator to client	`A <hex>` / `B <hex>`	a frame the IPMC transmits
emulator to client	`#ready <hex>`	at connect and after `#reset`: the IPMC's own address
emulator to client	`# <text>`	a trace line (with `--trace`)
emulator to client	`#error <line>: <reason>`	a rejected client line; the session continues

7.5 Pins

Signal	Pin	Header	Owner	Direction
IPMB-A SCL / SDA	PB8 / PB9 (I2C1, AF4)	D15 / D14	`ipmb_pins.rs`, `i2c_hw.rs`	open-drain, no pull-up
IPMB-B SCL / SDA	PA8 (I2C3, AF4) / PB4 (I2C3, AF9)	D7 / D5	`ipmb_pins.rs`, `i2c_hw.rs`	open-drain, no pull-up; parked when not fitted
GA0..GA4, GAP, NVMRO	PC0..PC4, PC5, PC6	morpho	`ga.rs`	input, pull-up
Console TX / RX	PA2 / PA3 (USART2, AF7)	ST-LINK VCP	`uart.rs`	
APP_UART TX / RX	PB6 / PB7 (USART1, AF7)	CN5-3 (D10) / CN7-21	`uart.rs` (claim "app_uart")	TX output; RX input, pull-up
APP_UART CTS / RTS	PA11 / PA12 (USART1, AF7)	CN10-14 / CN10-12	`uart.rs`	claimed only with `IPMC_APP_UART_FLOW` 1; CTS input with pull-up (an open CTS reads "not clear to send"), RTS output
BOOTLOADER	PC10	CN7-1	`ga.rs` (claim "boot_pin")	input, pull-down; logic 1 at reset keeps the bootloader (Phase 12); read for the banner and `SYS STATUS`
LD2	PA5	D13	`led.rs`	output, high = on
Rails 1..4	PA0, PA1, PA4, PB0 (ADC1 ch 0, 1, 4, 8)	A0..A3	`adc.rs` (claim "adc")	analog
PWR_EN	PB5	D4	`gpio.rs` (claim "board_io")	output, active high
PAYLOAD_RST_OUT_N	PA9	D8	`gpio.rs` (claim "board_io")	output, open drain, active low, released (high impedance) at start-up; held low `IPMC_PAYLOAD_RESET_PULSE_MS` by a Cold Reset of the payload (commercial release build)
PGOOD	PB10	D6	`gpio.rs`	input, active high, pull-up
FAULT_N	PA10	D2	`gpio.rs`	input, active low, pull-up
OVERTEMP_N	PC7	D9	`gpio.rs`	input, active low, pull-up
PAYLOAD_RESET_N	PA6	D12	`gpio.rs`	input, active low, pull-up

`Documentation/nucleo_f401re_pinout.md` is the full pin table for people; a pin change edits the owning module and that table together.

7.6 Application port protocol (`ipmc-app`)

The application port connects the payload CPU on the same board to the IPMC over APP_UART (USART1, PB6 TX, PB7 RX, `IPMC_APP_UART_BAUD` 115200, 8N1; RTS/CTS on PA12/PA11 with `IPMC_APP_UART_FLOW` 1). The protocol is plain ASCII text:

- One request per line, ended by CR LF or LF, at most `IPMC_APP_LINE_MAX` (128) bytes. A byte other than printable ASCII (20h..7Eh) or a tab makes the line a `BadChar` line.
- Every request line gets exactly one answer line, ended by CR LF, that starts with `OK` or `ERR <CODE>`. A line that cannot be a request still gets its answer: `ERR LINE_TOO_LONG max=128` for a long line, `ERR ARGS only printable ASCII is accepted` for a bad byte.
- Commands are not case sensitive. A request may start with a tag, `#` and 1 to 8 letters or digits; the answer starts with the same tag and a space, so a payload CPU with several requests in flight can match answers to requests. A bad tag is answered `ERR ARGS` without a tag.
- The IPMC answers at most one request line per main-loop pass, so a payload CPU that sends many requests cannot hold the IPMB service off. An answer that does not fit the transmit ring is dropped whole and counted (`SYS STATUS`, `DEBUG STATUS`); the payload CPU asks again.
- Every request line and every answer line goes into the capture log as `app request <line>` and `app answer <line>` (section 10.11).

Command	Arguments	Answer	Help text
`HELP`	`[keyword]`	`OK HELP HELP HELLO STATUS FRU GET SET FIELDS IPMI SAVE`; with a keyword, `OK HELP <keyword> <args> - <help>`	list the commands, or describe one
`HELLO`	none	`OK IPMC-APP <protocol version> <firmware name> v<version> tier=<n>`	firmware name, version, Tier and protocol version
`STATUS`	none	`OK STATUS ipmb=<addr>h ga=<n> fru=M<n> power=<state> health=<state> nvmro=<0 or 1> uptime_s=<n>`	IPMB address, GA, FRU state, power, health, NVMRO, uptime
`FRU`	`STATE`	`OK FRU M<n> <state name>`	the Operational State of FRU 0 (M0, M1, M4, M5, M6)
`GET`	`<field>`	`OK <FIELD> <value>`; an empty field prints nothing after the name	read a per-unit FRU field as stored
`SET`	`<field> [value]`	`OK <FIELD> <value>` with the value as stored now; no value clears a text field	write a per-unit FRU field; no value clears it
`FIELDS`	none	`OK FIELDS <FIELD>=<max> ...`	the per-unit FRU fields and the longest value of each

Command	Arguments	Answer	Help text
`IPMI`	`<netfn> <cmd> [data...]`	`OK IPMI CC=<cc> DATA=<hex bytes>`	run an IPMI request locally; hex bytes; nothing goes on the IPMB
`SAVE`	none	`OK SAVED`	OK once the FRU image is in the flash log

`HELP` lists the table `COMMANDS`, so the table is the one description of what the port accepts. `HELLO` reports protocol version 1 (`PROTOCOL_VERSION`); a change that breaks an existing request or answer raises it, a new command does not. In `STATUS`, power is `on`, `off` or `fault`, health is `ok` or `fault`, and `power=off` also covers a power-on that still waits for PGOOD; `health=fault` covers a bad reading and a missing one (before the first sample). `IPMI` takes every number as a hex byte (`06`, `0x06` and `6` are the same) and runs the request through the dispatcher as if the Chassis Manager had sent it, the same function as `DEBUG IPMI`; a completion code other than 00h is still an `OK` answer, because the request ran.

The unit fields (`FIELDS`; `max` is the room `IPMC_FRU_*_MAX` reserves, 24 by default):

Field	FRU area	Longest value
`BOARD_SERIAL`	Board Info, serial number	24
`BOARD_PART`	Board Info, part number	24
`MFG_DATE`	Board Info, manufacturing date and time, written as `YYYY-MM-DD HH:MM` (UTC, from 1996-01-01)	16
`PRODUCT_PART`	Product Info, part or model number	24
`PRODUCT_SERIAL`	Product Info, serial number	24
`ASSET_TAG`	Product Info, asset tag	24

`SET` writes the RAM image through `IpmbLink::set_unit_field`, the editor of section 6.2, then reruns the self test; the dirty mark and the quiet time of `IPMC_FRU_PERSIST_DELAY_MS` carry the change to the flash, as for a Write FRU Data from the Chassis Manager. The text of a field is the rest of the line after the one space that follows the field name, so spaces inside the value stay. Error answers:

Answer	Cause
`ERR UNKNOWN <word>`	no command of that name
`ERR ARGS <text>`	missing or extra arguments, an unknown field, a bad tag, a bad hex byte, an empty request, a bad byte in the line
`ERR LINE_TOO_LONG max=<n>`	the line is longer than `IPMC_APP_LINE_MAX`
`ERR TOO_LONG max=<n>`	the value is longer than the field holds

Answer	Cause
`ERR BAD_LENGTH`, `ERR NOT_ASCII`	a text of one byte (a type/length field holds 0 or at least 2 bytes); a value that is not printable ASCII
`ERR NO_ROOM`	the rewritten fields do not fit the area (an image a Chassis Manager wrote may have no spare room)
`ERR NVMRO`	the NVMRO pin is asserted; nothing is written
`ERR BAD_DATE`	the date text is malformed or out of range
`ERR FRU_IMAGE <name>`	the area is missing or damaged, or the rewritten image would fail `verify`
`ERR WRITE_DISABLED`	`IPMC_APP_FRU_WRITE` is 0 and the request is `SET` or an `IPMI` Write FRU Data; nothing runs
`ERR LINE_DAMAGED <n> bytes lost on the port`	the UART lost `<n>` bytes (overrun or receive ring full) since the previous line ended; the line does not run, and the answer carries the tag when the line had a valid one
`ERR IPMI_FAILED <why>`	the request did not run: `TOO_LONG` (more data than one frame holds) or `DISPATCH` (a NetFn that is odd or above 3Eh)
`ERR PENDING a save is waiting`	`SAVE` while a change waits for its flash save
`ERR LOG_OFF`	`SAVE` with the flash log off (`IPMC_FRU_PERSIST` 0 or the store did not start)
`ERR ANSWER_TOO_LONG`	the answer did not fit the answer buffer (`ANSWER_MAX` 256 bytes)

A refused `SET` leaves the FRU image as it was. A payload CPU that wrote unit data repeats `SAVE` until it reads `OK SAVED` before it removes power.

A worked exchange (`>` is the request, `<` the answer; public Tier-1 build, GA 1):

```
> HELLO
< OK IPMC-APP 1 ipmc-nucleo-f401re v0.1.0 tier=1
> STATUS
< OK STATUS ipmb=82h ga=1 fru=M4 power=on health=ok nvmro=0 uptime_s=342
> #a7 SET BOARD_SERIAL 00:11:22:33:44:55
< #a7 OK BOARD_SERIAL 00:11:22:33:44:55
> SET MFG_DATE 2026-10-25 14:30
< OK MFG_DATE 2026-10-25 14:30
> SET BOARD_SERIAL 012345678901234567890123 4
< ERR TOO_LONG max=24
> SAVE
< ERR PENDING a save is waiting
> SAVE
< OK SAVED
> IPMI 06 01
< OK IPMI CC=00 DATA=01 80 00 01 02 09 00 00 00 4B 49 00 00 00 00
```

The second `SAVE` comes after the quiet time. The bench client `tools/app-port/app_cli.py` tags each request (`#t0001 STATUS`) and skips an answer with another tag.

8. Clocking, Reset and Start-Up

8.1 Clocks

Clock	Source	Frequency	Used by	Notes
HSE	ST-LINK MCO into OSC_IN, bypass	8 MHz	PLL input	HSI 16 MHz (M = 16) when HSE never becomes ready
SYSCLK, HCLK	PLL: 8 MHz / 8 x 336 / 4	84 MHz	core, DWT cycle counter, SysTick	`IPMC_SYSCLK_HZ`, compile-time check
PCLK1	HCLK / 2	42 MHz	I2C1, I2C3 (CR2.FREQ 42), USART2	APB1 maximum
PCLK2	HCLK	84 MHz	ADC1 (ADCCLK = PCLK2 / 4 = 21 MHz), USART1 (the application port)	
PLLQ	VCO / 7	48 MHz	unused	
SysTick	HCLK	1 kHz tick	`millis`, LED pattern	`catch_up` after a flash erase
DWT CYCCNT	HCLK	84 MHz	bit-bang timing, deadlines, erase stall time	started before the FRU log boot

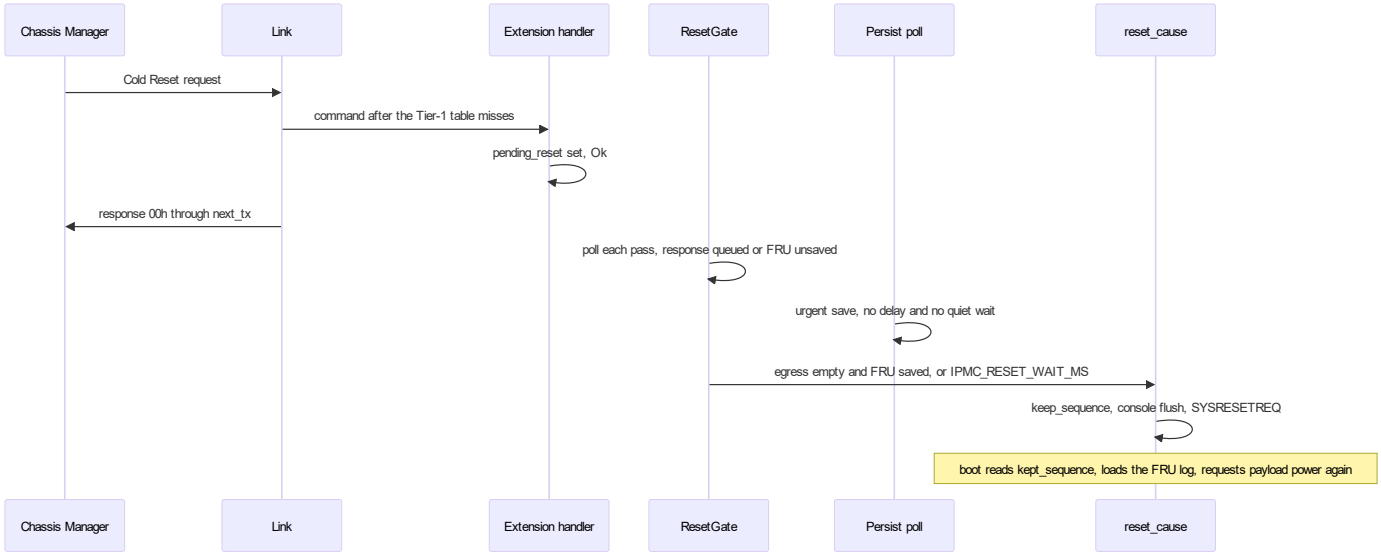
8.2 Reset

The firmware has one reset domain: the MCU reset (power-on, pin, software from `SYS RESET` or from a Cold Reset, watchdog flags reported but no watchdog is enabled). A Warm Reset and the Cold Reset of the payload (commercial release build) are not processor resets; both follow the Cold Reset path below. `bsp/reset_cause.rs` reads RCC_CSR once at start-up, reports the most specific cause in the banner and clears the flags. A HardFault or an unhandled interrupt records its state in a static, prints one line when the UART is up and parks in `wfi`; it does not reset, so the fault stays visible on the bench. The panic handler is `panic-halt`. RAM state starts from its Rust initialisers; the state kept across a reset is the FRU log in flash and, after a software reset, the request sequence words described below.

Cold Reset path (commercial release build)

A handler never resets the processor. The Cold Reset handler of the extension validates the request and sets `Context::pending_reset` to `Some(ResetRequest::Cold)`; the response (00h) is queued as any response. The owner of the processor, the firmware main loop or the emulator, performs the reset.

1. Each pass `IpmbLink::service_reset` calls `ResetGate::poll(pending, now_ms, egress_empty, fru_clean, trace)`. The gate returns true when the egress queue is empty (the response left the board or was dropped) and the FRU store holds no unsaved change, or when `IPMC_RESET_WAIT_MS` has passed since it first saw the request; in the second case one trace line says what was still open.
2. While a reset is pending the FRU save runs at once (`service_fru_log`): no persist delay and no wait for a quiet bus, because the IPMB is about to go away. The erase it may need is bounded (at most 500 ms, covered by `IPMC_RESET_WAIT_MS`). A failed attempt returns to the normal delay, so a failing flash does not stall every pass until the wait runs out.
3. When the gate says go, one console line goes out and the console is flushed. `reset_cause::keep_sequence` stores the next request sequence number and the mask of the sequence numbers sent inside T9 (`Link::recent_request_seqs`) in four `.uninit` words, which start-up code does not touch, the last a check word over the other three. `reset_cause::system_reset` then requests SYSRESETREQ, so the banner after the restart reads "software reset".
4. At the next start-up `reset_cause::kept_sequence` returns the stored values after a software reset, and zero after any other reset or when the check word does not match. `IpmbLink::take` hands them to `Link::with_first_request_seq`, which books the recent numbers as sent at start-up, so a request after the restart cannot repeat a sequence number the Event Receiver saw inside T9.
5. Everything else restarts as at power-up: link, context, sensors and the extension's state (IPMB-B is isolated again until Set IPMB State), and the FRU image loads from the flash log.



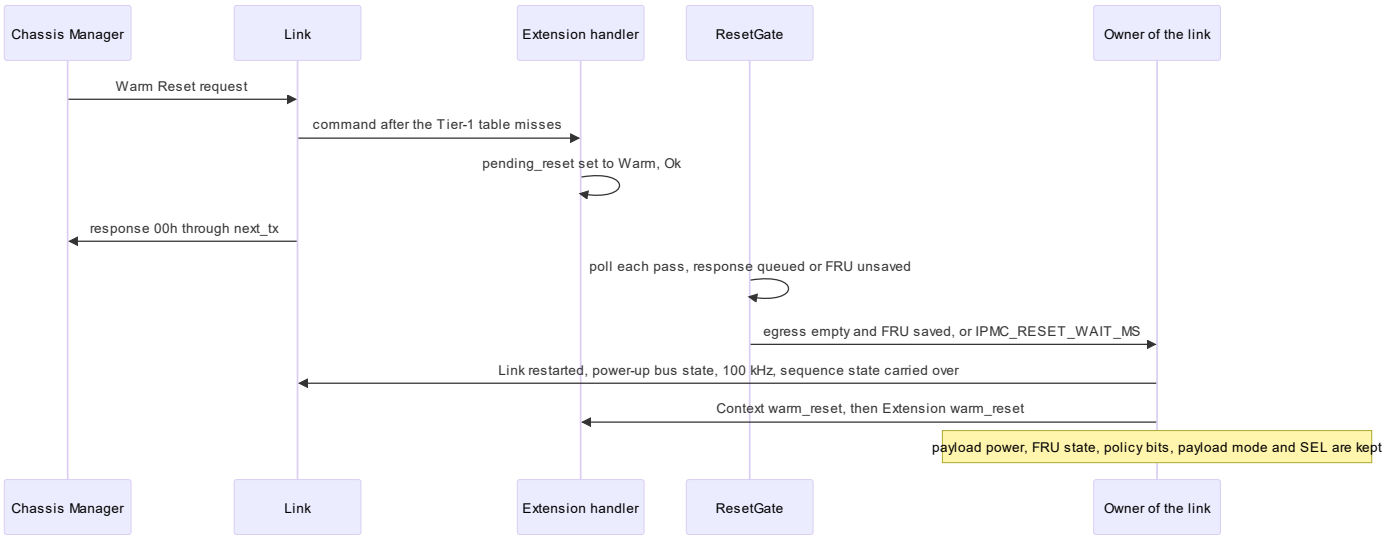
Limitation (owner decision 21, 10-24-2026): on the NUCLEO-F401RE a Cold Reset is a processor reset, which releases PWR_EN (PB5) with every GPIO, so the payload loses power for the reset and boot time. Recommendation 4.1.9-3, that a reset should not change the payload state, is not met on this board, which has no latch to hold the pin; a target board holds PWR_EN externally. The emulator models the board.

Warm Reset (commercial release build)

A Warm Reset (06h/03h, Tier-3) restarts the link state without a processor reset, so PWR_EN keeps its level and the payload keeps its power (the problem of decision 21 does not arise). The handler sets `Context::pending_reset` to `Some(ResetRequest::Warm)` and queues the 00h response; a pending Cold Reset wins over a Warm Reset request. The same `ResetGate` decides when it runs (the response has left and an accepted FRU write is saved, or `IPMC_RESET_WAIT_MS`), and the FRU save runs at once while it is pending. The owner (`IpmbLink::warm_reset`, or the emulator's `service_reset`) then:

1. builds a new Link with `Link::restarted`: the same configuration and address, the power-up bus state (IPMB-A Local Control, IPMB-B isolated until the next Set IPMB State, both buses back at 100 kHz, Rule 4.1.1-4), empty queues and duplicate cache, and the request sequence state carried over so no sequence number is reused inside T9 (the requester is rebuilt with it);
2. keeps a start-up fault of a bus, so a bus that failed at start-up stays disabled;
3. calls `Context::warm_reset`, which installs the new link view and clears every `pending_*` hand-off, the Warm Reset request included;
4. calls `Extension::warm_reset`, which forgets the event in flight;
5. prints `Warm Reset: link state re-initialised`; `apply_bus_speeds` returns the drivers to 100 kHz on the next pass.

Kept: payload power and PWR_EN, the FRU state, the FRU store and its settings byte (the policy bits), the SDR repository, the self test result, the sensors, the payload mode, the SEL, the Event Receiver address and the event enables. Not kept: the IPMB state (IPMB-B returns to isolated and both buses to 100 kHz, Rules 4.1.9-2 and 4.1.1-4.3, shown as Link events) and the duplicate cache, which starts empty after a Warm or a Cold Reset, so a Chassis Manager retry of a request answered just before the reset runs its handler again.



Cold Reset of the payload (commercial release build)

FRU Control option 00h (Cold Reset, Tier-3) resets the payload, not the IPMC (Rule 4.1.2-9). It is accepted in M4 or M5 and leaves the FRU state unchanged. The handler sets `Context::pending_payload_reset`; each main-loop pass `Sensing::serve_hand_offs` takes it and `PayloadResetPulse::pulse_payload_reset` drives PAYLOAD_RST_OUT_N (PA9, open drain, active low) low for `IPMC_PAYLOAD_RESET_PULSE_MS`, and the same call releases it when the time is up. The pulse is not

blocking: IPMB traffic continues during it. A build whose power group is simulated has no output and prints that; FRU Control still answers 00h. The emulator models the pulse as one trace line.

8.3 Start-up order (`main.rs`)

1. Reset cause (first, before anything can reset again); after a software reset the kept request sequence numbers (`kept_sequence`).
2. Clock to 84 MHz; a clock fault is reported later in the banner.
3. `ports::init` takes GPIOA..GPIOC; LED claim; SysTick start; USART2 console (a UART failure blinks LD2 at 5 Hz and parks; a pin-claim conflict prints both owners and parks); then, when `IPMC_APP_UART_ENABLE` is 1, APP_UART on USART1 (clocked from PCLK2). A failure of the application port other than a pin-claim conflict does not stop the board: the console prints `APP_UART not started: <reason>` after the banner. The evaluation image starts its TIM2 run-time counter after the UARTs (section 8.4).
4. IPMB pins parked as floating inputs.
5. Sensor hardware for the groups whose source is hardware (ADC1, board GPIO).
6. Strap pins (`ga::init`).
7. DWT cycle counter, then the FRU log boot (`fru_log::FruLog::boot`): checks the store region against the linker symbols, loads the newest valid record of sector 1, and erases the sector only when it holds damage and no valid record and NVMRO is deasserted. It runs before the I2C slaves exist, because an erase stalls the core (typical 250 ms, at most 500 ms).
8. `addr::resolve` on the raw straps; `ipmb_link::start_drivers`: the selected master takes the pins and the DWT counter, then, when the address is usable, each fitted bus gets `bring_up` (bus recovery when SDA is held low, then the I2C slave). A bus that fails stays disabled with its local status set, and the fault tracker restarts it once the lines read idle-high.
9. `IpmbLink::take` builds the static Link (first request sequence number from step 1) and dispatch Context (self test runs; a stored image replaces the built one and the self test runs on it).
10. Banner (the application port line and the `BOOTLOADER pin:` line included), the `APP_UART not started` line when it applies, the events setting (`IPMC_EVENTS_AT_BOOT` 0 makes the IPMC send itself Set Event Receiver FFh and print `Events: off from start-up`; a build without event messages answers C1h and prints nothing), sensor start (payload power requested when `IPMC_POWER_AT_BOOT` is 1 and the power-on check, `IpmbLink::power_on_check`, allows it; a refusal prints `Power at boot refused: ...`), prompt, main loop. Each pass runs `console.poll` (one console line, or a step of a running `DEBUG DUMP`), `console.service_app` (at most one application request line), `console.service_sensors`, `console.service_link` and `wfi`.

8.4 Evaluation build (commercial release build)

The evaluation image is a time-limited build of the commercial release build (owner decision 20).

- `gen.ps1 build -Eval` or `gen.ps1 flash -Eval` sets `IPMC_EVAL=1` and writes `ipmc-nucleo-f401re-eval.bin`; without `-Eval` the script passes `IPMC_EVAL=0`, so a build made after an evaluation build carries none of it. `-Eval` needs the extension folder.
- `ipmc-defs` accepts `IPMC_EVAL=1` only with the extension built in and the release profile, and fails the build otherwise. It sets `cfg(ipmc_eval)`, which the firmware crate takes through `dep_cfg.rs`. Every evaluation line sits behind that cfg (`eval.rs`, `bsp/eval_timer.rs`, the timer start and the period check in `main.rs`, `IpmbLink::stop_for_eval`), so no other image, a Tier-1 or normal release or debug build, contains any evaluation code or a flag that could stop the IPMB.

- `bsp/eval_timer.rs` runs TIM2 as a free-running 32-bit up-counter at 10 kHz (0.1 ms per tick, 7158 minutes of range), clocked from PCLK1 (twice PCLK1 when the APB1 prescaler is not 1, RM0368 section 6.2). It starts after the console and before the banner; a start failure stops the image, because it could not end the evaluation. SysTick is not used. The banner of the evaluation image carries the line `EVALUATION BUILD: the IPMB stops after N minutes`.
- The period is `IPMC_EVAL_MINUTES` of run time (section 5, Tier-2/3 header). A power cycle starts a new period.
- At expiry the main loop calls `stop_for_eval` once and prints one console line. It switches both I2C slaves off through the driver's suspend (a healthy bus waits up to the pause time for idle, a faulted one is switched off at once), drops what is queued and never calls `IpmbLink::service` again, so the whole IPMB, responses and requests, stops. The console and the sensor sampling continue, and the FRU save step still runs: a Write FRU Data accepted before the stop is saved, and the save never touches the stopped slaves.

9. Memory Map and Data Layouts

9.1 Flash and RAM (`memory.x`, `link.x`, `bsp/flash.rs`)

The STM32F401RE has 512 KB of flash in eight sectors (RM0368 Table 5). `memory.x` names three regions and `link.x` places the sections:

Sector	Base	Size	Region	Contents
0	`0x0800_0000`	16 KB	`VECTORS`	the vector table, nothing else
1	`0x0800_4000`	16 KB	`STORE`	the FRU flash store, nothing else
2, 3	`0x0800_8000`	2 x 16 KB	`FLASH`	program: `.text`, `.rodata`, the `.data` load image
4	`0x0801_0000`	64 KB	`FLASH`	program
5, 6, 7	`0x0802_0000`	3 x 128 KB	`FLASH`	program

Region	Base	Size	Contents
`RAM`	`0x2000_0000`	96 KB	statics and stack
System memory	`0x1FFF_7A2A` .. `0x1FFF_7A2F`	6 bytes	factory VREFIN_CAL, TS_CAL1, TS_CAL2 (read by `adc.rs`)

`link.x` is the script of cortex-m-rt 0.7.7 with these differences: the vector table sits in `VECTORS`, every other flash output section sits in `FLASH` (480 KB, sectors 2 to 7), `STORE` receives no output section, and three asserts fail the build when the vector table does not fit `VECTORS`, when the three regions overlap or are out of order, or when a section lands in `STORE`. The store is the one sector the FRU log erases and programs, so an erase never touches the vector table or the program. `memory.x` exports `__fru_store_start` and `__fru_store_end`, and `link.x` exports `__vectors_end` and `__program_start`; `Flash::new` hands them to `ipmc_fru::persist::store_region_ok`, which refuses to start the store unless the region is exactly sector 1 (`STORE_START`, `STORE_LEN`), the vector table ends before it and the program starts after it. A layout that drifts apart can therefore never erase code. `arm-none-eabi-size` in `gen.ps1 build` prints the section sizes of the image.

`gen.ps1` makes the `.bin` with `objcopy --gap-fill 0xFF`: the image spans the vector sector, the store sector and the program, and the gap over the store reads FFh (erased), not 00h. The two ways to program the board treat the store differently:

Way	What it writes	The FRU store
Drive copy to `NOD_F401RE` (`gen.ps1 flash -Drive`, or when OpenOCD is absent)	the whole `.bin`, sector 0 through the program	erased (FFh), so the stored FRU image and the unit fields go; the next start-up loads the built image

Way	What it writes	The FRU store
SWD program of the ELF (`gen.ps1 flash -Swd`, or when OpenOCD finds the probe)	the loadable sections only (`program <ELF> verify reset exit`)	sector 1 is not touched and keeps its records

A reflash may erase the whole flash, unit fields included, and the firmware has no guard against it (owner decision 10-25-2026).

The statics of the firmware are the `IpmbLink` (the duplicate cache of 64 entries of 76 bytes, the capture log of `IPMC_LOG_SIZE` bytes and the frame queues) and the four UART rings (`IPMC_UART_RX_BUF`, `IPMC_UART_TX_BUF`, `IPMC_APP_UART_RX_BUF`, `IPMC_APP_UART_TX_BUF`), which live in `.bss` and not in the flash image.

9.2 FRU 0 image (256 bytes)

Offset	Size	Content
00h	8	Common Header: `01 00 00 01 0E 00 00 F0` (version 1; Internal Use, Chassis Info, MultiRecord 00h; Board at 1 x 8, Product at 0Eh x 8, checksum)
08h	104	Board Info Area (length byte 0Dh): version, length, language, manufacturing date (3 bytes LE minutes), manufacturer, product name, serial, part number, FRU file ID, C1h, room for the serial and part number (00h), padding, checksum
70h	120	Product Info Area (length byte 0Fh): version, length, language, manufacturer, name, part/model, version, serial, asset tag, FRU file ID (empty), C1h, room for the part number, serial and asset tag (00h), padding, checksum
E8h	24	00h padding to 100h (`IMAGE_LEN` is 232)

The area sizes are those of the default strings and `IPMC_FRU_*_MAX` values; an area is always a multiple of 8 bytes, and a longer string or a larger `_MAX` grows it (the build fails when the image no longer fits `IPMC_FRU_SIZE`). The room after C1h is the `_MAX` value of each unit field less the length of its default string, so an update of a field up to its `_MAX` rewrites the field in place and the area keeps its length (section 6.2).

9.3 Get Sensor Reading data (bytes after the completion code)

Sensor	Type	Byte 2	Byte 3	Byte 4
0 FRU State	F0h	00h	40h	Current State Mask, bit n for Mn (M0 01h, M1 02h, M4 10h, M6 40h; M5 20h at Tier-3)
1 IPMB Link	F1h	B local control bit 7, B status 6:4, A local control bit 3, A status 2:0	40h	link state bit 3 both, 2 only B, 1 only A, 0 none; at Tier-3 bit 4 IPMB-A and bit 5 IPMB-B run at 400 kHz

Sensor	Type	Byte 2	Byte 3	Byte 4
2 FRU Health	F2h	00h	40h	01h functioning, 02h not functioning
3 FRU Voltage	02h	00h	40h	01h in range, 02h out of range
4 FRU Temperature	F3h	00h	40h	LNC bit 0, LC 1, UNC 3, UC 4, UNR 5
5 Payload Test Results	F4h	00h	60h (unavailable)	00h
6 Payload Test Status	F5h	00h	40h	01h not in progress
7 Payload Mode (Tier-3)	F7h	NVMRO bit 6, transition status bits 5:4 (00b in progress, 01b complete, 10b failed), previous mode bits 3:0; gDiscrete1 bit 7 reads 0	40h	current-mode mask, LS byte (bit n is mode n); a fifth byte holds the MS byte (modes 8..14 in bits 6:0), bit 7 reads 1

Status byte (response byte 3) 40h is "scanning enabled"; bit 5 (20h) marks a reading unavailable; bit 7 is the sensor's event enable (`Context::event_enables`, `sensor::apply_event_enable`): 0 in the Tier-1 build, so the table shows 40h, and 1 from power-up in a Tier-2/3 build (the byte reads C0h) until the enable changes. Sensors 2..4 read `00 60 00` before the first sample and while their source cannot be read. On the bench with only IPMB-A enabled sensor 1 reads `08 40 02`. Sensor 7 exists only at Tier-3 (Get Sensor Reading answers CBh for it below Tier-3 and for any number above 7); it is always available, and in mode 0 with the transition complete it reads `10 C0 01 80` (status C0h with the event enable). Set Payload Mode, in the commercial release build, is its only writer.

9.4 FRU flash record (`persist.rs`)

Offset	Size	Field
0	4	commit word 55464931h, programmed last
4	4	sequence number, from 1, never 0 or FFFFFFFFh
8	2	image length (= `IPMC_FRU_SIZE`)
10	1	record format version (1)
11	1	Tier-3 non-volatile settings: bit 0 Default-Activation-Locked, bit 1 Commanded-Deactivation-Ignored (Rule 4.1.2-10.2), other bits 0
12	256	FRU image
268	4	CRC-32 over bytes 4..268

A record is 272 bytes (68 words), so the 16 KB store sector holds 60 records (16,320 bytes; 64 bytes at the end stay unused). All fields are little-endian; the length, version and settings fields share one word, every other field is whole words. Erased flash reads FFh. The save that finds no free slot erases the sector and writes the record as the first one (section 10.8).

The settings byte is byte 11, the high byte of what the 16-bit version field of an earlier format held (0001h). A record written before the byte existed therefore reads as settings 00h, and a record with settings 00h is byte-identical to one of that format: a Tier-1 and a Tier-2 build never write a non-zero value. A stored byte is masked with 03h when loaded and a saved byte before it is written. A build from before the byte rejects a record whose settings are not 00h and loads the newest older record or the built image without a report, so a Tier-3 image that saved a policy bit is not downgraded in the field without a FRU check. The byte rides in the record with the image, so one dirty mark and one save rule cover a Write FRU Data and a policy-bit change; `FRU RESTORE` keeps the byte as it is.

9.5 Device SDR repository

Record ID	Type	Content
0000h..0006h	02h compact sensor	sensors 0..6 in number order: entity A0h (VPX module) instance 60h, scanning on, events off at Tier 1, reading masks per Table 10.2-1, ID strings from `IPMC_SDR_*_NAME`; initialization, capabilities and event masks by Tier (section 9.6)
0007h (Tier-3 only)	02h compact sensor	sensor 7, Payload Mode: sensor type F7h, event/reading type 6Fh, initialization 03h, capabilities 40h, reading and assertion masks `IPMC_PAYLOAD_MODES`, deassertion mask 0000h, ID string `IPMC_SDR_PAYLOAD_MODE_NAME`
0007h (0008h at Tier-3)	12h MC device locator	owner = own IPMB address, global initialisation 01b and device capabilities 0Bh at Tier 1 (section 9.6 for Tier 2 and 3), entity A0h instance 60h, ID string "IPMC"

Get Device SDR Info reports 7 sensors (or 8 records with operation bit 0), and 8 sensors (9 records) at Tier-3; Get Device Locator Record ID returns the locator's record ID, 0007h (0008h at Tier-3). The locator is always the last record.

9.6 Tier-dependent bytes

The Tier rules of `tier.rs` set these bytes; the build reports the Tier of `IPMC_TIER` (1 in the public build, section 4). A Tier-1 IPMC generates no events (Rule 7-4); a Tier-2 and a Tier-3 IPMC are Event Generators (Rule 7-3); a Tier-3 IPMC is also a SEL Device (Rule 7-8) with per-state event enables. A Tier-3 IPMC also controls payload power and activation (Rule 4.1.2-10.1, FRU State M5), has the Payload Mode sensor (sensor 7, Rule 6.1-17) and supports FAST-MODE, 400 kHz (Rule 4.1.1-4.4).

Item	Tier 1	Tier 2	Tier 3
Get VSO Capabilities, IPMC identifier byte bits 1:0 (`IPMC_TIER` - 1)	00b	01b	10b
Get Device ID byte 7, additional device support	09h	29h	2Dh
compact record, sensor initialization	01h	03h	03h
compact record, sensor capabilities	43h	42h	40h
compact record, assertion and deassertion event masks	0000h	per sensor, below	per sensor, below
locator record, global initialization (bits 1:0)	01b	00b	00b
locator record, device capabilities	0Bh	2Bh	2Fh
Get Sensor Reading status bit 7 at power-up (all sensors)	0	1	1
event generator, SEL device, per-state event enables	no, no, no	yes, no, no	yes, yes, yes
activation control (M5), Payload Mode sensor, FAST-MODE	no, no, no	no, no, no	yes, yes, yes
sensors, SDR records (Get Device SDR Info)	7, 8	7, 8	8, 9
locator record ID (Get Device Locator Record ID)	0007h	0007h	0008h
Get VSO Capabilities, IPMB capabilities byte bits 5:4 (maximum frequency)	00b, 100 kHz	00b, 100 kHz	01b, 400 kHz
sensor 0 reading mask (FRU State)	0053h	0053h	0073h
sensor 1 reading mask (IPMB Link)	000Fh	000Fh	003Fh
FRU Control Capabilities (2Ch/1Eh, commercial release build)	C1h	00h	11h
Set IPMB State speed byte accepts 01b (400 kHz)	no, CCh	no, CCh	yes, on a fitted bus

Byte 7 of Get Device ID carries the sensor and FRU inventory bits (09h) at every Tier, the Event Generator bit (bit 5) from Tier 2 and the SEL Device bit (bit 2) at Tier 3. The locator's device capabilities byte is that value plus the Device SDR bit (02h).

Sensor capabilities carry auto re-arm (bit 6) and the event message control in bits 1:0: 11b none (Tier 1), 10b global disable only (Tier 2), 00b per state (Tier 3).

Event masks at Tier 2 and 3, assertion and deassertion (`sdr.rs`, LS byte first in the record). Tier-3 adds M5 to sensor 0, the 400 kHz states (offsets 4 and 5, asserted and deasserted) to sensor 1, and sensor 7 with one event per supported mode (`IPMC_PAYLOAD_MODES`, here modes 0 and 1):

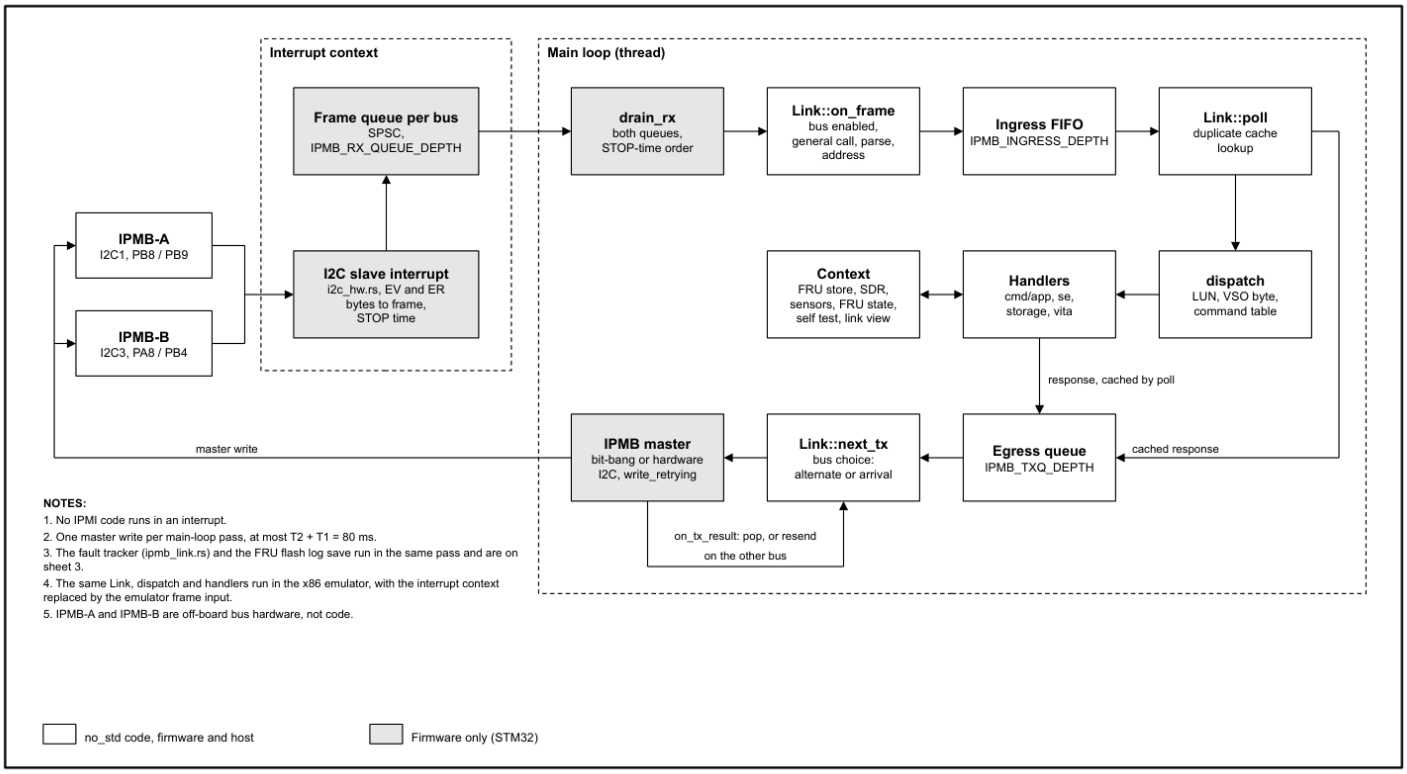
Sensor	Tier 2 assertion	Tier 2 deassertion	Tier 3 assertion	Tier 3 deassertion
0 FRU State	0053h	0000h	0073h	0000h
1 IPMB Link	000Fh	0000h	003Fh	0030h
2 FRU Health	0003h	0000h	0003h	0000h
3 FRU Voltage	0003h	0000h	0003h	0000h
4 FRU Temperature	0038h	0038h	0038h	0038h
5 Payload Test Results, 6 Payload Test Status	0000h	0000h	0000h	0000h
7 Payload Mode	none	none	0003h	0000h

The power-up event enables (`EventEnables::power_up`) take the same masks, so the records and the enables cannot disagree.

10. Data Flow and Protocol

10.1 Request path in the firmware

Figure 10-1. Firmware request path, receive to transmit (ipmc-software, sheet 2)



Source: `diagrams/src/ipmc-software.js`, sheet 2.

- On receive, the I2C EV interrupt (`i2c_hw.rs` `on_event`) clears ADDR, starts a frame with the matched address byte, appends each data byte, and at STOP posts the frame with the SysTick time to that bus's SPSC queue. A frame longer than 33 bytes, an address-only write or a full queue is dropped and counted. The ER interrupt counts and clears error flags and drops the partial frame, except for ARLO alone (the hardware master lost arbitration), which is counted and keeps the reception, because the winner's frame may be addressed to this slave. A read of the slave answers FFh bytes.
- In the main loop, `IpmbLink::service` samples NVMRO, drains both queues in STOP-time order into `Link::on_frame` (at most 2 x `IPMB_RX_QUEUE_DEPTH` frames per pass; each frame goes into the capture log before the parser sees it), runs `Link::poll`, books slave errors on the fault streak, transmits at most one response, clears faulted buses whose lines have been idle-high for `IPMB_FAULT_CLEAR_MS`, and runs the FRU log save step.
- On transmit, `Link::next_tx` gives the oldest response and its bus; the master writes it (`write_retrying`: free-bus wait, then the transfer, arbitration retries on the same bus inside one T2 + T1 budget); the frame and its outcome go into the capture log; `Link::on_tx_result` pops the frame on success or moves it to the other bus for a resend.

10.2 Link layer (`ipmc-core/src/link.rs`)

`on_frame(bus, frame, now_ms)`:

1. A bus that is disabled or not fitted drops the frame (`BusUnavailable`).
2. A leading 00h (general call) is stripped and the request marked broadcast.
3. `ipmb::parse`; a parse error is counted per kind; a response is offered to the requester and kept when it answers the outstanding request (`rx_answers`), else dropped (a Tier-1 IPMC sends no requests, so it drops every response).
4. rsSA must be the own address; a broadcast must also be Get Device ID (`NotForUs` otherwise).
5. The request joins the ingress FIFO with its bus and arrival time; a full FIFO drops the new frame. Every drop is logged for `poll` to trace; the counters keep the totals.

`poll`: traces the drops, then takes requests oldest first while the egress queue has room. A request whose `Key` is in the duplicate cache within the window gets the cached response queued again and runs no handler (`replays` counter). Any other request is dispatched against a private staging queue; a pending Set IPMB State change is applied; the response is cached and queued with the arrival bus and time. After the requests, `poll` steps the requester timing (a retransmission falls due, or the request ends with no response), traces a request that ended, refreshes the link view in the context and calls `Extension::poll` once.

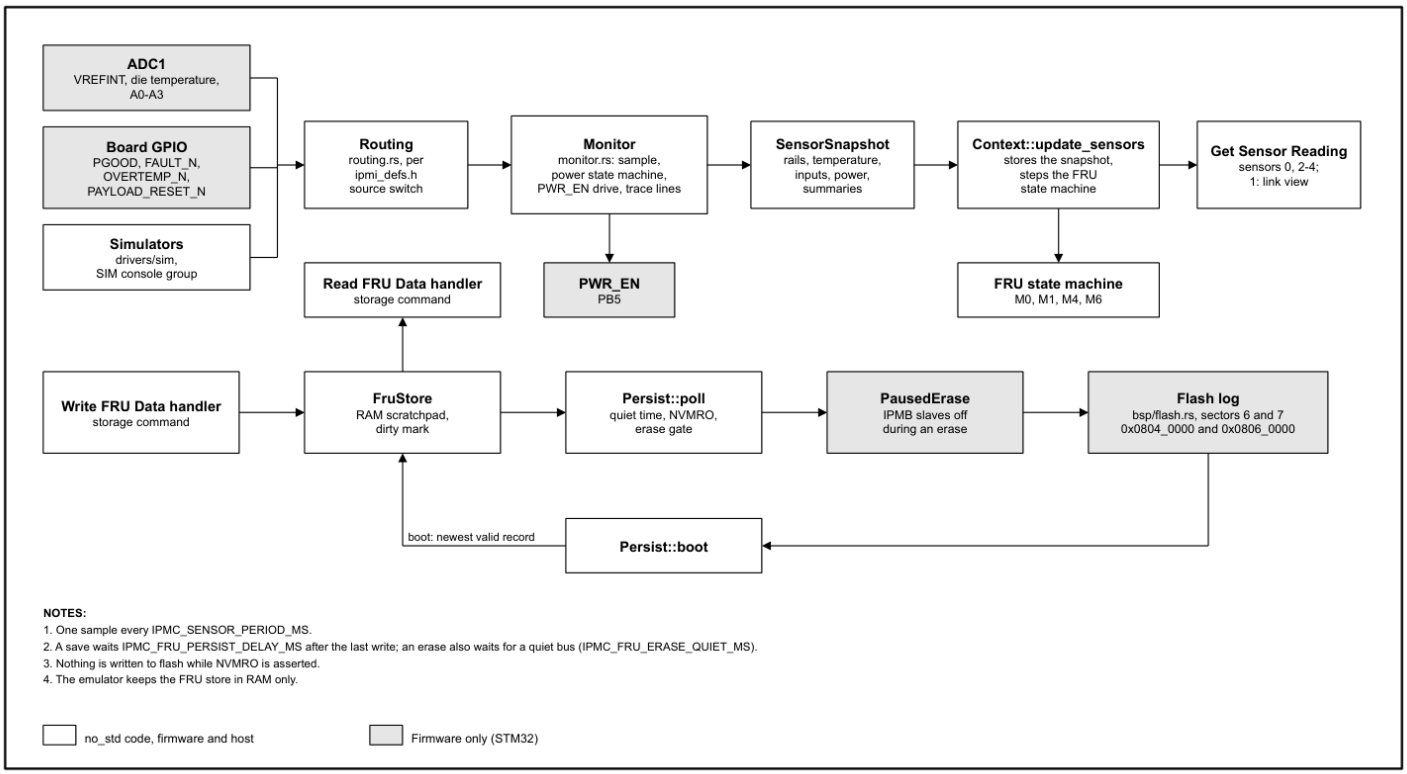
`next_tx`: a response handed out for the first time gets its bus now. ALTERNATE prefers the bus other than the latest transmission (resends count); ARRIVAL prefers the request's bus; when the preferred bus is not enabled the other enabled bus is used. A response whose assigned bus has since been disabled moves to the other bus. None when neither bus is enabled. A queued response always goes first; with none queued, a request that is due goes out on the bus other than the one of its own previous transmission (the other enabled bus when that one is disabled), and the same frame is handed out until its outcome is reported.

`on_tx_result(bus, ok, now_ms, trace)`: success pops the frame and counts a late response when the transmission completed more than 227 ms after the request's STOP. Failure counts `tx_fail` and the streak; when the other bus is enabled and the frame has fewer than `IPMB_TX_RETRIES` resends it moves there, otherwise it is dropped (`tx_dropped`) and the requester's retry gets the cached response. A request frame is booked by the requester instead: success starts its T3 wait, a failure waits at least T6 and retransmits on the other bus when it is enabled (`tx_requests` counts every request transmission).

Bus state per bus: `fitted`, `mode` (Isolated or Local Control, the Set IPMB State value) and `status` (local fault, `LocalStatus`). A bus is enabled when it is fitted, in Local Control and without a local fault.

10.3 Sensor data, FRU state and FRU persistence

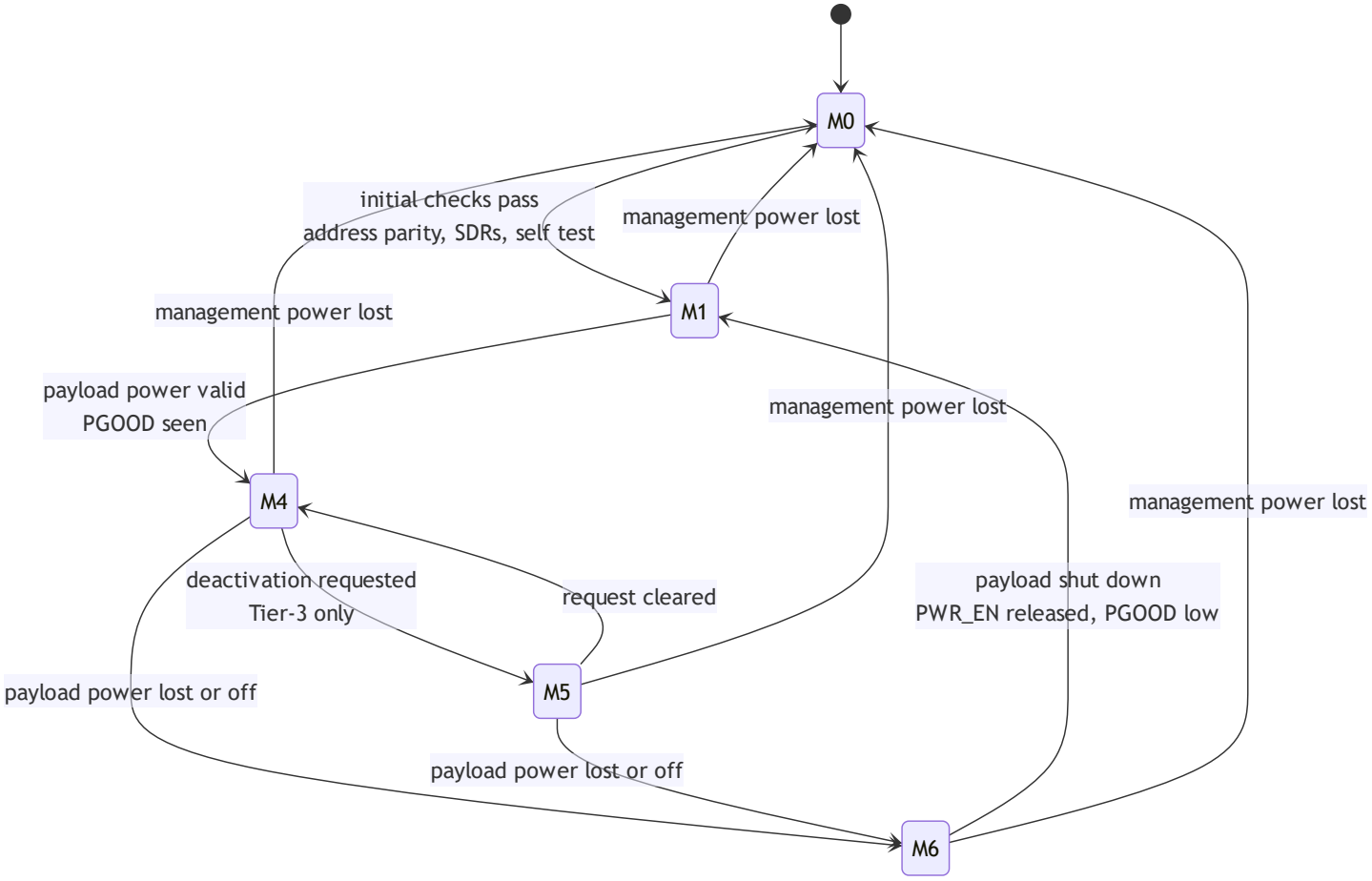
Figure 10-2. Sensor data, FRU state and FRU persistence (ipmc-software, sheet 3)



Source: `diagrams/src/ipmc-software.js`, sheet 3.

Once per `IPMC_SENSOR_PERIOD_MS` the `Monitor` reads the inputs, steps the power state machine, drives PWR_EN, builds a `SensorSnapshot` and prints the change lines. `IpmbLink::feed_sensors` hands the snapshot to `ext::feed_sensors`, which calls `Context::update_sensors` and then the extension's `on_sample` hook with the FRU state transition that sample made. `update_sensors` stores the snapshot for Get Sensor Reading and steps the FRU state machine with the initial checks, "payload power valid" (power state On, so PGOOD was seen), "payload shut down" (PWR_EN released and PGOOD read deasserted) and, at Tier-3 only, the standing deactivation request (`Context::deactivation_requested`).

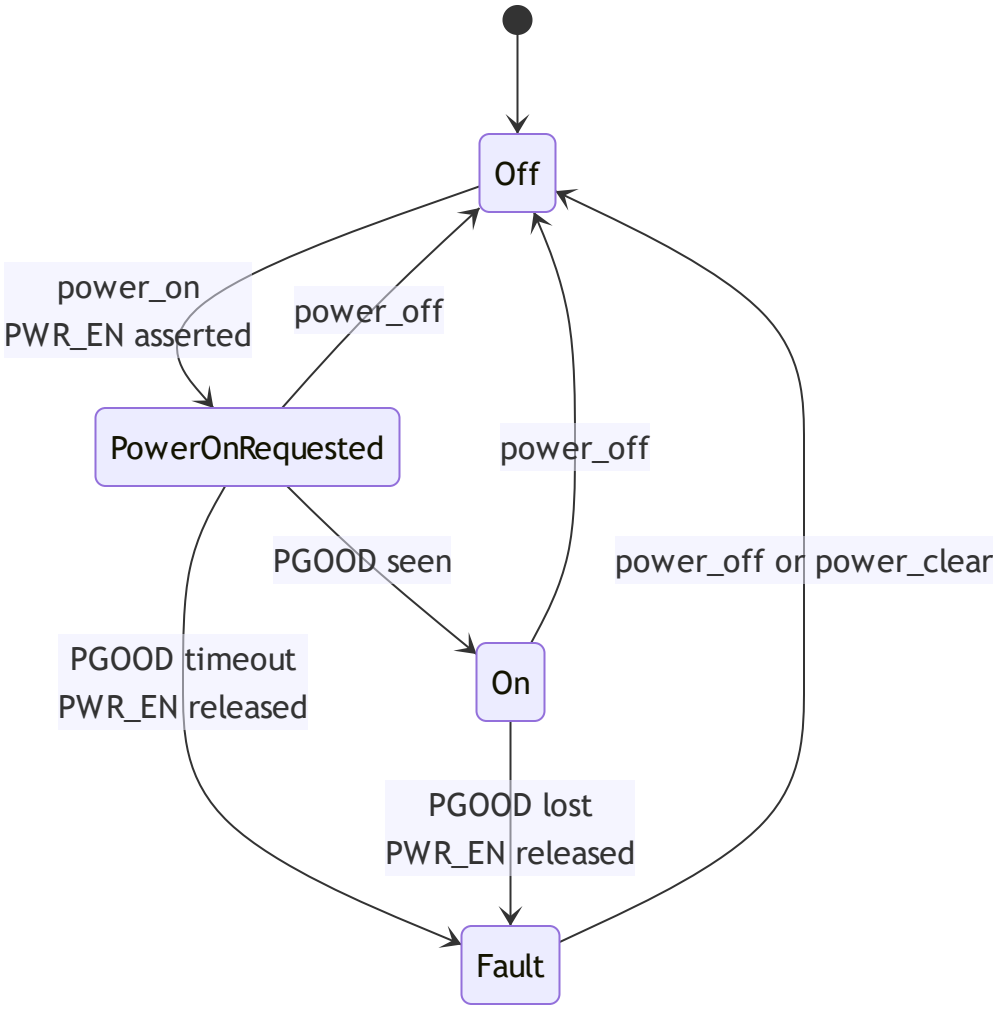
10.4 FRU 0 Operational State machine (`fru_state.rs`)



One transition per `step`. The state is reported only when Get Sensor Reading polls sensor 0 (a Tier-1 IPMC sends no events; the commercial release build also sends each change as a Platform Event, section 7.2). The order inside `step` is: management power lost (to M0), then the transition of the current state; from M4 and M5, lost payload power (to M6) is tested before the deactivation request. Power that returns during M6 does not cancel the deactivation. M5 (FRU Deactivation Request) is reachable only at Tier-3, where the IPMC controls payload power (`tier::activation_control`): `Inputs::deactivation_requested` is the Tier-3 handlers' standing request (Rules 4.1.5.4.3-2 and -6) and is false below Tier-3, so M4 never leaves for M5 there. M7 is declared by the Chassis Manager, never by the IPMC for its own FRU, so it is not reachable. A self-test failure after M0 does not leave M1, M4 or M5. Management power is valid by construction on this board (the IPMC runs from it).

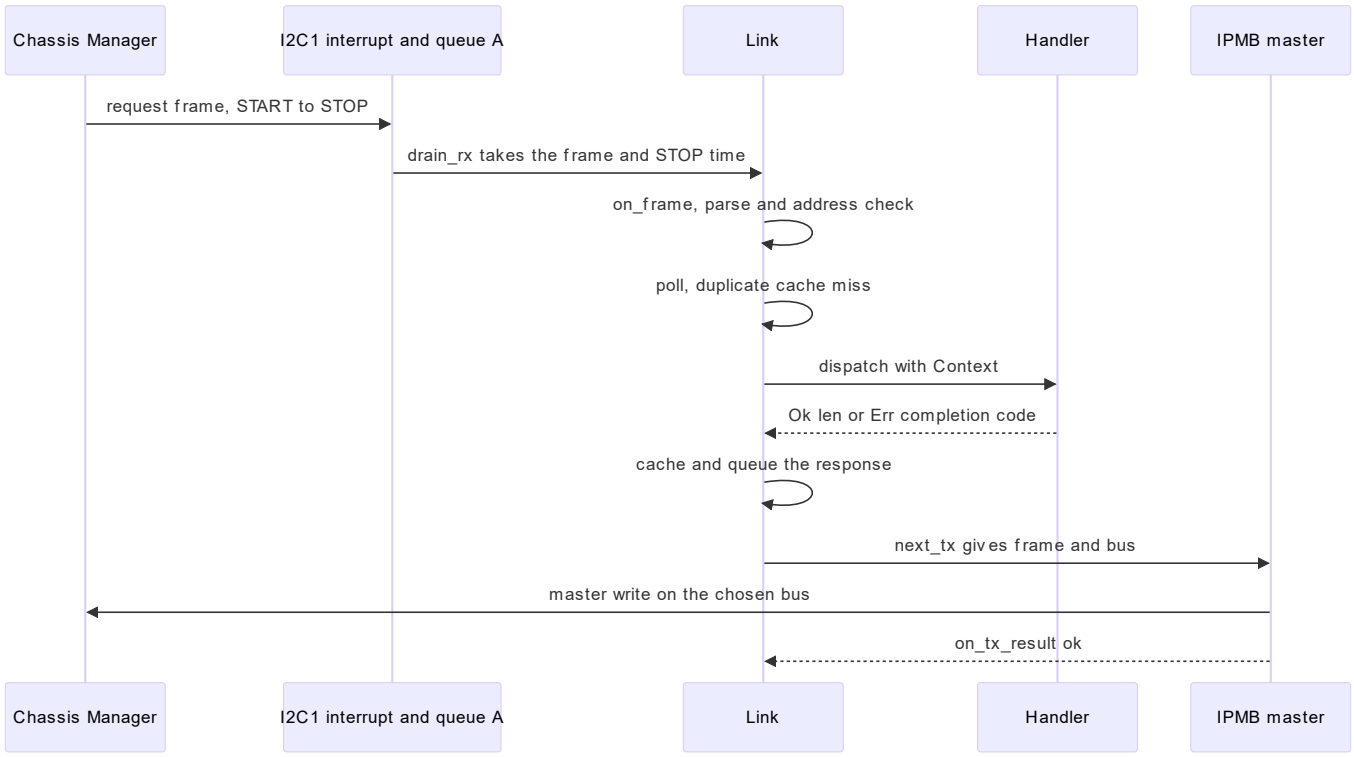
A payload power-on request is refused in M6 (`PowerOnRefused::DeactivationInProgress`) and, in the commercial release build, while the FRU is Activation-Locked (`PowerOnRefused::ActivationLocked`, from `Extension::power_on_allowed`, Rule 4.1.5.4.2-1). `ext::check_power_on` is the one check for the boot power-on, the `pending_power` On hand-off (`Monitor::apply_power_request`) and the console `POWER ON`; the first refusal is returned and printed with its `name()`.

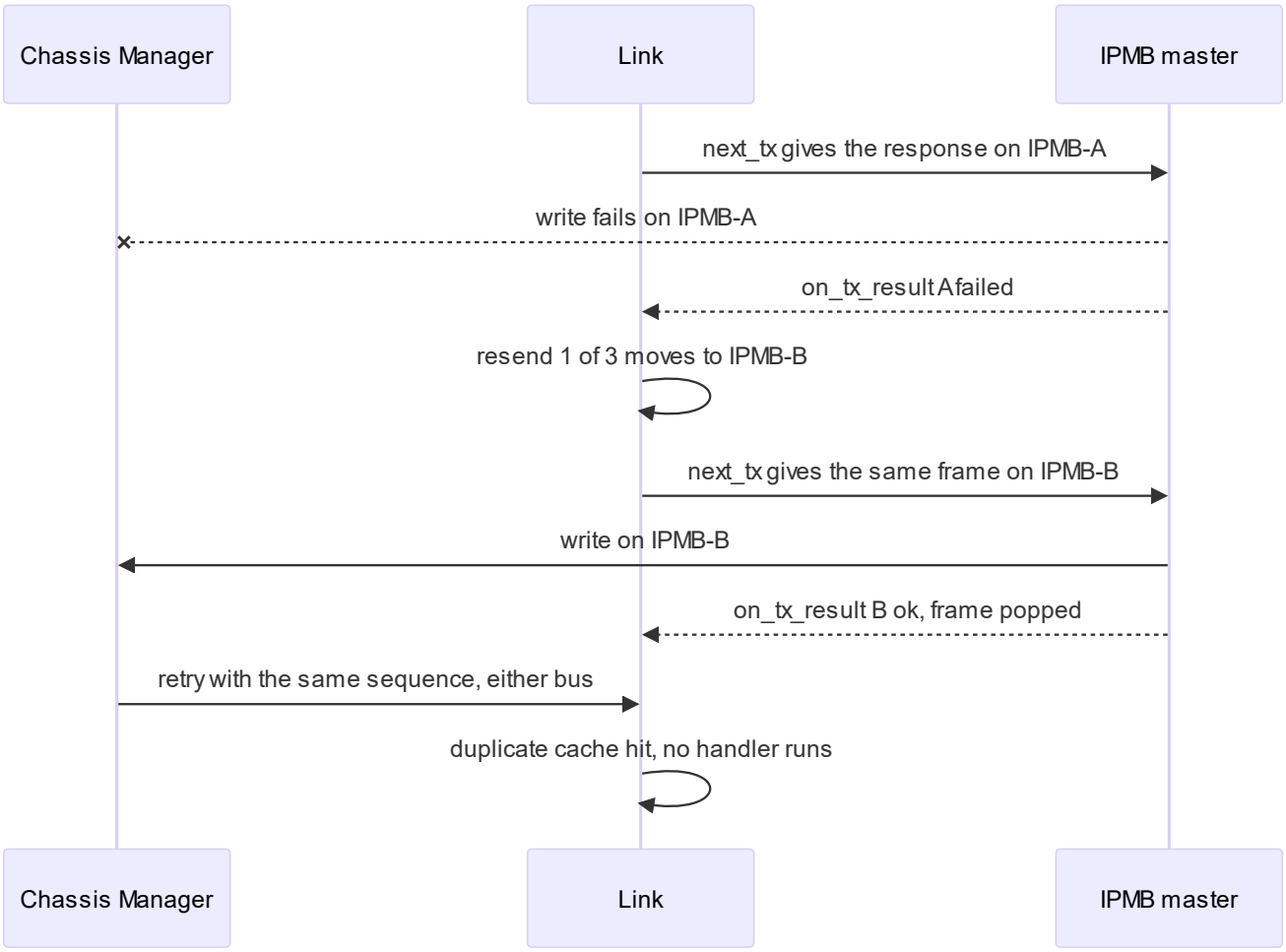
10.5 Payload power state machine (`ipmc-sensors/src/eval/power_fsm.rs`)



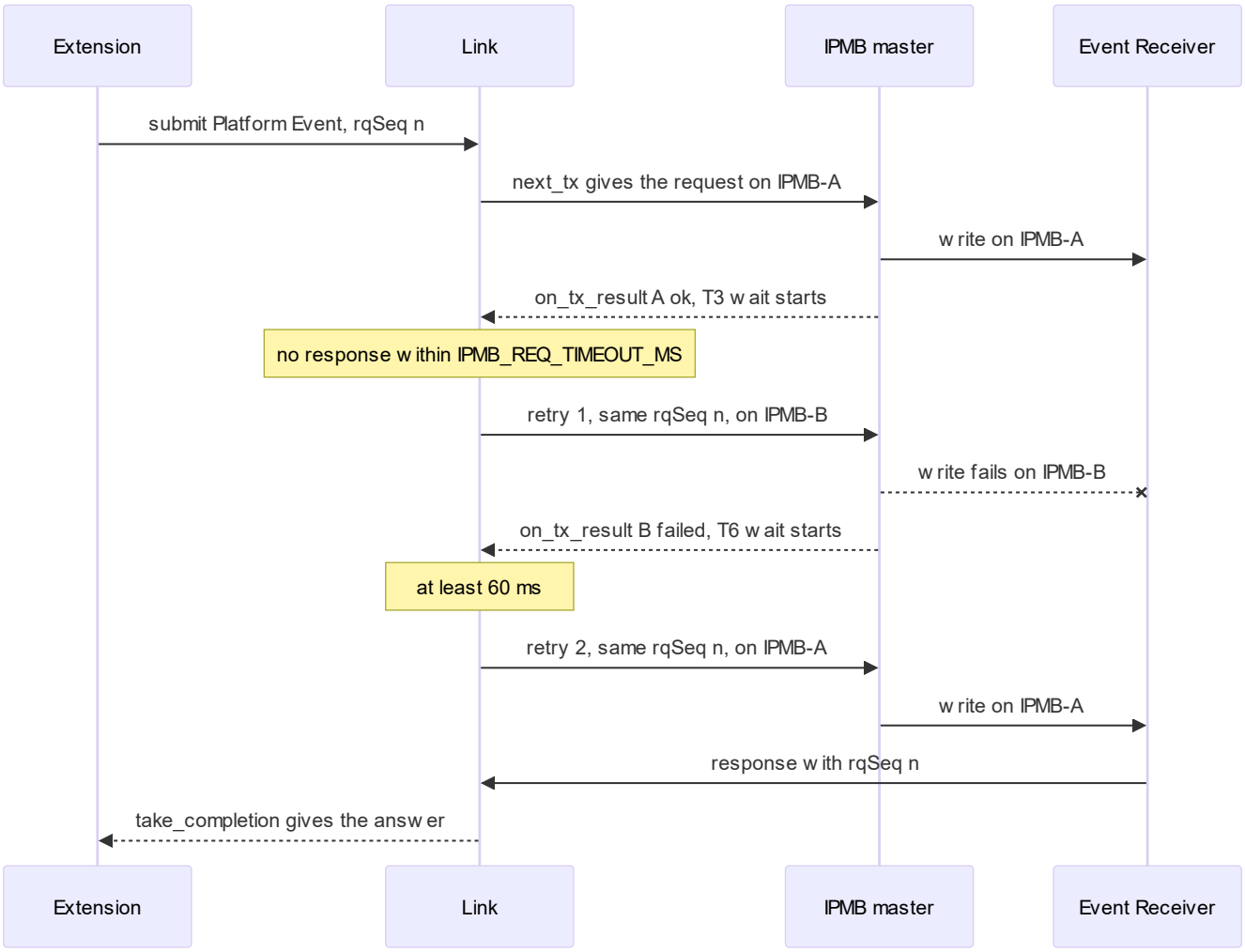
A fault stays latched until `POWER OFF` or `POWER CLEAR`, and `power_on` is refused while it is latched. A failed PWR_EN write latches `Fault` (EnableWrite) from any state. PGOOD seen in the same step as the timeout wins. Console `POWER ON` and a `pending_power` On request check `ext::check_power_on` first (section 10.4). The start-up request (`IPMC_POWER_AT_BOOT`) runs in M0, where the FRU machine's check always passes; the extension's activation lock can still keep the payload off at boot.

10.6 Request, response and resend sequences

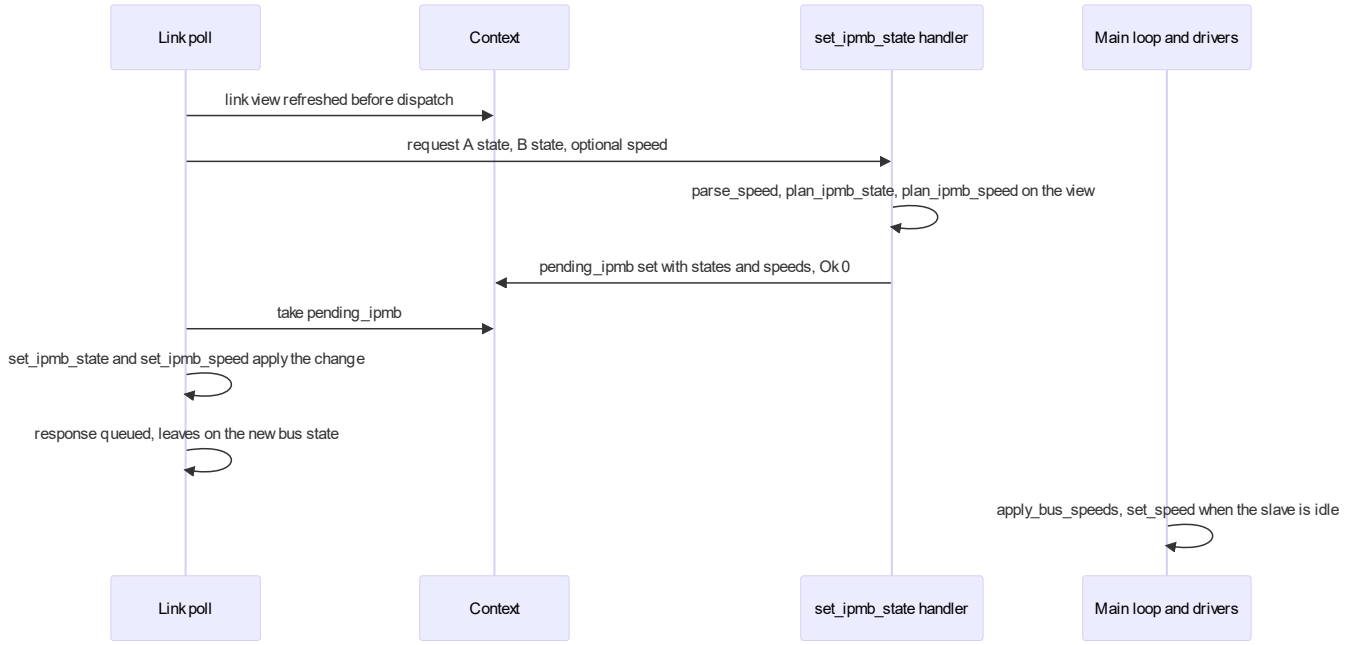




A request this IPMC sends (commercial release build) keeps one sequence number through its retries. The first transmission goes out on IPMB-A and is acknowledged, but no response arrives within `IPMB_REQ_TIMEOUT_MS`; the retry goes on IPMB-B, the bus other than the previous transmission's, and fails on the wire; after at least T6 (60 ms) the next retry goes on IPMB-A and the response arrives.


Extension
Link
IPMB master
Event Receiver
submit Platform Event, rqSeq n
next_tx gives the request on IPMB-A
w rite on IPMB-A
on_tx_result A ok, T3 w ait starts
no response w ithin IPMB_REQ_TIMEOUT_MS
retry 1, same rqSeq n, on IPMB-B
w rite fails on IPMB-B
on_tx_result B failed, T6 w ait starts
at least 60 ms
retry 2, same rqSeq n, on IPMB-A
w rite on IPMB-A
response w ith rqSeq n
take_completion gives the answ er
Extension
Link
IPMB master
Event Receiver

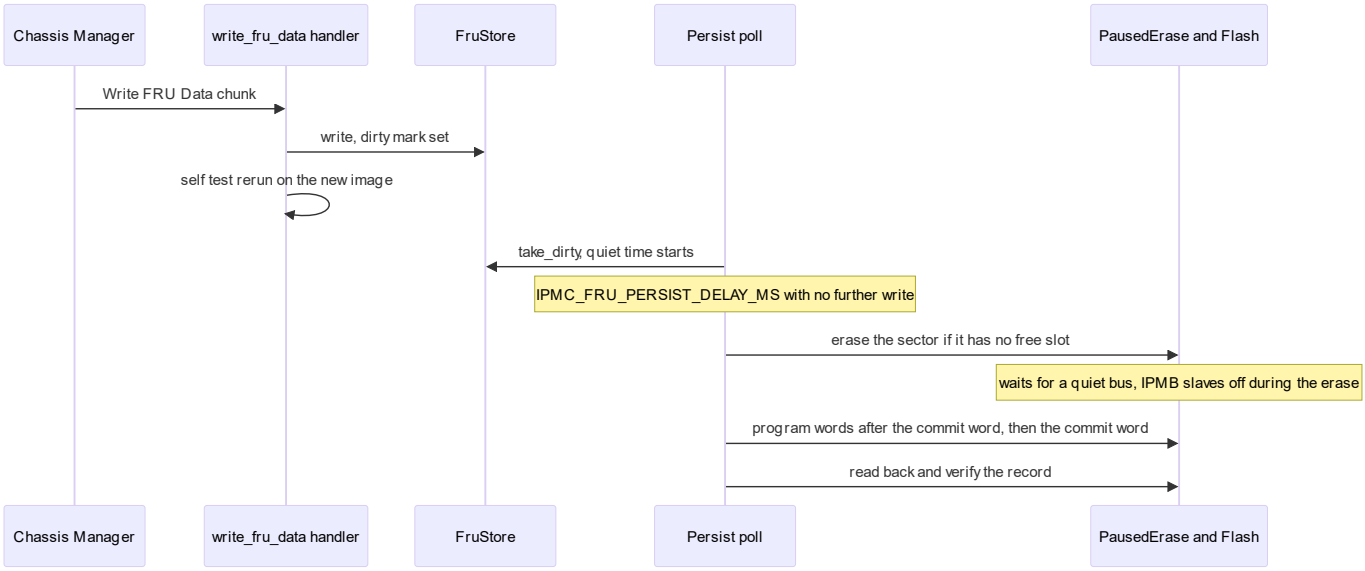
10.7 Set IPMB State hand-off



The speed byte is optional. Its bits 1:0 are the IPMB-A speed, bits 3:2 the IPMB-B speed and bits 7:4 are reserved: 00b is STANDARD-MODE, 100 kHz; 01b is FAST-MODE, 400 kHz, accepted only at Tier-3 (`tier::fast_mode`); 10b, 11b, a non-zero reserved nibble and 01b below Tier-3 answer CCh (Rule 4.1.1-14). `plan_ipmb_speed` also refuses 400 kHz for a bus that is not fitted (CCh); 100 kHz is always accepted. A request with an absent speed byte changes no speed. A refusal changes nothing, states included. `Link::poll` applies the state, then the speed, before the response is queued; the response leaves on the new state, still at the old speed.

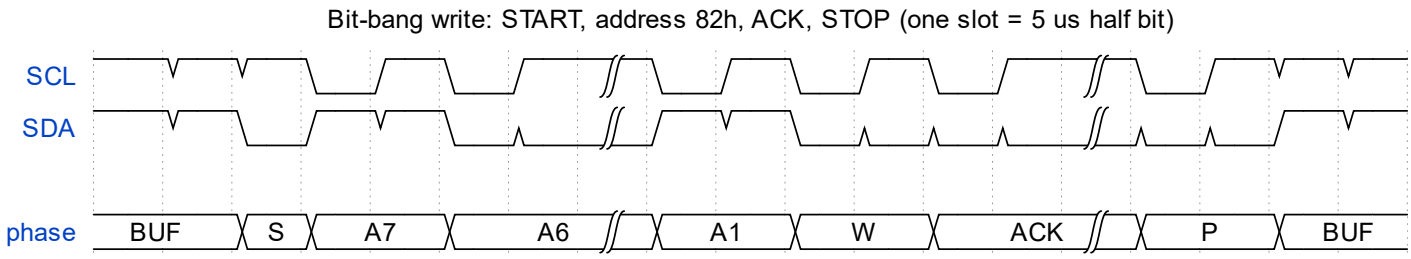
The firmware applies the speed between transfers. `IpmbLink::apply_bus_speeds` runs at the end of `service`, compares each running bus's `bus_state(bus).speed` with the speed applied, and calls `IpmbMaster::set_speed`. That calls `i2c_hw::set_speed`, which suspends the slave only when it is idle (a busy or stopped slave returns false and the next pass asks again), loads CCR, TRISE and F/S for the speed and resumes the slave; the master write uses the new phases from the next frame on, and one console line `IPMB-A: running at 400 kHz` marks the change. Every restart of the link starts at STANDARD-MODE (Rule 4.1.1-4.3): at power-up, after a Cold Reset and after a Warm Reset (`Link::restarted`), and the drivers follow on the next pass. The System IPMB Link sensor and its FAST-MODE events follow the link state as soon as Set IPMB State is applied; the driver changes the clock once its slave is idle, so for a pass or two the sensor can show 400 kHz before the master runs at it. Bus recovery at start-up runs at 100 kHz.

10.8 FRU save to the flash log



The store is one sector with 60 slots, so the erase comes once per 60 saves: the save that finds no free slot erases the sector and writes the newest image as the first record. That erase starts only when the egress queue is empty and no frame arrived for `IPMC_FRU_ERASE_QUIET_MS`; until then the image stays safe in RAM. `FRU RESTORE` erases at once when its record needs a sector, without that quiet gate. `PausedErase` suspends each running slave (a healthy one waits up to `IPMC_FRU_PAUSE_WAIT_MS` for idle; a faulted one is switched off at once), erases, resumes the slaves and reports the time the board was silent (a status line on the console and in the capture log), measured with the DWT counter. After the slaves are off the driver checks both receive queues; a frame accepted before that defers the erase (`FlashError::Deferred`): the slaves resume, nothing is printed, the write stays pending with the failure counter unchanged, and the next pass retries; `systick::catch_up` adds the stalled time to the clock. On the bench the sector erase paused the IPMB for 198 ms (DS10086 Table 45, x32 programming: typical 250 ms, maximum 500 ms); a request that arrives in that time is missed and the Chassis Manager retries it. A power cut between the erase and the end of the first record's programming loses the stored image (section 15). A failed save keeps the write pending and retries once per delay. A per-unit field written through the application port takes the same path: `set_unit_field` marks the store dirty and `service` saves it after the quiet time.

10.9 Bit-bang master timing (`bsp/i2c_bitbang.rs`)



One slot is `IPMB_BITBANG_HALF_NS` (5000 ns). The bus is free when both lines have read high for tBUF (4.7 us) and the slave is idle. START: SDA falls while SCL is high (tHD;STA timed from the SDA write). Each bit: SDA set while SCL is low, at least tSU;DAT (250 ns) before SCL is released; SCL released and read back until high (clock stretching, up to `IPMB_STRETCH_TIMEOUT_US`); SDA sampled; after every released 1 bit a low SDA means another master holds the bus (`ArbitrationLost`). ACK: SDA released for the ninth clock and sampled low (the slave acknowledges). STOP: SDA low, SCL released, SDA released. Lines are only ever released or driven low (open-drain output: latch 1 releases the line, latch 0 drives it low); a pin is never driven high. The SCL period measured with the DWT counter is 12.0 us (83 kHz), below the 100 kHz limit.

At 400 kHz (FAST-MODE, Tier-3) the two phases differ: SCL low lasts `IPMB_BITBANG_FAST_LOW_NS` (1400 ns, at least the I2C fast-mode tLOW of 1300 ns) and SCL high `IPMB_BITBANG_FAST_HIGH_NS` (1000 ns, at least tHIGH of 600 ns). The cycle counts come from `ipmc-defs` `BitPhases`; the low phase also times the bus-free wait after a STOP (at least 1.3 us in fast mode) and the high phase the START setup and hold and the STOP setup. The slack above the minimum high time absorbs a slow rise, and a stretched clock extends the phase so SCL stays high for the minimum after it was seen high. The SCL period measured with the DWT counter is 6795 ns (about 147 kHz): the two phase times plus the loop overhead of the pin operations, with SCL low and high both above the fast-mode minimums. The hardware master runs CCR 35 (F/S = 1, DUTY = 0) and TRISE 13 at PCLK1 = 42 MHz; its measured SCL period is 2981 ns (about 335 kHz).

Peripheral hand-off: `i2c_hw::suspend` re-checks the slave is idle inside a critical section, masks the bus's EV and ER interrupts and clears PE, which releases both lines; the pins go to open-drain GPIO; after the STOP `i2c_hw::resume` restores the alternate function and re-arms the slave exactly as `init` does. While PE is 0 the slave cannot hear its bus; a frame another master sends at that moment is lost and recovered by the requester's retry.

10.10 Error handling

Error	Detection	Reporting	Recovery
Bad frame (length, checksum, address bit)	`ipmb::parse` in `on_frame`	`RxErrors` per kind, trace line	dropped; the requester retries
Frame on a disabled bus, not for us, response	`on_frame`	`bus_unavailable`, `not_for_us`, `rx_response`	dropped

Error	Detection	Reporting	Recovery
Ingress or egress full	`Fifo::push`	`ingress_full`, `egress_dropped`, `dispatch_errors`	dropped; requester retries
Transmit failure (NACK, busy, stretch, stuck line)	`MasterError` from the master	`tx_fail`, per-kind master counters, trace line	resend on the other bus up to `IPMB_TX_RETRIES`
Arbitration loss	released SDA reads low, or ARLO	`arb_retries`	same-bus retry up to `IPMB_ARB_RETRIES` within the write budget
Persistent bus fault	`IPMB_FAULT_STREAK` consecutive bus-fault errors (stretch timeout, SCL/SDA stuck, bus busy, slave BERR or TIMEOUT)	console `IPMB-A fault: <name>`, IPMB Link sensor local status	bus disabled (Rule 9.3-10); cleared after both lines read high for `IPMB_FAULT_CLEAR_MS`, with the BUSY workaround and a slave re-arm
Stuck BUSY flag	both lines high longer than T7 (3 ms) with BUSY set and no frame	`busy-errata-resets` counter	`clear_stuck_busy` (I2C peripheral reset sequence)
Bus fails at start-up	`bring_up` (recovery or slave init)	console line, local status	bus disabled; restarted after the lines read idle-high
Request without a response (commercial release build)	`Requester` T3 timeout after each transmission	trace line `request seq .. no response after N transmissions`	retried up to `IPMB_REQ_RETRIES` times with the same sequence number; then the completion is no response and the extension decides
Payload power-on refused	`ext::check_power_on`: the FRU is in M6, or the extension reports the FRU Activation-Locked (commercial release build)	console line `power on request refused: ...`, `Power at boot refused: ...`, or the refusal text of console `POWER ON`	no power change; allowed again once the FRU is in M1 or activated
Cold or Warm Reset blocked (commercial release build)	`ResetGate`: response still queued or FRU write unsaved	trace line after `IPMC_RESET_WAIT_MS`	the reset runs anyway at that time
Corrupt FRU image	`selftest::run`	Get Self Test Results 57h with bit 5	FRU stays in M0 when found at start-up; Write FRU Data or `FRU RESTORE` repairs it

Error	Detection	Reporting	Recovery
Flash program, erase or verify failure	`FlashError`	console `FRU log: save failed: <name>`, `FRU STORE`	write stays pending, retried once per delay
PGOOD timeout or loss	`power_fsm`	console event line, `POWER STATUS`; FRU Health is bad only while PWR_EN is asserted without PGOOD, so it returns to good once the fault releases PWR_EN	latched until `POWER OFF` or `POWER CLEAR`
Pin-claim conflict, UART failure	`ports::claim`, `uart::init`	console line naming both owners, or 5 Hz LED	fail-stop (park); an application port failure other than a pin conflict prints `APP_UART not started` and the board runs without it
UART receive ring full, overrun, transmit timeout	`uart.rs` `service`, `wait_for_room`	per-port counters in `SYS STATUS`	the byte is dropped (receive) or the bounded wait ends (transmit)
Trace line without room in the console transmit ring	`print_line_or_drop`	`tx lines dropped` in `SYS STATUS`	the line is dropped whole and never stalls the IPMB; the capture log still holds it
Application answer without room in the transmit ring	`app_port.rs` `send`	`dropped` in `SYS STATUS`, `answers dropped` in `DEBUG STATUS`	the answer is dropped whole; the payload CPU asks again
Capture log full	`CaptureLog::add`	`overwritten` in `DEBUG STATUS` and in the dump header	the oldest whole records are dropped

10.11 Capture log (`capture.rs`, `ipmb_link.rs`, `debug_cmds.rs`)

The capture log is the field-service record of the IPMB. A person enables it on a target system, reproduces a fault and mails the dump back; the dump reads the same on the console, in a spreadsheet and, through `tools/capture/capture2pcap.py`, in Wireshark. The log lives in RAM (one `ByteRing` of `IPMC_LOG_SIZE` bytes, 8191 usable by default) and is empty after a reset.

What goes in:

- every whole IPMB frame taken from a driver queue, before the parser (so a frame the parser rejects is there too), with status `rx`;
- every frame handed to the master, with its outcome: `ok`, or the cause of the failure;
- every frame handed to the link by `DEBUG INJECT`, with status `inject`;
- the trace lines of the link, the dispatcher and the requester, the status lines (FRU log saves, bus faults and speed changes, resets), the lines of `DEBUG IPMI` (`console request ...`, `console response ...`) and the application port lines (`app request ...`, `app answer ...`).

`IPMC_IPMI_TRACE` 0 removes the link and dispatcher trace lines at compile time, so they are absent from the capture log as well as from the console; the frames, the status lines and the console and application lines stay. `DEBUG TRACE IPMI OFF` stops only the console copy: the log keeps the lines. `DEBUG TRACE I2C ON` (or `IPMC_I2C_TRACE_AT_BOOT` 1) prints one console line per frame, the same row `DEBUG DUMP TRACE` prints, even while the log is off.

Record layout (little-endian): a 6-byte header, then the payload.

Byte	Content
0	flags: bit 0 kind (0 frame, 1 text), bit 1 bus (0 A, 1 B), bit 2 direction (0 in, 1 out), bits 6:3 status code
1	payload length: a frame at most `IPMB_MSG_MAX` (32) bytes, a text line at most `IPMC_LOG_TEXT_MAX` (160)
2..5	`time_ms`, u32, the SysTick time
6..	the payload: the IPMB frame as it is on the wire, or the line as ASCII

The status code is 0 received, 1 injected, 2 sent, then 3 + n for a failed transmission with the cause n: address NACK, data NACK, arbitration lost, bus busy, stretch timeout, line stuck, unspecified. The record index is not stored: the i-th held record is `total - records + i`. A record that does not fit drops the oldest whole records first and counts them. A 23-byte frame takes 29 bytes of the ring.

The DEBUG group (help text as in `console.rs`):

Command	Help text
`DEBUG STATUS`	Debug switches, capture log use, event receiver, duplicate cache
`DEBUG LOG {STATE}`	Capture log ON or OFF; no argument shows its state

Command	Help text
`DEBUG DUMP {FORMAT}`	Print the capture log: CSV (default) or TRACE; any key stops it
`DEBUG CLEAR`	Empty the capture log
`DEBUG TRACE {WHAT} {STATE}`	Console trace: IPMI (request, link and event lines) or I2C (one line per frame), ON or OFF
`DEBUG EVENTS {STATE}`	Event messages ON (receiver 20h) or OFF (receiver FFh)
`DEBUG IPMI [NETFN] [CMD] {DATA...}`	Run one IPMI request locally, as from 20h; nothing goes on the bus
`DEBUG INJECT [BUS] {BYTES...}`	Hand a whole IPMB frame to the link as if received on bus A or B
`DEBUG CACHE {ACTION}`	Duplicate cache: no argument shows the entries, CLEAR empties it

`DEBUG IPMI` and `DEBUG INJECT` take hex bytes (`06`, `0x06` and `6` are the same). `DEBUG IPMI` runs the request through the dispatcher (`IpmbLink::local_request`) and prints `-> CC 00h, 15 bytes: 01 80 ... (App Get Device ID)`. `DEBUG INJECT` hands the frame to the link as received on bus A or B; the link answers a request it accepts on the bus like any other.

`DEBUG DUMP` prints one header line, the rows and an end line. The header line is

```
# IPMC capture: ipmc-nucleo-f401re v0.1.0, ipmc-core v0.1.0, IPMB address 82h, Tier 1, uptime 123456 r
```

and the end line is `# end of capture`. A dump runs as a job: a few rows per main-loop pass, while the console transmit ring has room for the longest row, so the main loop never waits for the wire. It prints the records held when it started; a record the ring overwrote meanwhile is skipped and counted in `# <n> records overwritten during the dump`.

The CSV format has one header row and one row per record, with the columns

Column	Content
`rec`	the record index since the last clear
`time_ms`	the SysTick time of the record
`type`	`FRAME` or `TEXT`
`bus`	`A` or `B`; empty for text
`dir`	`IN` or `OUT`; empty for text
`status`	`rx`, `inject`, `ok`, `addr-nack`, `data-nack`, `arb-lost`, `bus-busy`, `stretch`, `line-stuck` or `failed`; empty for text
`len`	the frame length in bytes; empty for text

Column	Content
`summary`	quoted; for a frame `<src>h-><dest>h <NetFn> <Req or Rsp> <command> seq <n>h` and `CC <cc>h` for a response, or `unparsed: <error>`; for text the line
`bytes`	the frame bytes in hex, upper case, one space apart; empty for text

```
rec,time_ms,type,bus,dir,status,len,summary,bytes
12,123456,FRAME,A,IN,rx,7,"20h->82h App Req Get Device ID seq 01h",82 18 66 20 04 01 DB
13,123459,TEXT,,,,,"FN(06h) CMD(01h) Get Device ID -> CC 00h, 15 bytes",
14,123461,FRAME,A,OUT,ok,23,"82h->20h App Rsp Get Device ID seq 01h CC 00h",20 1C C4 82 04 01 00 01 8
```

`DEBUG DUMP TRACE` prints the same records as Wireshark-style text rows: `#<index>`, the time in seconds with milliseconds, `IPMB-A` or `IPMB-B` and the direction, `<src>h -> <dest>h`, the NetFn name with `Req` or `Rsp` in a 14-character column, the command name in a 22-character column, the tail (`seq <n>h`, `CC <cc>h`, and the status word of an injected or failed frame), and after `|` the frame bytes. A text record is `#<index>`, the time, `TEXT` and the line.

```
#12   123.456  IPMB-A  IN   20h -> 82h  App Req       Get Device ID          seq 01h          | 82 18
#13   123.459  TEXT    FN(06h) CMD(01h) Get Device ID -> CC 00h, 15 bytes
#14   123.461  IPMB-A  OUT  82h -> 20h  App Rsp       Get Device ID          seq 01h CC 00h   | 20 1C
```

A command name comes from the Tier-1 table, then from the extension, then from Platform Event, the one request the IPMC sends; an unknown command prints as `CMD 5Ah`, a NetFn without a name as `NetFn 3Ah`.

The Wireshark path: save the terminal log of a CSV dump and run `python -I tools/capture/capture2pcap.py dump.txt -o dump [--csv]`. The tool reads only the lines between `# IPMC capture:` and `# end of capture` (a log may hold several dumps and other terminal text) and skips a `TRACE` block. It writes `dump.pcap`, a little-endian pcap with link type 260 (`LINKTYPE_IPMI_HPM_2`): one packet per record, stamped with the record's `time_ms`. Each packet holds one block of the Wireshark IPMI trace format: byte 0 the block type (0 frame, 2 text line) in bits 7:4 and the channel (0) in bits 3:0, bytes 1..4 the seconds and bytes 5..6 the milliseconds, byte 7 the data type 01h (IPMB 1.0), bytes 8..9 the flags (bit 6 IPMB-B, bit 7 direction to the controller, an `IN` frame), byte 10 the length and then the data, the whole IPMB frame from rsSA or the ASCII line. A failed `OUT` frame is the frame followed by a text packet `tx failed: <status>`. With `--csv` the rows are also copied to `dump.csv`.

11. Software API

11.1 Link layer (`ipmc_core::link::Link`)

API	Inputs	Output	Notes
`Link::new(config, &addr)`	`LinkConfig { b_fitted, tx_policy, requests_allowed }`, `AddrInfo`	`Link`	`const`; power-up bus state; unusable address isolates both buses; `with_first_request_seq` sets the first rqSeq, `next_request_seq` reads it; `restarted(&addr, now_ms)` builds the Link of a Cold or Warm Reset
`on_frame(bus, frame, now_ms)`	bus, whole frame (own address or 00h first), STOP time	`Result<(), FrameDrop>`	main loop only; copies at most `IPMB_MSG_MAX` bytes; never traces
`poll(ctx, ext, now_ms, trace)`	dispatch `Context`, `&mut dyn Extension`, time, trace sink	requests taken	runs handlers (the extension for a command the table lacks); stops when the egress queue is full; steps the requester and calls `ext.poll`
`next_tx()`		`Option<(Bus, &[u8])>`	response first, else a due request; frame stays handed out until `on_tx_result`
`on_tx_result(bus, ok, now_ms, trace)`	outcome of the frame handed out	`Result<(), LinkError>`	`NoTxInFlight`, `TxBusMismatch` change nothing
`set_ipmb_state(a, b)`	`Option<BusMode>` per bus	`Result<(), LinkError>`	through `plan_ipmb_state`, all or nothing
`set_ipmb_speed(a, b)`	`Option<BusSpeed>` per bus	`Result<(), LinkError>`	through `plan_ipmb_speed`, all or nothing; `BusNotFitted` for 400 kHz on a bus that is not built in
`set_local_status(bus, status)`	`LocalStatus`		the firmware's persistent-fault decision
readers			`view`, `bus_state`, `counters`, `config`, `own_address`, `evictions`, `late_responses`, `dispatch_errors`, `egress_empty`, `egress_dropped`, `sensor_data`

11.2 Dispatch (`ipmc_core::dispatch`)

```
pub type Handler = fn(&mut Context, &Frame, &mut [u8]) -> Result<usize, u8>;
```

`Ok(n)`: n response data bytes written after the completion code. `Err(code)`: that completion code with no data. For NetFn 2Ch the frame data and the output slice both exclude the VSO byte.

`Context` field	Owner (only writer)	Read by
`addr: AddrInfo`	`addr::resolve` at start-up; `nvmro_asserted` by the NVMRO sample	Write FRU Data, Get FRU Address Info, initial checks
`sensors: SensorSnapshot`	`update_sensors`	Get Sensor Reading 2..4
`fru: FruStore`	Write FRU Data; `set_fru` at boot and `FRU RESTORE`	Read FRU Data, self test, `Persist::poll`
`sdr: SdrRepository`	Reserve Device SDR Repository (reservation)	Get Device SDR, Get Device SDR Info, locator
`self_test: SelfTest`	`Context::new`, `set_fru`, Write FRU Data	Get Self Test Results, initial checks
`fru_state: FruMachine`	`update_sensors`	Get Sensor Reading 0
`link: LinkView`	`Link::poll` before every dispatch	Get VSO Capabilities, Set IPMB State, Get Sensor Reading 1
`pending_ipmb: Option<PendingIpmbState>`	Set IPMB State handler	`Link::poll` (applies and clears)
`event_enables: EventEnables`	`Context::new` (power-up values by Tier); the Tier-3 Set Sensor Event Enable handler	`read_sensor` (status bit 7), the event generator
`pending_reset: Option<ResetRequest>`	the Cold Reset and Warm Reset handlers (extension)	`ResetGate` owner: the firmware main loop, the emulator (cleared by the reset; `Context::warm_reset` clears it for a Warm Reset)
`pending_power: Option<PowerRequest>`	the Tier-3 Set FRU Activation handler (extension)	the owner of the payload power, through `Monitor::apply_power_request` (On asks `ext::check_power_on`)
`pending_payload_reset: bool`	the Tier-3 FRU Control Cold Reset handler (extension)	the owner of PAYLOAD_RST_OUT_N: `PayloadResetPulse` (section 8.2)
`deactivation_requested: bool`	the Tier-3 activation handlers (extension)	`update_sensors`, at Tier-3 only (M4 to M5)
`payload_mode: PayloadMode`	the Tier-3 Set Payload Mode handler (extension)	`read_sensor` (sensor 7)

`Context` field	Owner (only writer)	Read by
settings byte, in `fru` (`nv_settings`, `set_nv_settings`)	the Tier-3 Set FRU State Policy Bits handler (extension), the boot load (`set_fru`)	the extension, `Persist::poll` (saved with the image)

Other entry points: `Context::new(addr)`, `set_fru(store)`, `initial_checks()`, `update_sensors(now_ms, &snapshot, trace) -> Option<Transition>`, `read_sensor(number, &buses, out) -> Result<usize, SensorReadError>` (`read_sensor_at(tier, ..)` for a given Tier; `out` holds `sensor::READING_LEN_MAX` = 4 bytes), `warm_reset(link_view)`, `dispatch(ctx, req, arrival, arrival_ms, queue, trace)` (the Tier-1 table and `NoExtension`), `dispatch_with(table, ext, ctx, ...)` and `build_response_with` (the one frame builder behind `dispatch_with` and the tests).

11.3 Hardware-facing traits

Trait	Crate	Methods	Implemented by
`FlashLog`	`ipmc-fru::persist`	`sector_len`, `read`, `program` (whole words), `erase`, `millis`	`bsp::flash::Flash`, `PausedErase`, the test flash in `persist/tests.rs`
`IpmbMaster`	firmware `bsp::ipmb_master`	`state`, `state_mut`, `write_once`; provided `write`, `pins`, `recover_if_stuck`, `lines_high`, `clear_bus_fault`, `stats`, `speed`, `set_speed`	`BitbangMaster`, `HwMaster`
`AnalogSource`	`ipmc-sensors::hal`	`vdda_mv`, `die_temperature_dc`, `rail_mv`	ADC1, simulator, INA226, MCP3208
`DigitalInputs`	`ipmc-sensors::hal`	`level_high`	board GPIO, simulator, PCA9555
`PowerSwitch`	`ipmc-sensors::hal`	`set_enable`, `is_enabled`	board GPIO, simulator, PCA9555
`ResetOutput`	`ipmc-sensors::hal`	`set_reset(asserted)`	board GPIO (PA9, open drain), the test pin of `reset_pulse.rs`
`Trace`	`ipmc-core::trace`	`line(fmt::Arguments)`	firmware console, emulator stdout, `NoTrace`, `RecordingTrace`
`Extension`	`ipmc-core::ext`	`command`, `poll`, `on_sample`, `warm_reset`, `power_on_allowed`	`NoExtension`; the Tier-2/3 extension through `ipmc_ext::Active`
`MenuOutput`	`ipmc-menu`	`write_str`	firmware UART output

Trait	Crate	Methods	Implemented by
`AppHost`	`ipmc-app::host`	`version`, `status`, `fru_state`, `unit_field`, `set_unit_field`, `ipmi`, `save_state`	`LinkHost` in the firmware `app_port.rs`, `FakeHost` in the tests
`Names`	`ipmc-core::capture`	`command(netfn, cmd)`	`CoreNames` (Tier-1 table, extension, sent requests)

11.4 Concurrency model

- Single core, one thread (the main loop) plus interrupts. Interrupt handlers (I2C EV/ER at NVIC priority 1 << 4, above USART1 and USART2 at 2 << 4, and SysTick) move bytes, stamp times and bump counters; the SysTick handler also evaluates the LED pattern and writes PA5. The I2C handlers run above the UART so a slow handler does not stretch SCL past T8.
- Hand-off from interrupt to thread is lock-free: one `heapless::spsc` queue per bus, producer in the interrupt, consumer in `ipmb_link.rs`. Driver statics use `Mutex<RefCell<Option<T>>>` through `static_slot.rs`, every access inside a short critical section.
- Counters shared between interrupt and thread are single-word atomics with Relaxed ordering.
- `Link`, `Context`, handlers, the console and the trace sink never run in an interrupt. Nothing in `ipmc-core` blocks; the blocking calls are the master write (bounded by T2 + T1), a console message while the transmit ring is full (86.8 us per byte, bounded per byte; otherwise the message goes into the ring and the interrupt sends it), a sensor sample (about 2 ms of polled ADC conversions, 23.4 us each), a flash word program (up to 100 us per word) and a flash erase (at most 500 ms, IPMB slaves off).
- A flash erase or program stalls every flash access, interrupt handlers included, because code runs from the same bank.

12. Build, Integration, and Configuration

12.1 Toolchain

Rust stable with the GNU host toolchain, targets `thumbv7em-none-eabihf` and `x86_64-pc-windows-gnu`; `gen.ps1` uses `arm-none-eabi-objcopy` for the `.bin` and `arm-none-eabi-size` for the section sizes. cargo is not on PATH in every shell (Bash: `export PATH="$HOME/.cargo/bin:$PATH"`). The firmware release profile uses `opt-level = "s"`, `lto = true`, `codegen-units = 1` and keeps debug information.

```
# host crates: format, lint, test
cargo fmt --all -- --check
cargo clippy --workspace --all-targets -- -D warnings
cargo test --workspace
# firmware (its own workspace)
powershell -NoProfile -ExecutionPolicy Bypass -File crates/ipmc-nucleo-f401re/gen.ps1 build
powershell -NoProfile -ExecutionPolicy Bypass -File crates/ipmc-nucleo-f401re/gen.ps1 flash
powershell -NoProfile -ExecutionPolicy Bypass -File crates/ipmc-nucleo-f401re/gen.ps1 console "ipmb st
# emulator
cargo run -p ipmc-emu -- replay vectors/sbc_ab.vec --b-fitted 1 --b-enabled 1 --policy alternate --qui
cargo run -p ipmc-emu -- serve --b-fitted 1
cargo run -p ipmc-emu -- sdr
```

The two build modes (section 4) use the same commands. With the extension folder next to the repository the build is the commercial release build; `IPMC_TIER23=0` forces the public Tier-1 build and `IPMC_TIER23=1` demands the folder. `gen.ps1 build` sets `IPMC_TIER23` from the folder, and `gen.ps1 build|flash -Eval` makes the evaluation image (section 8.4). The numbers of the capture replay (section 13.2) belong to the Tier-1 build.

`gen.ps1 flash` takes SWD when OpenOCD is installed and finds the ST-LINK probe (`openocd -f interface/stlink.cfg -f target/stm32f4x.cfg -c "program <release ELF> verify reset exit"`) and says which path it took. It copies the `.bin` to the `NOD_F401RE` drive when OpenOCD is absent or reports no probe, and prints `Flash OK` only after the copy, the drive check and the `FAIL.TXT` check pass. `-Swd` and `-Drive` force a path (both together are refused; `-Swd` never falls back), and a SWD program that starts and fails is an error, never a silent fallback. SWD needs the WinUSB driver on the ST-LINK debug interface. The two paths leave the FRU store in different states (section 9.1). `gen.ps1 console <command>` reads a reply until the console has been quiet for 600 ms, up to 60 s and 131072 bytes, so a full `DEBUG DUMP` (about 25 KB, 2.3 s at 115200 baud) comes back whole.

12.2 Changing a tunable

Edit the define in `ipmi_defs.h` and rebuild; `ipmc-defs` reruns `build.rs` whenever the header changes. A new define also gets its type line in `ipmc-defs/src/types.rs` and, when it has a range rule, a guard in `ipmc-defs/src/lib.rs`. A value fixed by a specification is not a define: it is a named constant in the owning module with its rule beside it.

12.3 Adding an IPMI command

1. Write the handler in `ipmc-core/src/cmd/<netfn>.rs` with the `Handler` signature; validate the request length first (C7h), write only data bytes, return `Err(code)` for refusals.
2. Add one row to `COMMANDS` in `commands.rs`.
3. Run `python -I tools/sync_ipmi_commands.py` so `Documentation/ipmi_commands.md` matches (`cargo test` fails until it does).
4. Add a table-driven test in `ipmc-core/tests/tier1_handlers.rs`; check the replay totals still read match=46 same_cc=40 cc_diff=134 not_implemented=89, or state why they change.

12.4 Adding an application port command

1. Write the handler in `ipmc-app/src/commands.rs` with the `Handler` signature (request, host, output): it writes one line, `OK ...` or `ERR <CODE> <text>`, and nothing before it knows which.
2. Add one row to `COMMANDS`: the keyword in capitals, the argument summary, one sentence of help and the handler. `HELP` lists the row at once.
3. A value the host does not offer yet is a new method of `AppHost` (`host.rs`), implemented in `LinkHost` (`app_port.rs`) and in `FakeHost` (`tests.rs`); the compiler lists the places that miss it.
4. Add one row to `REQUEST_TABLE` in `tests.rs` (the request line and the answer the fake host must produce) and run `cargo test -p ipmc-app`.
5. Run `cargo clippy --workspace --all-targets -- -D warnings` and `python -I tools/dupcheck.py` (a handler that repeats another handler's code fails it), and add the row to section 7.6.

12.5 Adding a sensor part

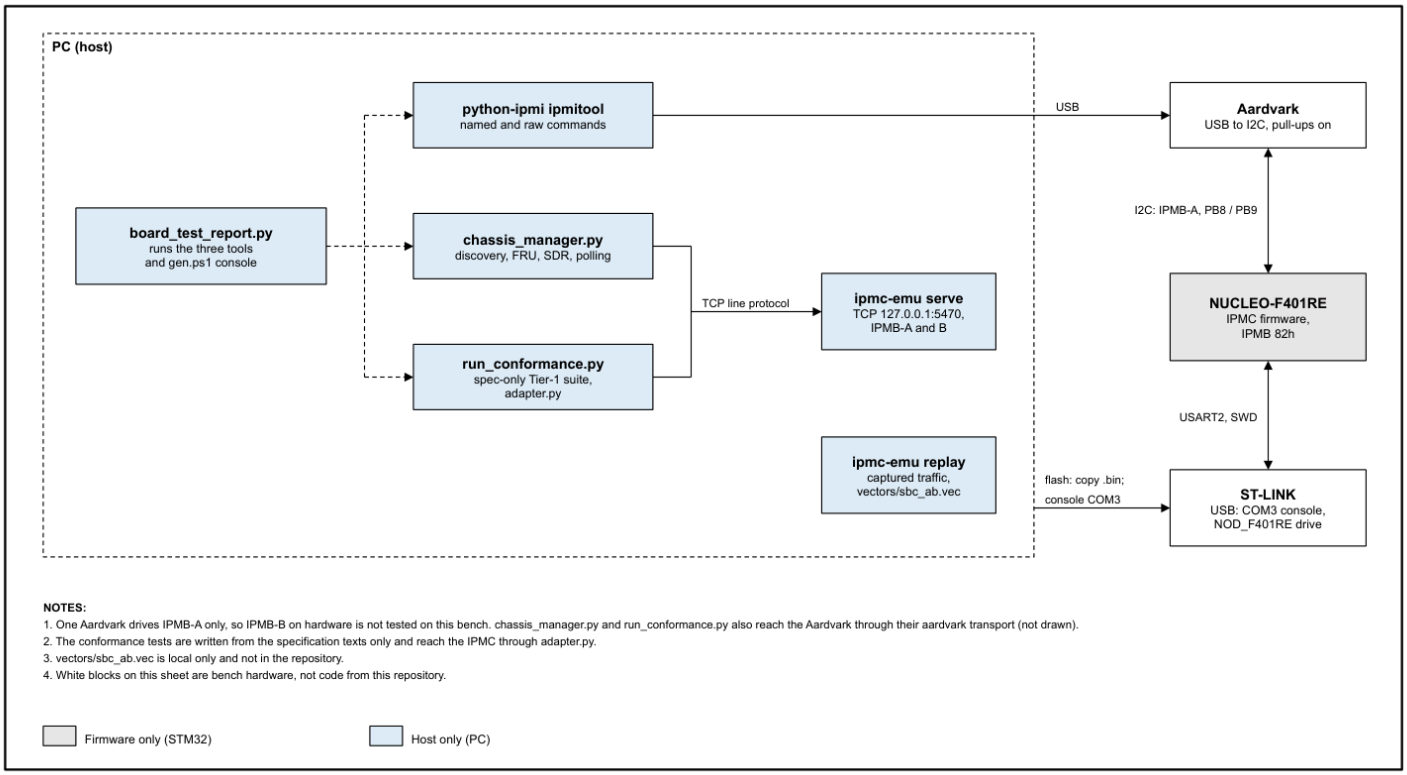
Write `drivers/<bus>/<part>.rs` implementing the matching `hal` trait against the embedded-hal 1.0 bus trait, test it against the crate's mock bus, hold it in the router in `routing.rs`, and select it in `ipmi_defs.h` with `IPMC_SRC_I2C` or `IPMC_SRC_SPI`. The firmware build refuses those values until a bus that is not an IPMB bus is assigned (`sensing.rs` guard).

12.6 Gates before handing work back

`cargo fmt`, `cargo clippy -D warnings` and `cargo test` on the workspace; firmware `gen.ps1 build`, `cargo clippy --release -- -D warnings` and `cargo fmt -- --check` inside `crates/ipmc-nucleo-f401re`; `sync_ipmi_commands.py --check`; `captured_sdr.py --check`; `dupcheck.py`; the name check against the local banned-name list; and for link or handler changes the replay totals above. With the extension folder present, a tenth gate repeats `cargo fmt`, `cargo clippy` and `cargo test` with `IPMC_TIER23=0` (the public Tier-1 build), runs the replay totals with `IPMC_TIER23=0` (they belong to the Tier-1 build), checks the extension's library source with `rustfmt --check --edition 2021`, adds the extension folder as a second path to the name check, and lints the evaluation code with `IPMC_EVAL=1` in the firmware clippy gate; `dupcheck.py` scans both repositories by itself.

13. Verification

Figure 13-1. Build, test and bench setup (ipmc-software, sheet 4)



Source: `diagrams/src/ipmc-software.js`, sheet 4.

13.1 Host tests (585 in the public build, `cargo test --workspace` with `IPMC_TIER23=0`, 10-25-2026)

Crate	Covers
`ipmc-core`	codec, address decode, link layer (bus speed, restarted link, the failure lines of a transmission), requester timing, reset gate, Tier rules, FRU state machine including M5, Payload Mode sensor, speed byte per Tier, event enables, extension hook and the power-on check, duplicate cache, queues, byte ring, capture log (records, wrap, drop order, CSV and trace rows, dump cursor, cut lines), dispatcher, handlers (`tests/tier1_handlers.rs`), Tier-1 link rules, SDR walk, command document in step (`tests/commands_doc.rs`)
`ipmc-app`	line assembler (CR LF, long line, bad byte), every request of the protocol against the test host (`REQUEST_TABLE`), tags, every error code, `WRITE_DISABLED`, the `HELP` listing
`ipmc-sensors`	evaluation, power state machine, routing, monitor and the power-request path, payload reset pulse timing, pin claims, LED level, example drivers against mock buses

Crate	Covers
`ipmc-emu`	vector reader, captured frames decode and re-encode byte-identically, replay runs including the Event Receiver role, Cold and Warm Reset rebuild, the flash copy across a power cycle, serve protocol over TCP, command line
`ipmc-fru`	image build, decoder, `verify`, store, the unit field editor (every field, every refusal, a MAC address in a serial), the single-sector flash log including a power cut at every word, the erase of a full sector and the settings byte
`ipmc-menu`	line editor, matching, help, table validation
`ipmc-defs`	parser rules (every malformed line form is rejected with its line number), the Tier-2/3 detection rules (`tests/tier23.rs`) and the IPMB timing arithmetic (`i2c_timing.rs`)
`ipmc-ext`	the default `Active` answers nothing for an unowned command; the bridge and `ipmc-defs` agree on the extension
`ipmc-wire`	checksum, builder, CRC-32, hex parser

The per-crate counts are in the `cargo test --workspace` output. The Chassis Manager request policy and the Event Receiver have 100 Python unit tests (`tools/chassis-manager/test_cm_link.py`); `tools/capture/test_capture2pcap.py` checks the pcap blocks (with tshark, when `C:\Program Files\Wireshark\tshark.exe` exists) and `tools/app-port/test_app_cli.py` checks the bench client against a fake transport. Tests are table-driven; `tools/dupcheck.py` fails near-identical test pairs as well as production pairs.

13.2 Capture replay

`ipmc-emu replay` feeds the 309 captured requests of a live Chassis Manager and SBC IPMC (IPMB-A and IPMB-B on one time base). With IPMB-B fitted and enabled and the alternate policy the result is match 46, same CC 40, CC differ 134, not implemented 89, every request answered exactly once. The differences are designed: the capture's IPMC is an earlier-revision controller that answers commands the 2022 specification makes not applicable to Tier 1 (C1h here), and the Device ID, FRU and SDR contents are this IPMC's own.

13.3 Conformance suite and Chassis Manager

`tools/conformance/` holds one test per testable Tier-1 requirement (157 rows in `REQUIREMENTS.md`), written from the specification texts only, against `TRANSPORT.md`, by an author who did not read this code. A failure is fixed in the IPMC or disputed with a specification citation, never by editing a test. Results on 10-23-2026: emulator 102 PASS, 0 FAIL, 55 NOT-TESTABLE; board 97 PASS, 0 FAIL, 60 NOT-TESTABLE (pins, electrical checks, second bus, optional features and NVMRO pin control cannot be observed on the one-Aardvark bench). The Chassis Manager emulator completes discovery, FRU and SDR reads, the locator check and polling on both targets, and its cross-check against the captured Chassis Manager reports 0 unexpected differences. The Phase 10 results are in section 13.6.

13.4 Board test report

`python tools/board_test_report.py` runs against the board over I2C and writes `Documentation/board_test_report.md`: environment (7 checks), command line through python-ipmi (27), Chassis Manager (12), conformance (157 tests and a totals row: 98 PASS with the totals row, 60 NOT-TESTABLE), link health from the console counters (8) and FRU persistence across a reset with `FRU RESTORE` (8). The 10-23-2026 run on the release image (81,360 bytes) records 160 PASS, 0 FAIL, 60 NOT-TESTABLE; 950 requests received and 950 responses sent, every master and driver error counter at 0.

13.5 Checks not yet run on hardware

The bench wiring of APP_UART uses no handshake (`Documentation/nucleo_f401re_pinout.md`, section 7), so RTS/CTS (`IPMC_APP_UART_FLOW` 1) has not run on hardware. Scope measurement of SCL/SDA in both master modes, at 100 kHz and in FAST-MODE (the FAST-MODE periods of section 10.9 are DWT-counter measurements); clock-stretch timeout and stuck-line jumper tests; arbitration loss with a second master; start-up fault recovery with a line held low; IPMB-B on hardware, with events and failover and the Tier-3 rules for two buses, FAST-MODE on both buses (needs a second Aardvark, one per IPMB); the Tier-3 activation cases that need PGOOD to follow PWR_EN (a jumper from PWR_EN, PB5, to PGOOD, PB10; the emulator models it); NVMRO from the strap pin; heartbeat LED by eye; power and fault input jumpers; Broadcast Get Device ID through a second independent tool. `.claude/PLAN.md` section 7 lists the procedure and expected result of each.

13.6 Phase 10 acceptance (10-24-2026)

Check	Build	Result
Tier-1 conformance suite on the emulator	public Tier-1	102 PASS, 0 FAIL
Chassis Manager on the board: events, withheld answers and a Cold Reset	commercial release build	14 of 14 checks pass
Tier-2 behaviour (section 7.2)	commercial release build	accepted on the emulator and the board
Tier-3 behaviour (section 7.2)	commercial release build	accepted on the emulator; on the board every case the one-Aardvark bench can reach (IPMB-A only)
Tier-3 by hand on the board	commercial release build	FAST-MODE on IPMB-A answers at 400 kHz (bit-bang SCL period 6795 ns, hardware master 2981 ns); FRU Control Cold Reset 00h; Set Payload Mode 1; Warm Reset keeps the payload mode, the SEL and the FRU state without a processor reset
Evaluation image with a 1-minute period	commercial release build, `-Eval`	answers until 52 s, IPMB stopped at 59.7 s; the normal image has no evaluation code (equal `.text`, no evaluation symbols or strings)

13.7 Phase 11 bench facts (10-25-2026)

Item	Result
Run-time sector erase of the FRU store	the IPMB was silent for 198 ms (console line `FRU log: erased the store sector at run time (IPMB paused 198 ms)`), against DS10086 Table 45 typical 250 ms and maximum 500 ms for a 16 KB sector at x32 programming; one erase per 60 saves, and only when the bus is quiet
python-ipmi request sequence	every request goes out with sequence number 01h, so identical requests inside the duplicate-cache window (`IPMB_DUP_WINDOW_MS`) are answered from the cache, which is the IPMB rule; a bench loop varies the request
`gen.ps1 console` reply	read until 600 ms of silence, at most 60 s and 131072 bytes per reply; a full `DEBUG DUMP` (about 25 KB, 2.3 s at 115200 baud) comes back whole
Capture log	`IPMC_IPMI_TRACE` 0 removes the link and dispatcher lines from the log as well as from the console

14. Example Sequences

14.1 Bring-up with a stored FRU image

1. Reset; `reset_cause::take` reads RCC_CSR; the clock reaches 84 MHz.
2. Pins claimed, console up, application port up, IPMB pins parked, strap pins read: GA forced to 01h, IPMB 82h.
3. `FruLog::boot`: the store region is checked against sector 1 and the sector is scanned; the valid record with the highest sequence number is loaded, and nothing is erased.
4. I2C1 slave armed at 7-bit address 41h (OAR1 = 4082h), general call on; I2C3 stays off.
5. `Context::new` builds the image and SDRs, `set_fru` installs the stored image and the self test runs on it: banner `Self test: 55h 00h (passed)`.
6. First sensor sample: M0 to M1 (initial checks pass); one step makes one transition, so the next sample, one period later, takes M1 to M4 once PGOOD is seen.

14.2 Get Sensor Reading of sensor 1

1. Request on IPMB-A: `82 10 6E 20 14 2D 01 9E` (NetFn 04h, sequence 05h, sensor 1).
2. The EV interrupt posts the frame; `on_frame` accepts it; `poll` misses the cache.
3. `run_request`: LUN 0, NetFn 04h; `commands::find` gives `get_sensor_reading`.
4. `Context::read_sensor(1, &link.buses, out)` encodes the link view: `08 40 02`.
5. Response `20 14 CC 82 14 2D 00 08 40 02 F3` queued, `next_tx` picks IPMB-A (the only enabled bus), the bit-bang master writes it, `on_tx_result(A, true)` pops it.

14.3 Write FRU Data, reset, read back

1. `raw 0x0a 0x12 0x00 0x35 0x00 0x32` writes one serial byte (response `00 01`); the self test reads 57h until step 2.
2. A second write corrects the board area checksum; the self test returns to 55h 00h.
3. After `IPMC_FRU_PERSIST_DELAY_MS` without another write the save appends a record (`FRU log: saved record N at offset M`).
4. `SYS RESET`; boot loads the new record; Read FRU Data returns the written bytes.
5. `FRU RESTORE` saves the built image as a new record; after a reset the original serial returns.

14.4 Failover in the emulator

1. `ipmc-emu serve --b-fitted 1`; the Chassis Manager enables IPMB-B with Set IPMB State.
2. The client sends `#bus A down`; the next response handed out on IPMB-A fails (`on_tx_result(A, false)`), moves to IPMB-B and is delivered there.
3. Requests the Chassis Manager sends on IPMB-A are lost; it retries on IPMB-B with the same sequence number; a retry of an answered request is replayed from the cache.

14.5 Capture a failure and open it in Wireshark

1. The log records from start-up (`IPMC_LOG_AT_BOOT` 1); `DEBUG LOG ON` starts it otherwise, and `DEBUG CLEAR` empties it before a run.
2. Reproduce the problem.
3. Start a terminal log and type `DEBUG DUMP`; the console prints the header line, the CSV rows and `# end of capture`. A key press stops a dump early.
4. `python -I tools/capture/capture2pcap.py dump.txt -o dump --csv` writes `dump.pcap` and `dump.csv`.
5. Open `dump.pcap` in Wireshark (link type 260). A frame that did not leave the board shows as the frame followed by a text packet `tx failed: addr-nack`, naming the cause.

14.6 Write the unit fields at the first boot

1. The payload CPU sends `HELLO` and reads `OK IPMC-APP 1 ...`, then `FIELDS` for the longest value of each field.
2. It sends `SET BOARD_SERIAL 00:11:22:33:44:55`, `SET PRODUCT_SERIAL ...`, `SET BOARD_PART ...`, `SET PRODUCT_PART ...`, `SET ASSET_TAG ...` and `SET MFG_DATE 2026-10-25 14:30`; each answer echoes the value as stored.
3. After `IPMC_FRU_PERSIST_DELAY_MS` without another write, the FRU log saves the image (`FRU log: saved record N at offset M`).
4. The payload CPU repeats `SAVE` until it reads `OK SAVED` and may then remove power.
5. A Chassis Manager's Read FRU Data returns the new values and the self test still reads 55h 00h.

15. Constraints and Limitations

- **Tier 1 in the public build.** Optional and Tier-2/3 commands answer C1h; there is no event generation, no FRU activation control by the Chassis Manager and no subsidiary FRU. The commercial release build adds Tier-2 and Tier-3 (section 7.2), Tier-3 by default.
- **Tier-3 runs on IPMB-A only on this bench.** Rule 4.1.1-1.1 asks a Tier-3 IPMC for both buses; the one-Aardvark bench has IPMB-A, so a Tier-3 build with `IPMB_BUS_B_ENABLE 0` is an owner deviation. FAST-MODE is proven on IPMB-A only; the dual-bus Tier-3 build waits for the second Aardvark.
- **FAST-MODE rates.** The bit-bang master reaches about 147 kHz in FAST-MODE and the hardware master about 335 kHz; both keep the fast-mode minimum low and high times and stay below 400 kHz.
- **SEL in RAM.** The Tier-3 SEL (`IPMC_SEL_ENTRIES` records) does not survive a power cycle or a Cold Reset; a Warm Reset keeps it. A full SEL stops logging and sets the overflow flag.
- **Cold Reset drops payload power on this board.** The reset releases PWR_EN with every GPIO (section 8.2, owner decision 21). A Warm Reset and the Cold Reset of the payload do not. The Activation-Locked bit is volatile: after a Cold Reset it takes the value of Default-Activation-Locked again (Rule 4.1.5.4.1-11), so a FRU deactivated by Set FRU Activation powers its payload again after the restart unless Default-Activation-Locked is set.
- **One outstanding request.** The requester sends one request at a time; a second `submit` returns `Busy` until the first completes or is cancelled.
- **One master transmission per main-loop pass.** A response burst drains one frame per pass; the egress queue holds `IPMB_TXQ_DEPTH` (4) responses, and further requests wait in the ingress FIFO.
- **Bit-bang deafness.** While the bit-bang master transmits, the slave peripheral is off; a frame another master sends then is lost and recovered by its retry.
- **Flash erase stall.** An erase stops the core for up to 500 ms (typical 250 ms; 198 ms measured); the IPMB slaves are off the bus for that time (the Chassis Manager sees timeouts and retries). Erases are rare: the store sector holds 60 records, the erase comes once per 60 saves, and only when the bus has been quiet for `IPMC_FRU_ERASE_QUIET_MS`.
- **One-sector FRU store.** The store has no second copy (owner decision 10-25-2026). A power cut between the erase of a full sector and the end of the first record's programming loses the stored image, and the next start-up loads the built image, unit fields included.
- **Reflash.** A drive copy of the `.bin` writes FFh over the store sector and erases the stored FRU image; an SWD program of the ELF keeps it (section 9.1). The firmware has no guard against either.
- **No hardware watchdog.** A flash busy flag that never clears, or any other hang, has no automatic recovery.
- **Console output.** Output goes into a 4096-byte ring that the interrupt sends (356 ms at 115200 baud). A message waits only while the ring is full (86.8 us per byte); a trace line that does not fit is dropped whole and counted, so tracing cannot stall the IPMB. A long listing (`DEBUG DUMP`) runs as a paced job and is stopped by a key press.
- **Capture log.** The log lives in RAM and is empty after a reset or power cycle; it holds the newest `IPMC_LOG_SIZE` - 1 bytes of records (a 23-byte frame takes 29 bytes). A dump of a full log is about 25 KB.
- **Application port.** The IPMC answers one request line per main-loop pass and drops an answer the transmit ring cannot take whole. The port has no emulator model: it runs on the board only (COM5 on the bench). The six unit fields are the only FRU content it writes; `IPMI` runs requests locally and puts nothing on the IPMB. The bench runs without RTS/CTS.
- **Straps forced on the bench.** GA and NVMRO come from `ipmi_defs.h` ; the pin paths are built but not exercised end to end, and the NVMRO electrical polarity is not yet confirmed against VITA 46.0.
- **Manufacturer ID 000000h** (unspecified) until an IANA enterprise number is registered.

- **IPMB-B** is built and tested in the emulator only; the one-Aardvark bench drives IPMB-A.
- **Duplicate window 3000 ms**, not the VITA 5 s sequence expiry: a Chassis Manager that reuses a sequence number with identical data inside 3 s gets the cached response.
- **Emulator FRU store is RAM only.** Flash persistence is tested in `ipmc-fru` against a test flash and on the board. The emulator keeps a RAM copy of the flash log (`FlashCopy`) across a reset and a `#reset` power cycle.
- **No custom registers.** The firmware uses the STM32F401 peripherals through the `stm32f4` PAC; each driver names the RM0368 section of every register field it writes (I2C chapter 18, flash section 3.5, RCC chapter 6, ADC chapter 11, GPIO chapter 8).